



Alignment of the Compact Muon Solenoid's new Gas Electron Multiplier detector

Devin Aebi
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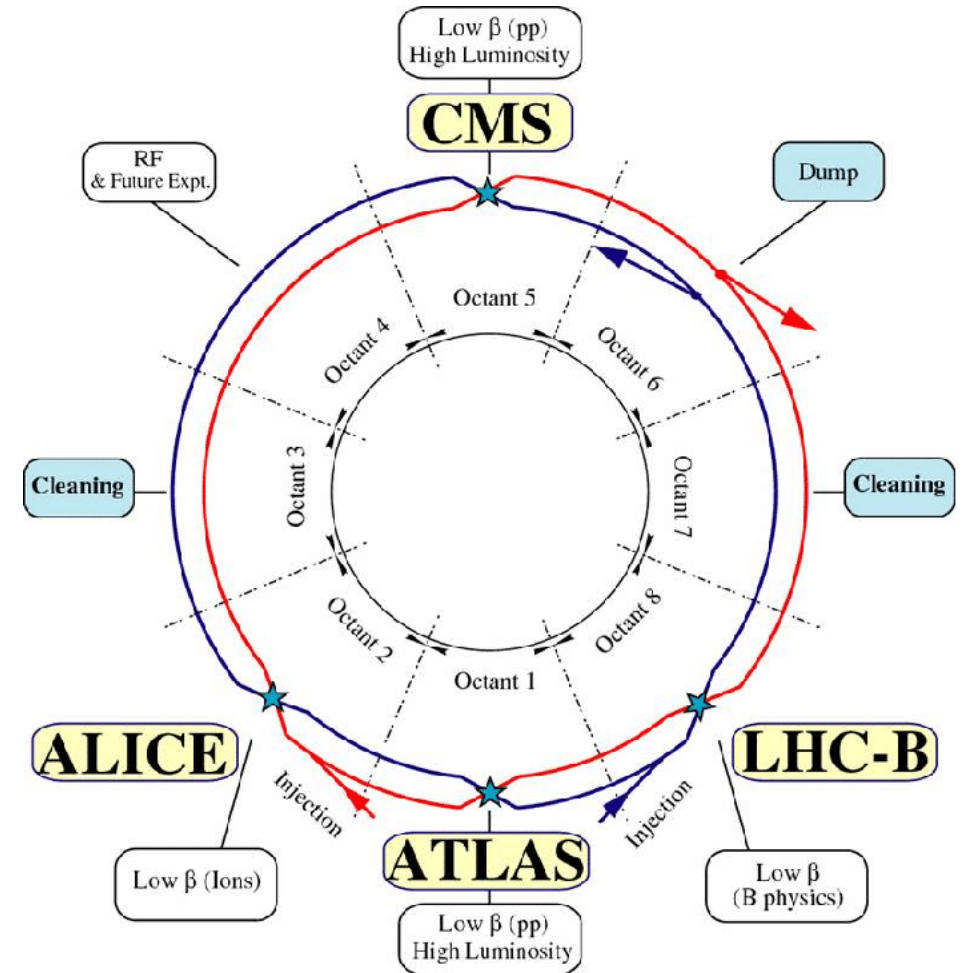
Large Hadron Collider



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The Large Hadron Collider (LHC)

- 27 km circumference ring
- Collides beams of protons and lead ions at near the speed of light
- Proton-proton collisions
 - 600,000,000 collisions per second
 - 13 TeV center of mass energy
- Built to study the Standard Model and Beyond Standard Model, some examples:
 - Higgs Boson
 - Dark Matter
 - Baryon Asymmetry
 - Extra Dimensions



4 Large Experiments

- ATLAS (A Toroidal LHC Apparatus) and CMS (Compact Muon Solenoid)
 - General purpose particle detectors
 - Same goal but different design
 - ATLAS has a solenoid magnet (2T) and toroidal magnets for muons
 - CMS built around a large solenoid magnet (3.8T)
- ALICE (A Large Ion Collider Experiment)
 - Designed to study lead ion collisions
- LHCb (Large Hadron Collider beauty)
 - Designed to study b quarks

CMS - Layers



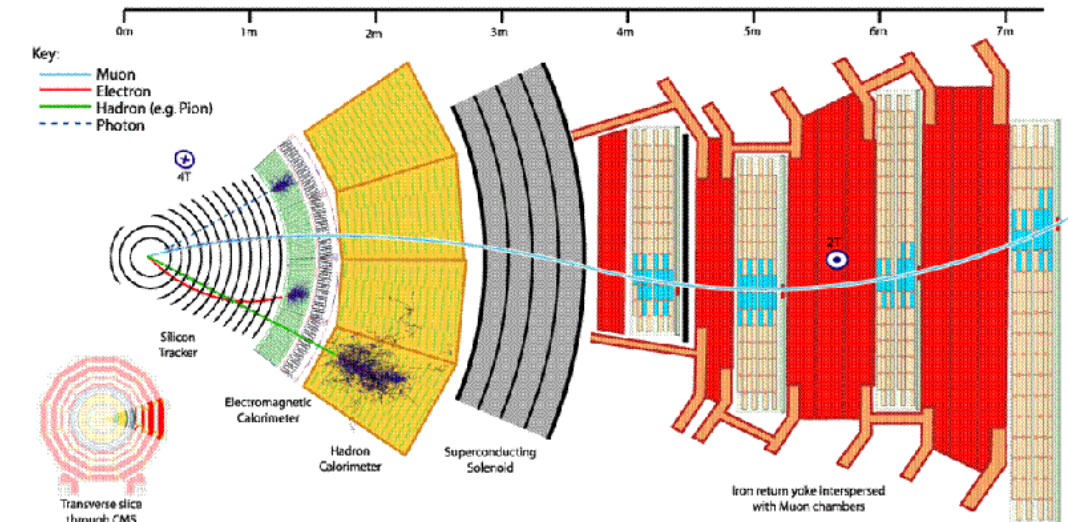
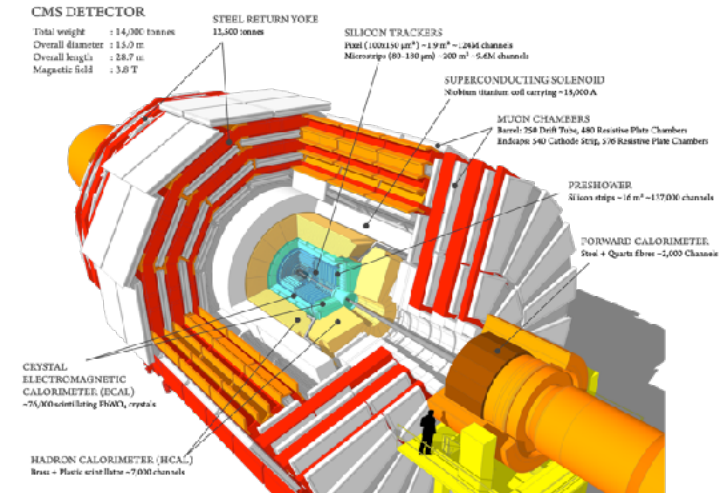
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Goal of CMS is to fully reconstruct all parts of a collision, to do this CMS was designed in layers of different detectors

- Layers are in a specific order to detect different particles most effectively

Layers from innermost to outermost

- Silicon Tracker
 - Pixel and strip technology to maximize resolution
 - Primary momentum measurement
- ECAL
 - Electromagnetic Calorimeter measures energy of charged particles (electrons, positrons, photons)
- HCAL
 - Hadronic Calorimeter measures energy of hadrons (protons, neutrons, pions, kaons)
- Solenoid (3.8T)
 - Magnetic field makes charged particles bend through the detector dependent on their momentum
- Muon System (DTs, CSCs, RPCs, GEMs*)
 - Outer most layer — muon detection is very important as many processes involve muons
 - Hits on multiple stations help reconstruct the entire muon path

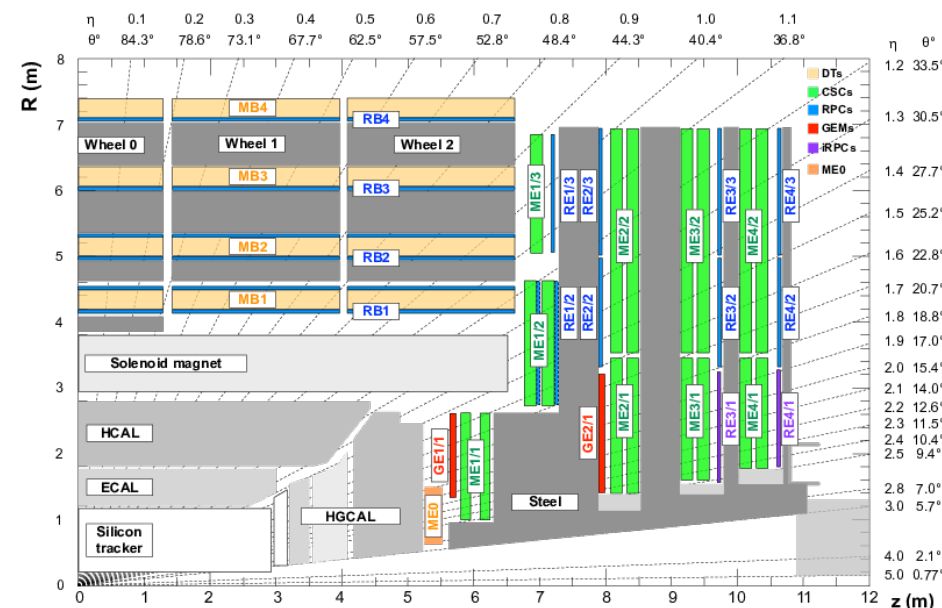
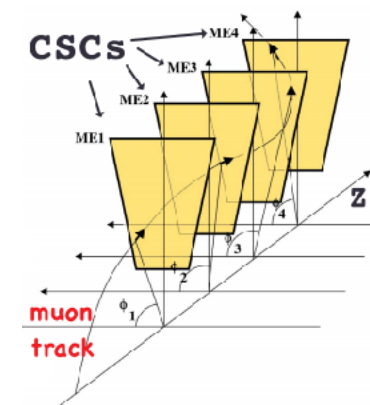
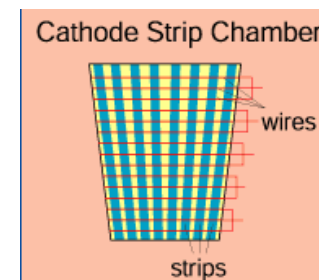
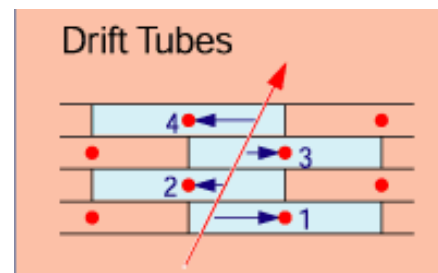


CMS - Muon System



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- Muon System — 1400 total chambers (+72)
- 250 **Drift Tubes**
 - Measure ϕ and z in barrel region
 - Stations 1-3 have 8(4) x layers (y layers)
 - Station 4 has 8(0) x layers (y layers)
- 540 **Cathode Strip Chambers**
 - Measure ϕ in endcap region
 - 6 layers
- 610 **Resistive Plate Chambers**
 - Measure ϕ in barrel and endcap regions
 - 2 layers
 - Very fast compared to DTs and CSCs
- 72 **Gas Electron Multipliers (GE1/1)**
 - New chambers for LHC planned upgrades
 - GE2/1 and ME0 will be installed in the future
- Muon System's goal
 - With hits in multiple stations, an entire muon track trajectory can be reconstructed

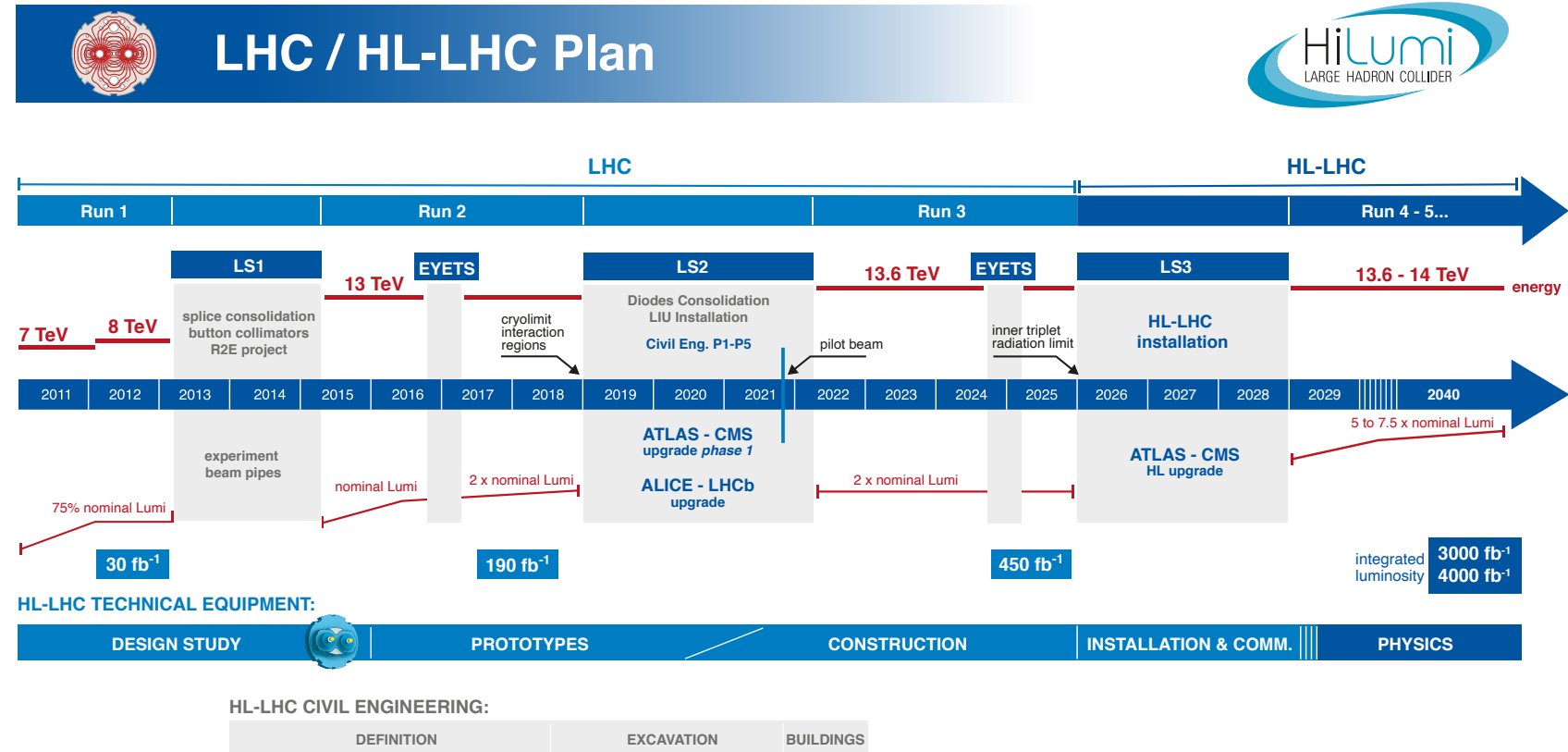


LHC Plans and HL-LHC



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- LHC has been running since 2010 with some breaks
 - Long Shutdown 2 **GE1/1** installed
 - Run 3 collisions started today, 5 July 2022**
- Long Shutdown 3 will upgrade to High Luminosity LHC (HL-LHC)
 - During LS3 upgrades are planned to increase luminosity by a factor of 5
 - Increased luminosity will increase discovery power
 - More data is good for analysis, but will pressure our current bandwidth
 - Will need upgrades to keep trigger rate manageable



Gas Electron Multiplier Detector

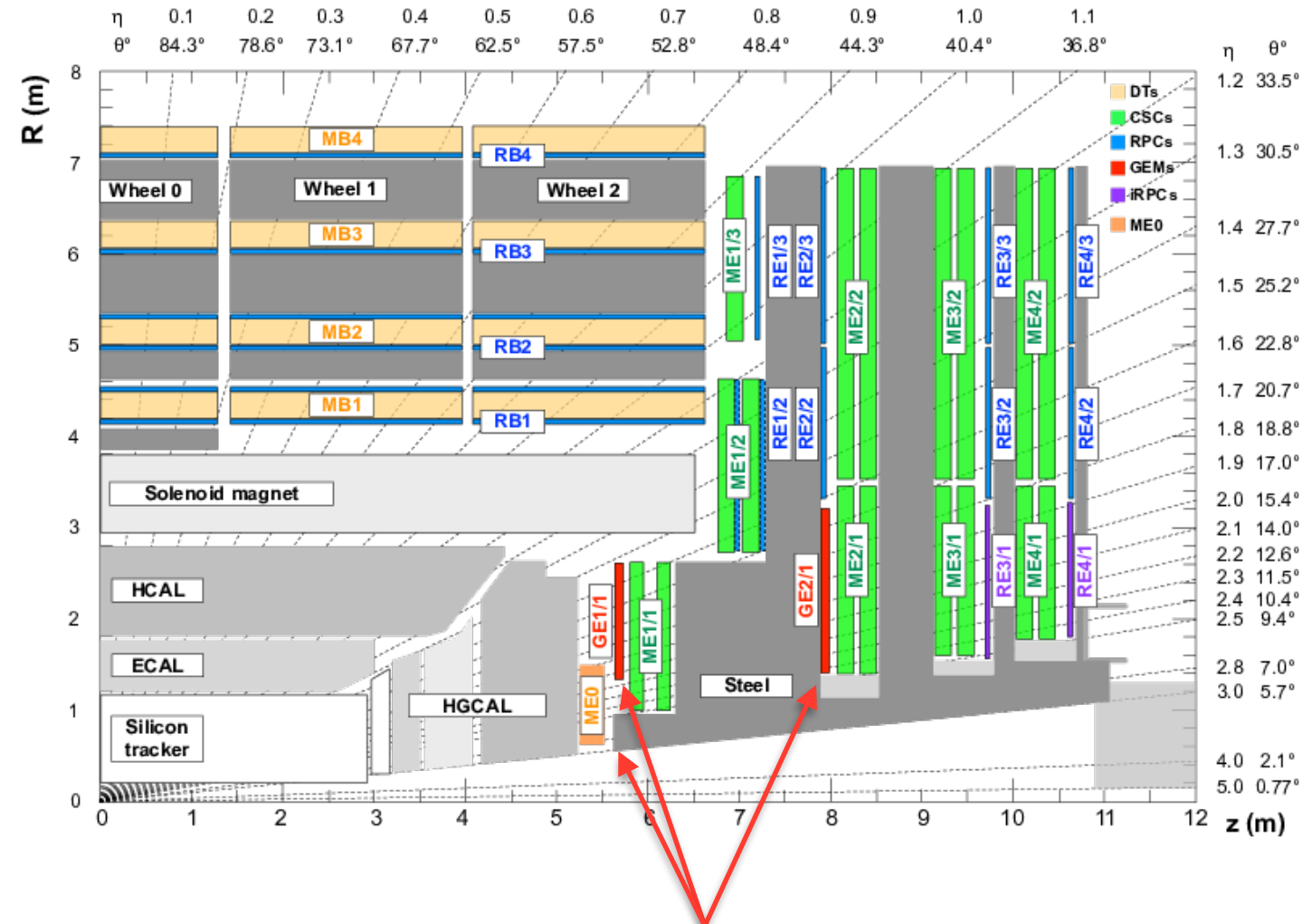
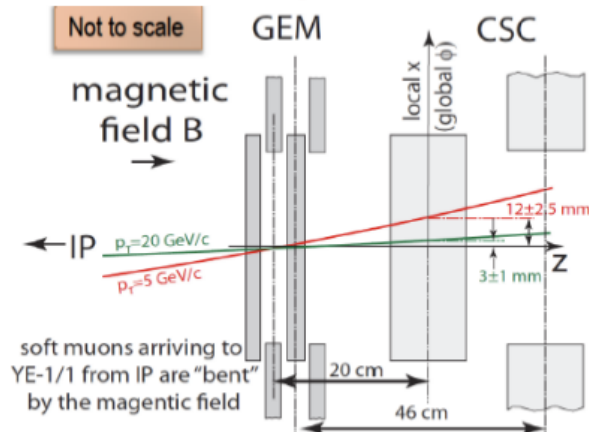


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- Gas Electron Multiplier (GEM)

- New detector to prepare for HL-LHC
- Main upgrade for the CMS Muon System
- Placed in high-background endcap region
- Close to CSC station ME1/1
- Will work as an extra 2 layers of ME1/1 chambers

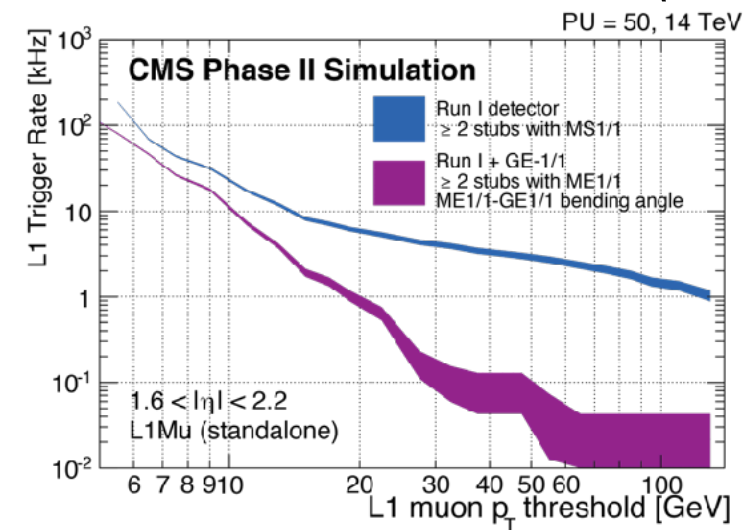
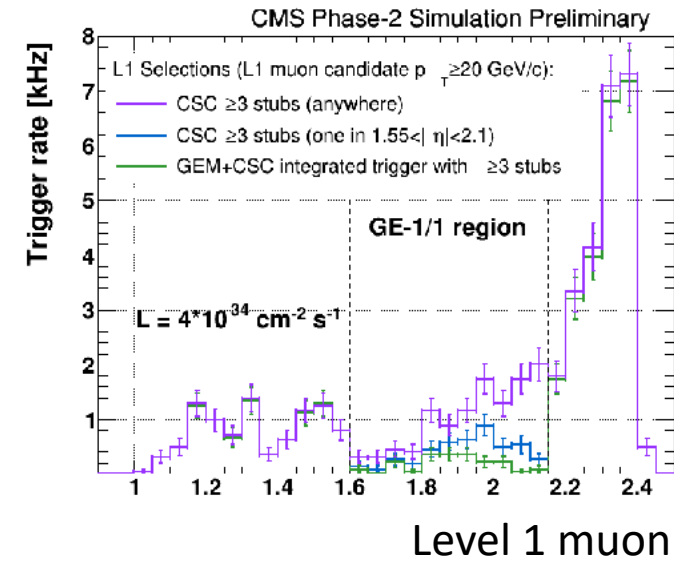
view from the top of the CMS down



GEM Motivation



- Current PP collisions occur at a rate of 40 MHz
 - Entire Level 1 trigger system fires at 100 kHz
 - Forward region ($1.6 < |\eta| < 2.2$) fires at 10 kHz
- With the increased luminosity, this will go up
 - Must add something to the trigger to keep this rate manageable
- New GEM detector in forward region will reduce trigger rate by a factor of 8 for $p_T > 20$ GeV
 - **How? By using the bending angle**

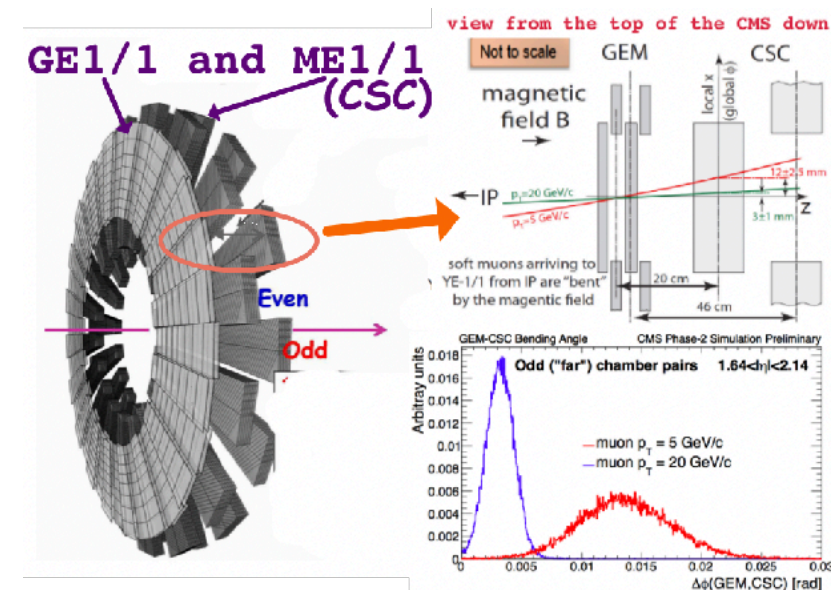
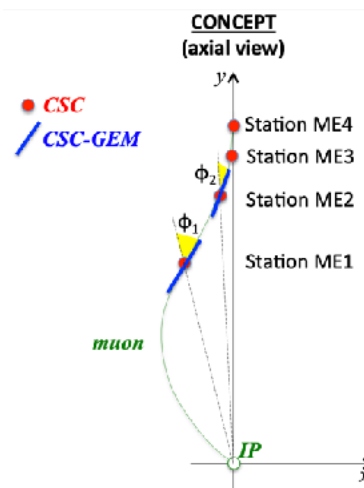
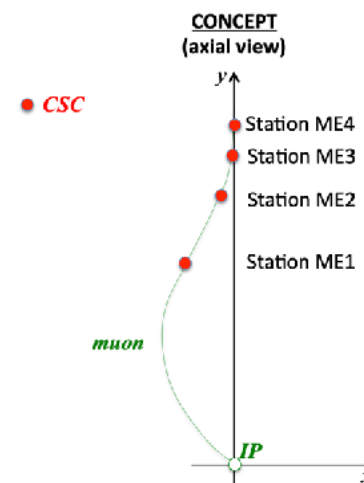


GEM-CSC Bending Angle



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- Level 1 muon trigger can be improved by using GEM-CSC bending angle
 - Bending Angle $\approx \frac{\Delta\phi}{\Delta z}$
 - Charged particles (muons) traveling through magnetic field bend
 - Current CSC detectors are too thin to measure muon bending
 - Bending between stations would go through too much material (scattering)
 - Making single chambers more thick would allow a measurement of this bending (muon momentum)
 - Lever arm between GEM-CSC is long enough that we can measure this bending (momentum) more precisely
 - GEM-CSC distances: Even (Odd) 21.94 (49.32) cm
 - No material between GEM-CSC — Reduces scattering
- The GEM system includes
 - 36 Super Chambers on each endcap
 - 2 Layers per Super Chamber
 - Measures ϕ position
 - Bending angle is a measurement of GEM ϕ and CSC ϕ
 - Any misalignment in position measurement will directly effect this bending angle

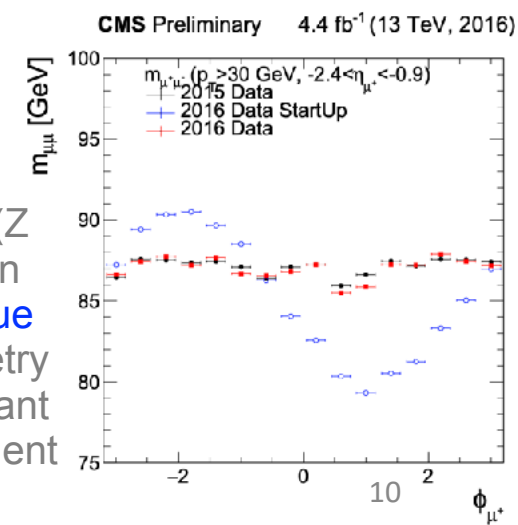
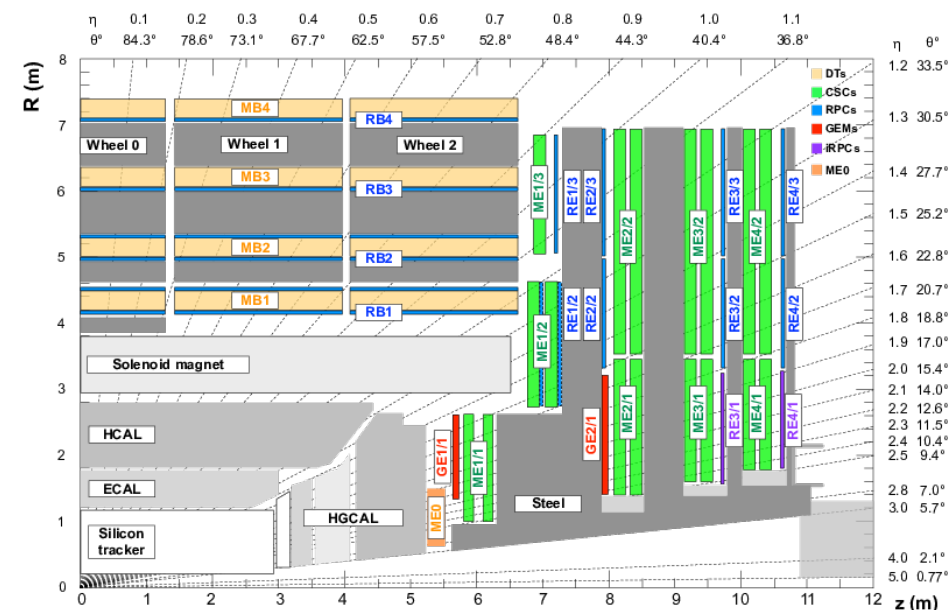


Muon System Alignment



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- Alignment is the process of adjusting geometry in software to match true detector position.
 - Alignment is not physically updating the detector position.
- Misalignment can cause incorrect particle reconstruction — bottom plot
- Texas A&M is responsible for Muon System Alignment
 - GEM Alignment method will be different due to importance of GEM-CSC relative position and bending angle



Muon System Alignment



- **Track-based muon alignment (TBMA)**

- Propagate the tracker part of muons into the muon system
- Measure residual, the difference between measured hit position and propagated hit position from propagation
- Calculate most likely geometry to match the residuals

- **The TBMA technique has been efficient, robust, and stable in Run1 and Run2**

- Sources of possible systematic uncertainties have been investigated and various improvements to reduce their effect are being developed
- **Muon system alignment is very important for muon reconstruction and TBMA has good accuracy of 100-200 μm order**

- **Sources of residuals**

- **Misalignment**

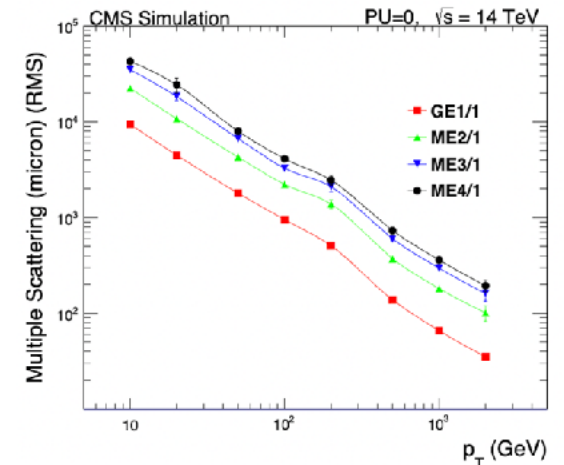
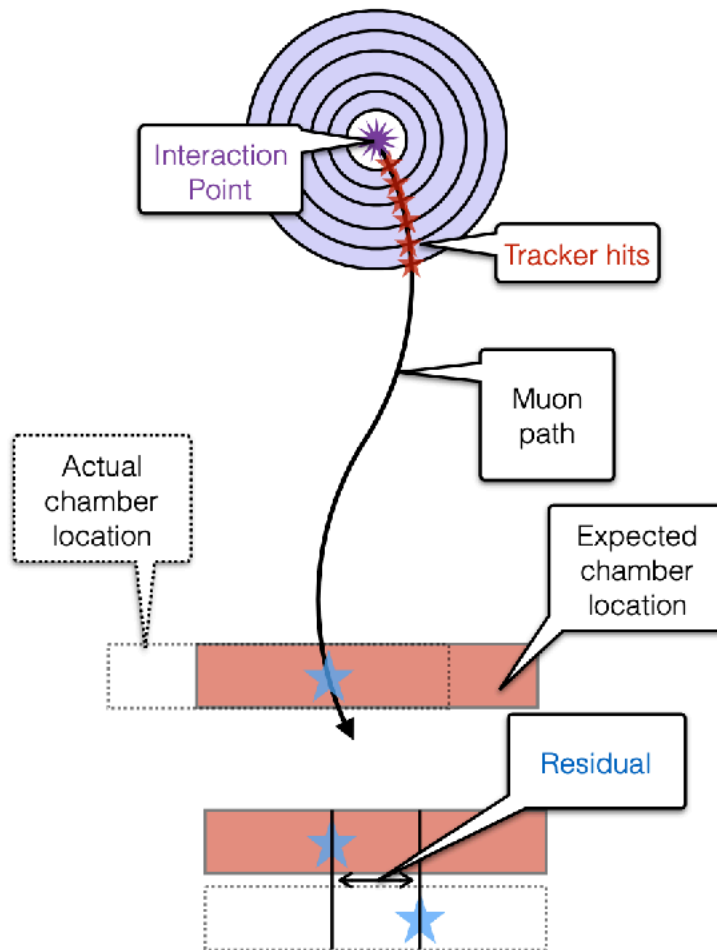
- Endcaps are moved about once per year
- Magnetic field can move detectors

- **Multiple Scattering**

- Amount of scattering depends on momentum of particle
- low p_T muons will scatter more than high p_T muons

- **GEM-CSC trigger requires good GEM alignment accuracy**

- The relative GEM-CSC position is more important than the true GEM position due to its use in the trigger
- **GEM alignment needs a new method to focus on relative position**



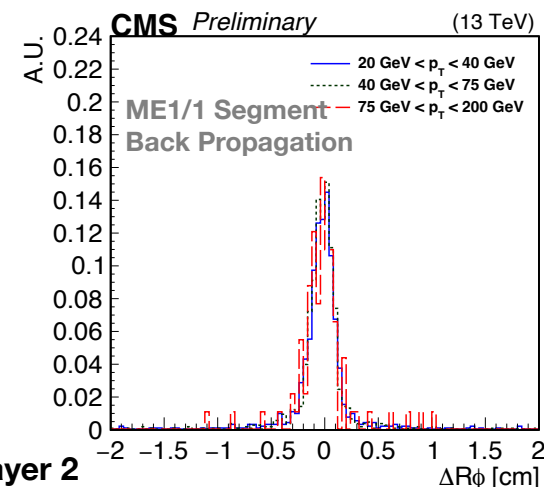
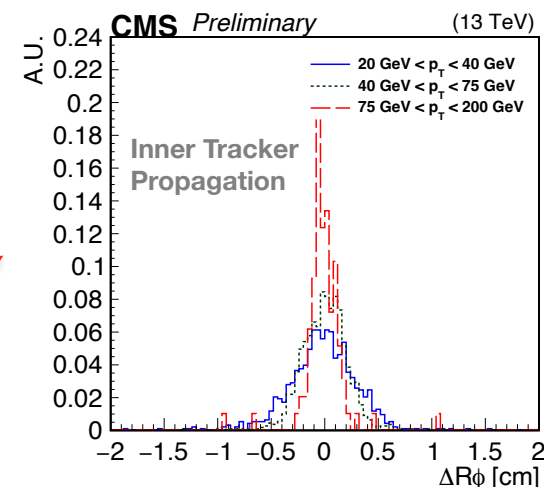
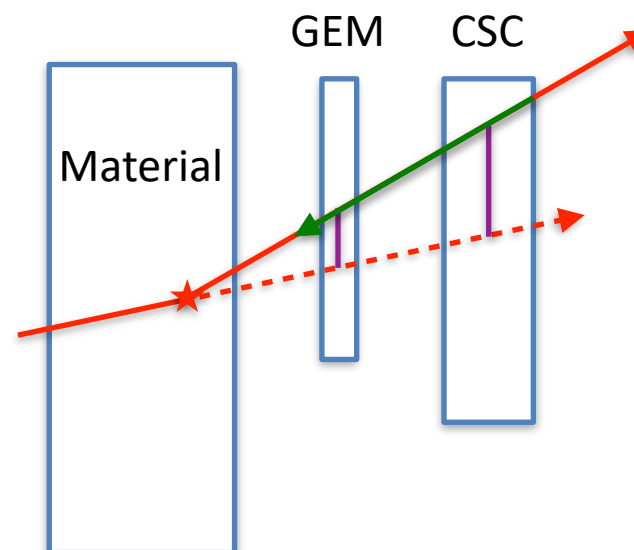
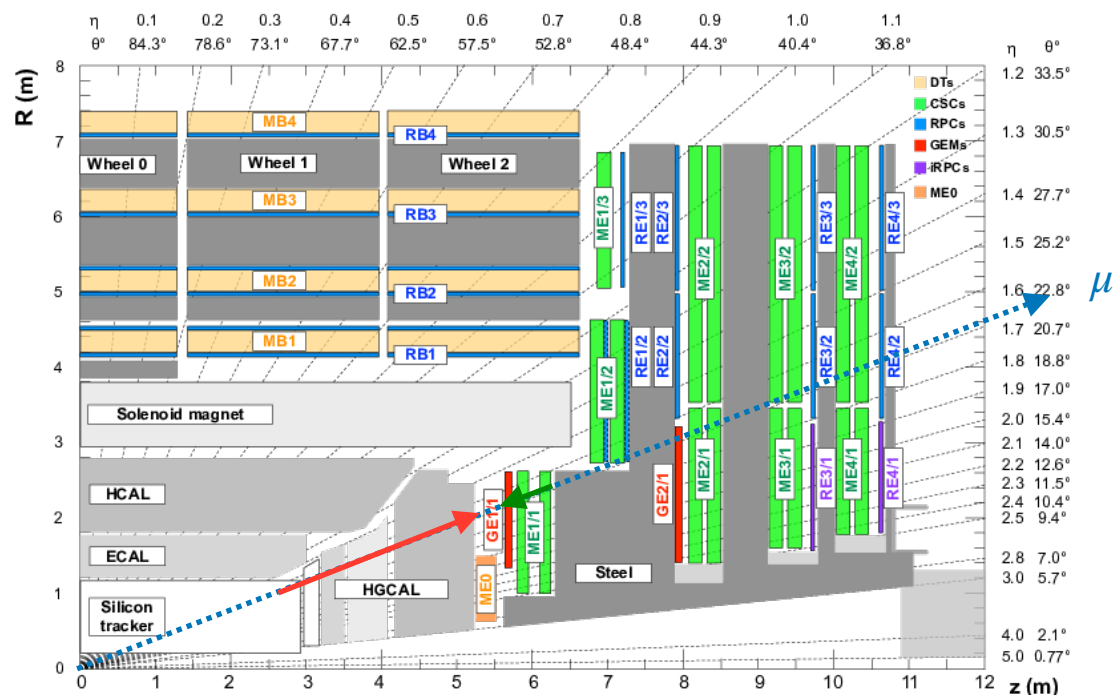
Muon Propagation to GEM



Since GEM-CSC bending angle is critical, the relative alignment of ME1/1 to GE1/1 is most important.

Two propagation methods:

- **Inner tracker propagation:** pass through large mass, reference position is inner tracker
- **ME1/1 Segment back propagation:** less scattering, reference position is ME1/1
 - Works better for GEM-CSC bending angle measurement



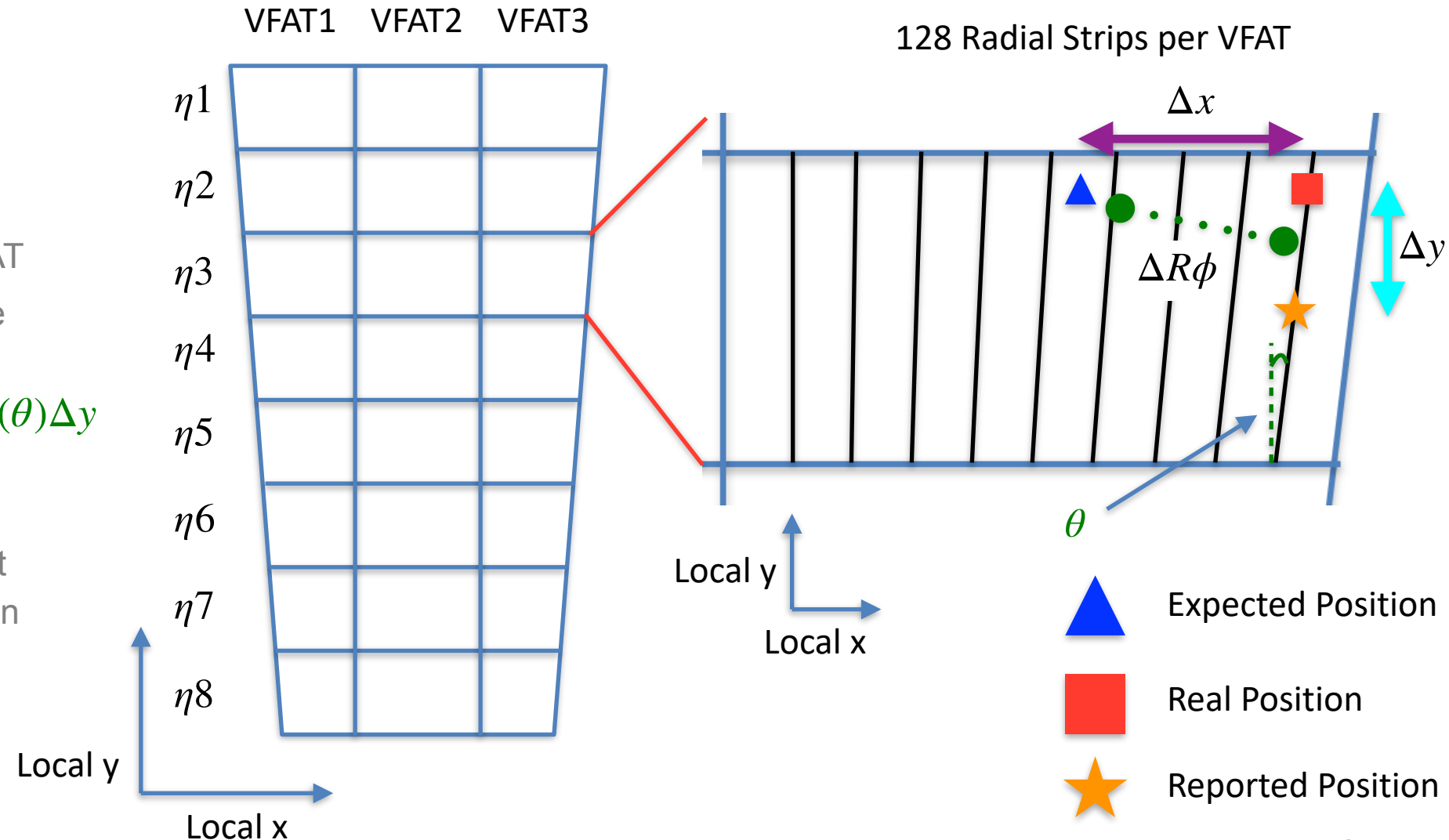
2018 Slice Test Data:
Region -1, Chambers 27-30, Layer 2
Already aligned
Colors show pT dependency in 3 bins

GEM Residual Measurement



- GEM Detectors

- 8 η partitions
- 3 VFATs per η partition
- 128 Radial Strips per VFAT
- Cannot use Δx alone due to strip angles
- $\Delta R\phi = \cos(\theta)\Delta x + \sin(\theta)\Delta y$
Where θ is the strip angle
- GEM local y is reported at the center of the η partition — poor y resolution (η partitions ~ 20 cm tall)

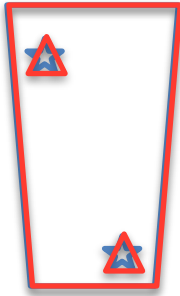


GEM Residual Measurement

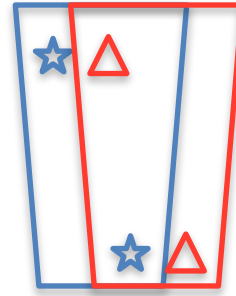


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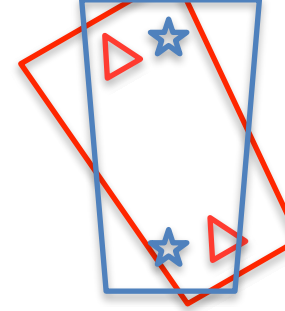
Ideal



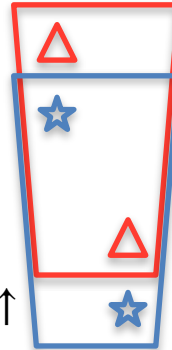
1cm $\delta x \rightarrow$



0.02rad $\delta\phi_z \curvearrowright$



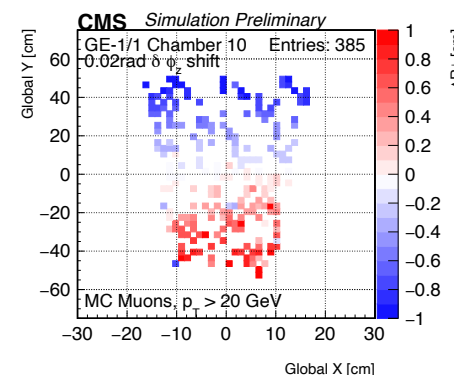
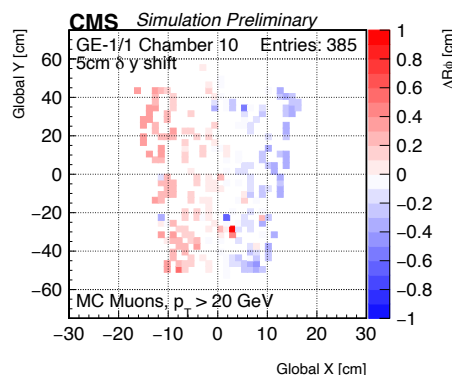
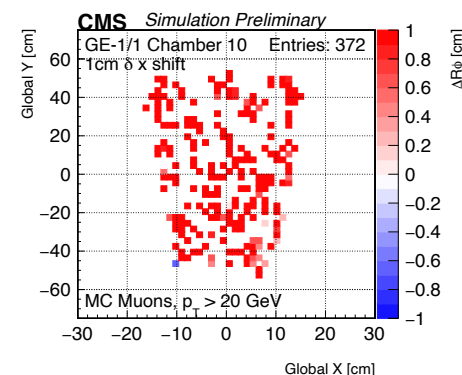
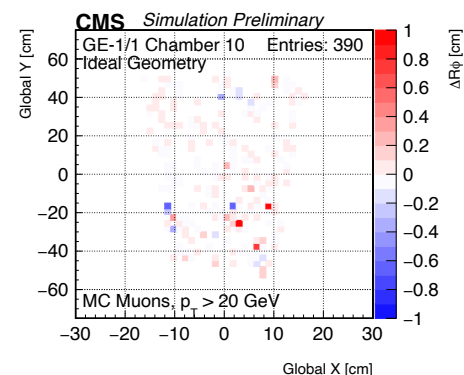
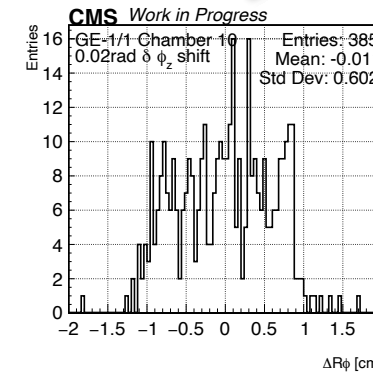
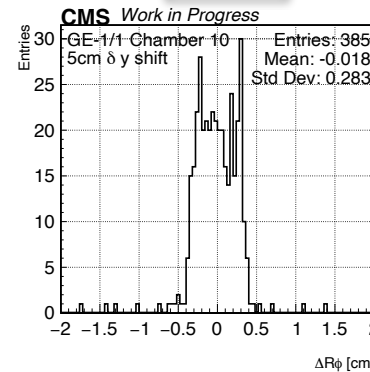
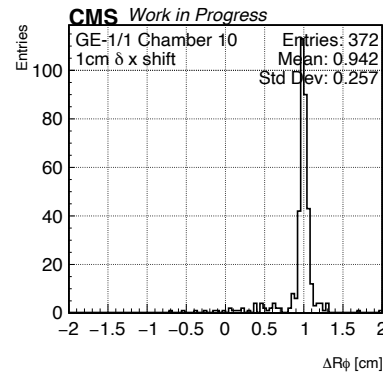
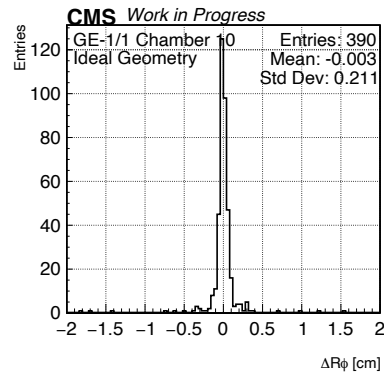
5cm $\delta y \uparrow$



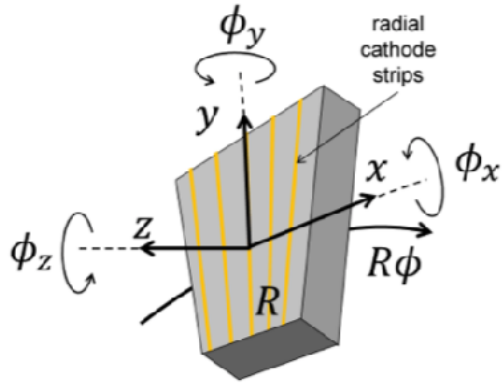
- △ Measured Hit Location
- ★ Expected Hit Location

- Some basic misalignment patterns and their $\Delta R\phi$ shapes are shown

- Plots show simulation with ideal, 1cm δx , 5cm δy , and 0.02rad $\delta\phi_z$ misalignments
- Middle plots show histogram of residuals $\Delta R\phi$
- Bottom plots show residuals $\Delta R\phi$ and where they are on the detector



GEM Alignment Variables



$$\Delta R\phi = \cos\theta * \Delta x + \sin\theta * \Delta y$$

θ : strip angle of recHit

Δx : $X_{\text{prediction}} - X_{\text{recHit}}$

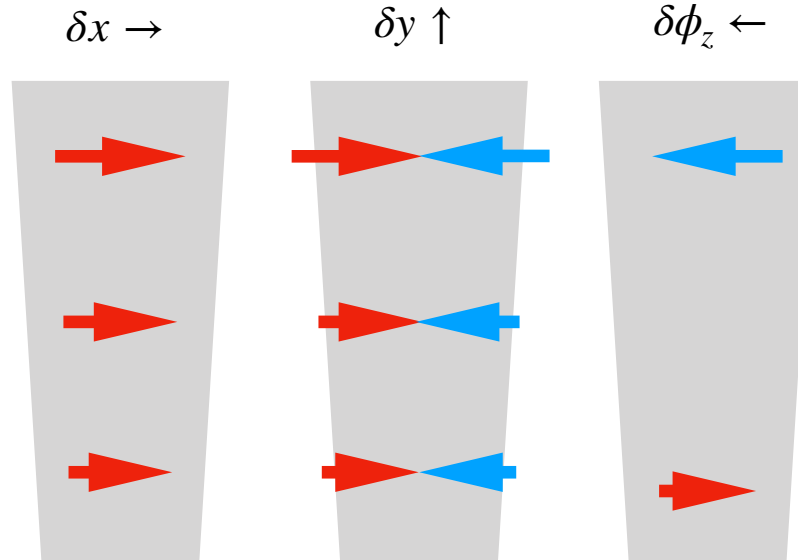
Δy : $Y_{\text{prediction}} - Y_{\text{recHit}}$

Measured Position

$$\text{Residual} \begin{pmatrix} \Delta(R\phi) \\ \Delta y_o \\ \Delta \frac{d(R\phi)}{dz} \\ \Delta \frac{dy}{dz_o} \end{pmatrix} = \begin{pmatrix} 1 & \left[-\frac{x}{R} + 3\left(\frac{x}{R}\right)^3\right] & -\frac{dx}{dz} & -y\frac{dx}{dz} & x\frac{dx}{dz} & -y \\ 0 & 1 & -\frac{dy}{dz} & -y\frac{dy}{dz} & x\frac{dy}{dz} & x \\ 0 & -\frac{1}{2R}\frac{dx}{dz} & 0 & \left[\frac{x}{R} - \frac{dx}{dz}\frac{dy}{dz}\right] & 1 + \left(\frac{dx}{dz}\right)^2 & -\frac{dy}{dz} \\ 0 & 0 & 0 & -1 - \left(\frac{dy}{dz}\right)^2 & \frac{dx}{dz}\frac{dy}{dz} & \frac{dx}{dz} \end{pmatrix} \begin{pmatrix} \delta x \\ \delta y \\ \delta z \\ \delta\phi_x \\ \delta\phi_y \\ \delta\phi_z \end{pmatrix}$$

Misalignment

$$\Delta(R\phi) = \underline{\delta x} + \left[-\frac{x}{R} + 3\left(\frac{x}{R}\right)^3 \right] \underline{\delta y} - \frac{dx}{dz}\delta z - y\frac{dx}{dz}\delta\phi_x + x\frac{dx}{dz}\delta\phi_y - y\underline{\delta\phi_z}$$



- Treat chamber as 6 degree of freedom rigid body

- Each DOF causes different residuals
- Cartoon shows simple DOF-residual relationship

- GEM currently uses 3 main misalignment DOFs δx , δy , and $\delta\phi_z$

- Equation shows DOF-residual relationship

- Misalignment measured precision order:

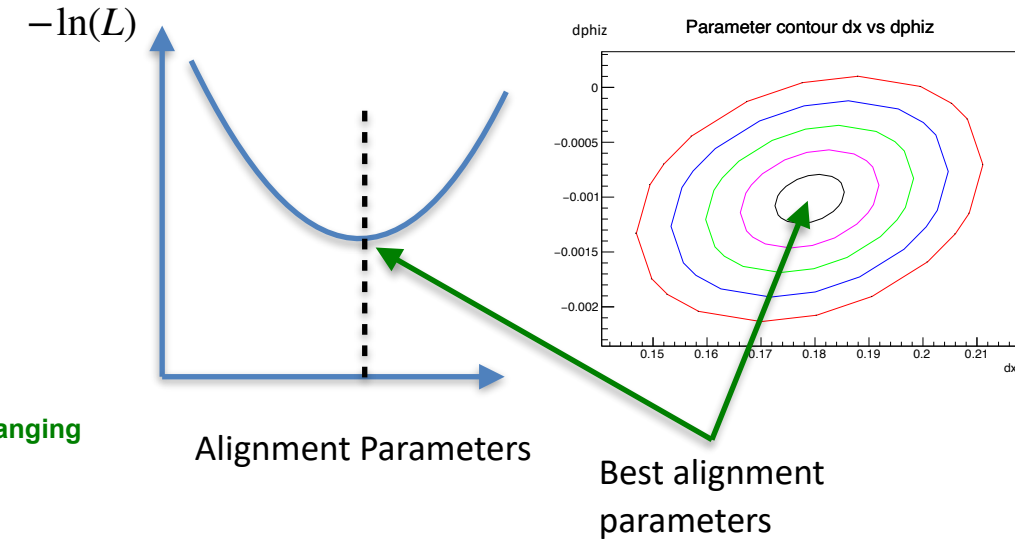
- $\delta x \rightarrow \delta\phi_z \rightarrow \delta y$

MINUIT Fitter



We use MINUIT to minimize our unbinned likelihood function to get best fit

- Input which DOFs to fit (δx , δy , $\delta \phi_z$ for GEMs)
- Input our measured Residuals for a chamber
- MINUIT calculates the Likelihood for a set of alignment parameters to return our measured Residuals
- MINUIT returns the parameters that gave the minimum log-likelihood for that chamber
- Residual $\Delta R\phi$ distributions can be complicated
 - Multiple misalignment parameters can be combined
 - **Minimizing the log-likelihood finds a geometry that best matches our $\Delta R\phi$ distributions by changing all 3 parameters**
- GEM alignment is done for individual layers
 - One SuperChamber contains 2 layers, treated separately — recent cosmic data shows differences up to $\sim 500 \mu\text{m}$ (0.05 cm)



$$L = \prod_{i=1}^{n_{\text{Entries}}} \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left[-\frac{(\text{MeasuredResidual}_i - \text{ExpectedResidual})^2}{2\sigma^2} \right] \quad -\ln(L) = - \sum_{i=1}^{n_{\text{Entries}}} \left[\left(-\frac{(\text{MeasuredResidual}_i - \text{ExpectedResidual})^2}{2\sigma^2} \right) - \frac{1}{2} \ln(2\pi\sigma^2) \right]$$

$$\text{Expected Residual} = \Delta(R\phi) = \delta x + \left[-\frac{x}{R} + 3 \left(\frac{x}{R} \right)^3 \right] \delta y - y \delta \phi_z$$

GEM Alignment Procedure



- Steps for GEM alignment

1. Muon Selection

- A. Stand Alone Muon (cosmic muons) or Global Muon (collision) — Requires a reconstructed muon track
- B. Must contain only 1 ME1/1 segment (avoid showers)
- C. Muon p_T cut is dependent on each dataset (cosmic samples are lower p_T than collision samples) — Higher p_T muons are more accurate for alignment

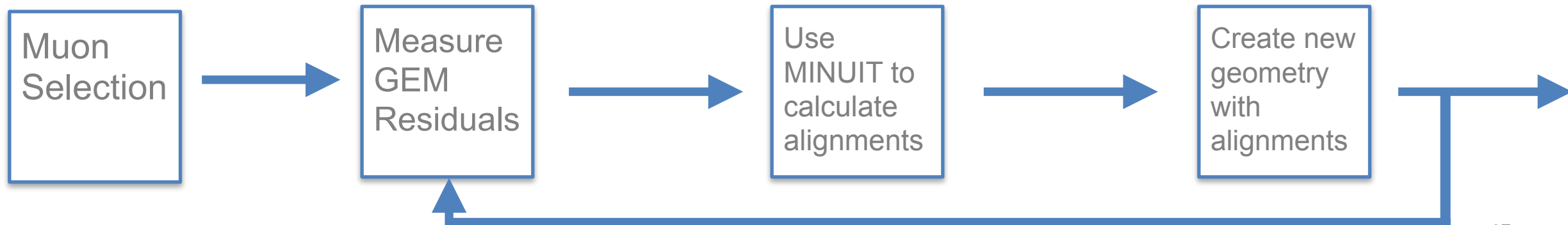
2. We measure residuals ($\Delta R\phi$) on all GEM layers

3. Use a MINUIT 3 degree of freedom ($\delta x, \delta y, \delta \phi_z$) minimizer to create suggested alignment values for each layer

- Minimizes the log-likelihood of our expected residual to our measured residuals by changing the 3 alignment parameters

4. Create a new internal Geometry Database tmp_file.db with this fixed alignment values

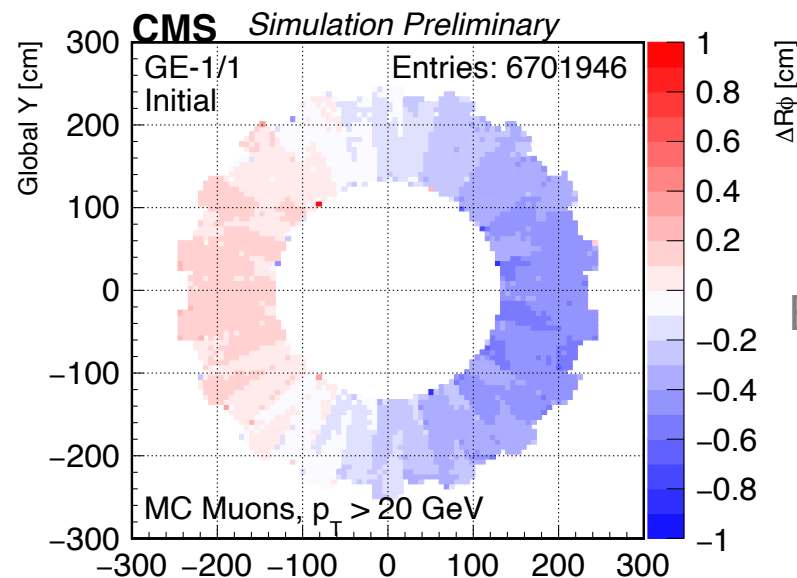
5. Repeat to step 2 until desired accuracy is achieved



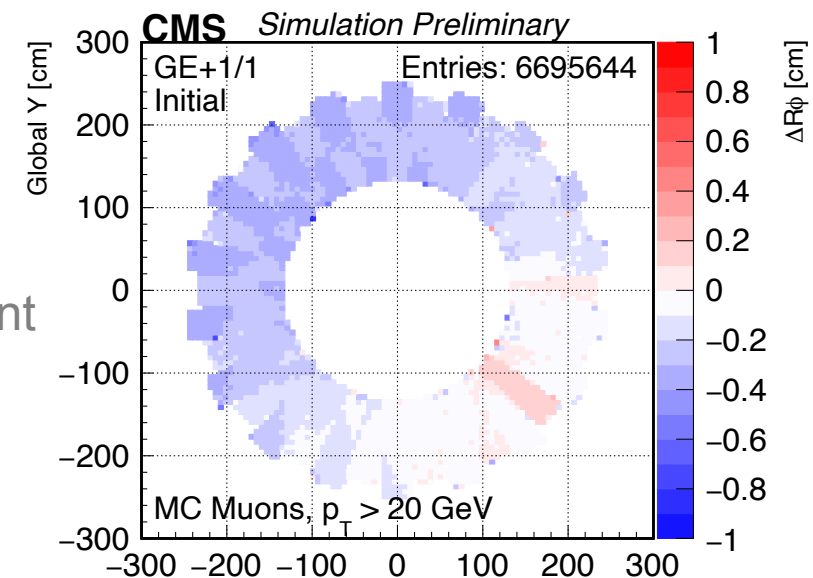
GEM Alignment validation with Simulation



- To validate we created a simulation collision dataset with 50M events (5000 pb^{-1})
- Simulated geometry was ideal (zero misalignments)
- Misaligned GEM ourselves to test our alignment procedure
- Simulated Realistic Misalignment Scenario
 - Initial GEM misalignment copied from the 2018 ME1/1 misalignment
- **With high statistics MC sample, we can test different alignment scenarios by comparing alignment output to true positions**



Before Alignment



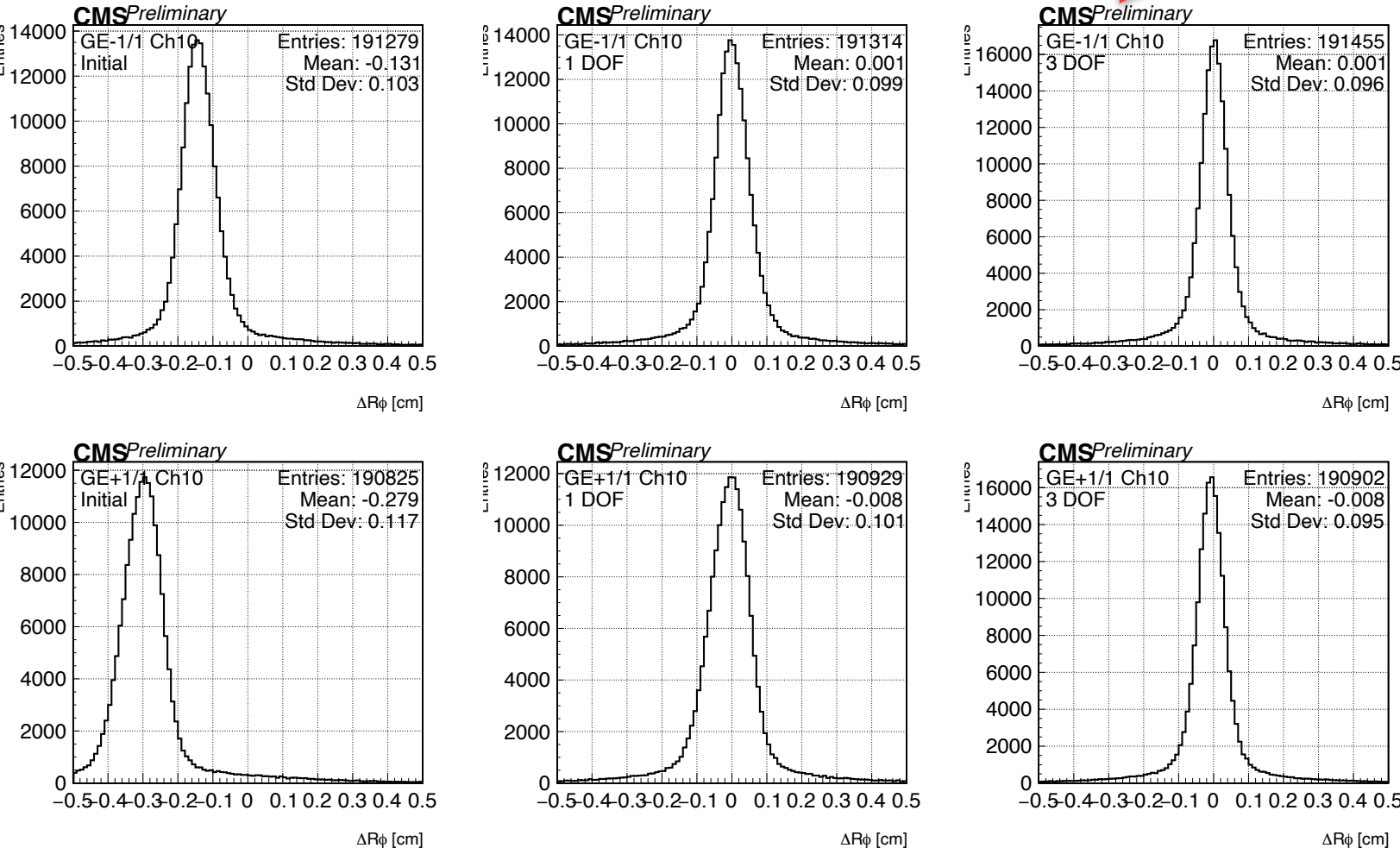
Global X [cm] Plot Cuts: $|\Delta R\phi| < 5 \text{ cm}$, $p_T > 20 \text{ GeV}$, nME1/1 segments = 1 Global X [cm]

Simulation — 1D Residuals

Initial

1dof

3dof



- Plots show residuals on Chamber 10 Layer 1 for negative(top) and positive(bottom) endcaps
- 1 dof (δx)
 - Shows a linear shift of distribution closer to 0
- 3 dof ($\delta x, \delta y, \delta \phi_z$)
 - Shows a slight improvement in the shape of distribution

$$\Delta(R\phi) = \delta x + \left[-\frac{x}{R} + 3 \left(\frac{x}{R} \right)^3 \right] \delta y - y \delta \phi_z$$

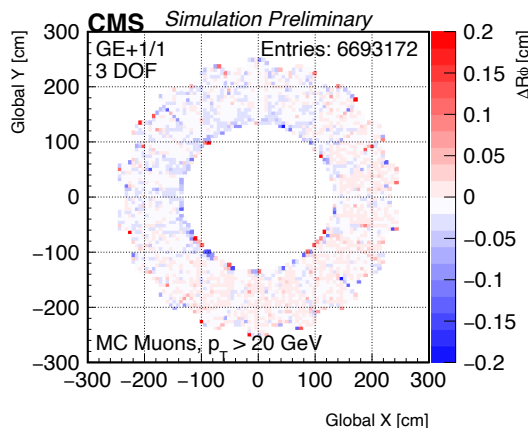
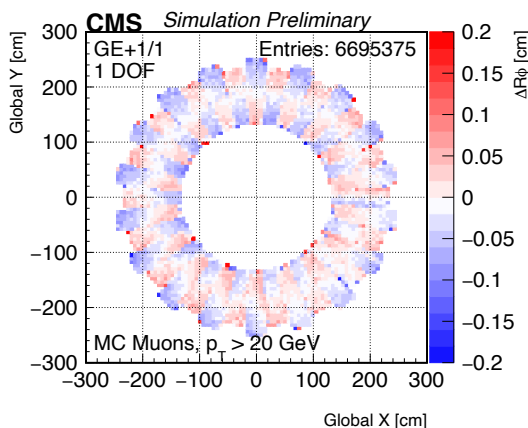
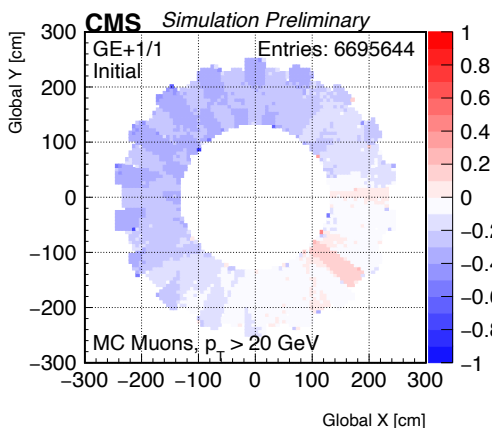
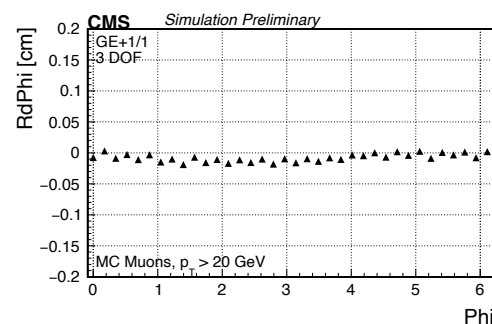
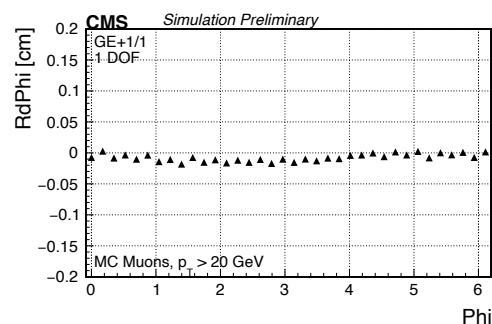
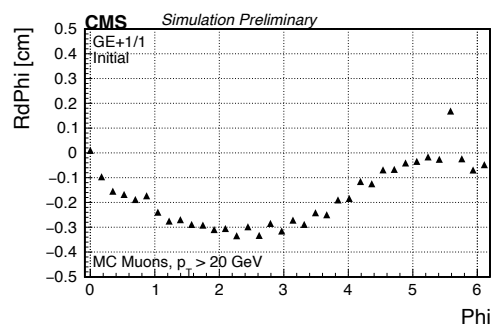
- Plot Cuts: $|\Delta R\phi| < 5\text{cm}$, $p_T > 20\text{GeV}$, nME1/1 segments = 1

Simulation — All Chamber Residuals

Initial

1dof

3dof



- Top plots: Average residual $\Delta R\phi$ for each chamber vs Global ϕ
- Bottom plots: Entire positive endcap of GE1/1 residual $\Delta R\phi$ vs Global Propagation Location

- 1 dof (δx)

- Shows alignment, but causes top/bottom effect on each chamber

- 3 dof ($\delta x, \delta y, \delta\phi_z$)

- Shows a removal of this top/bottom effect

- Top/bottom effect is a $\delta\phi_z$ rotation

- $\delta\phi_z$ is second order alignment on $\Delta R\phi$
- δy is third order alignment on $\Delta R\phi$

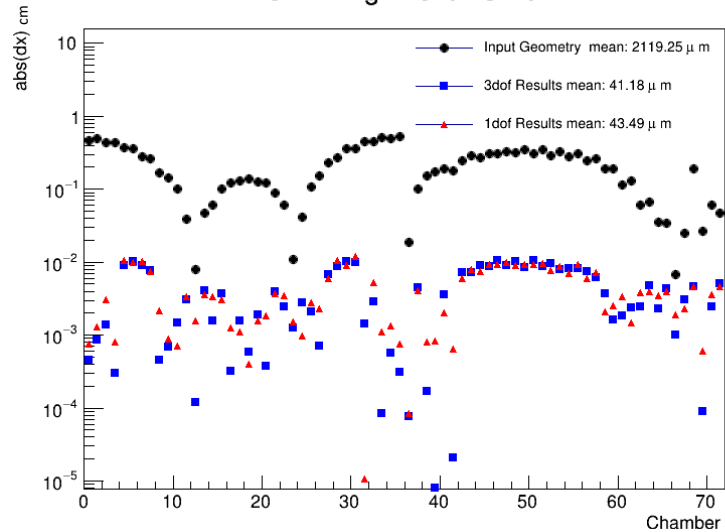
- Plot Cuts: $|\Delta R\phi| < 5\text{cm}$, $p_T > 20\text{GeV}$, nME1/1 segments = 1

Simulation — Alignment Results

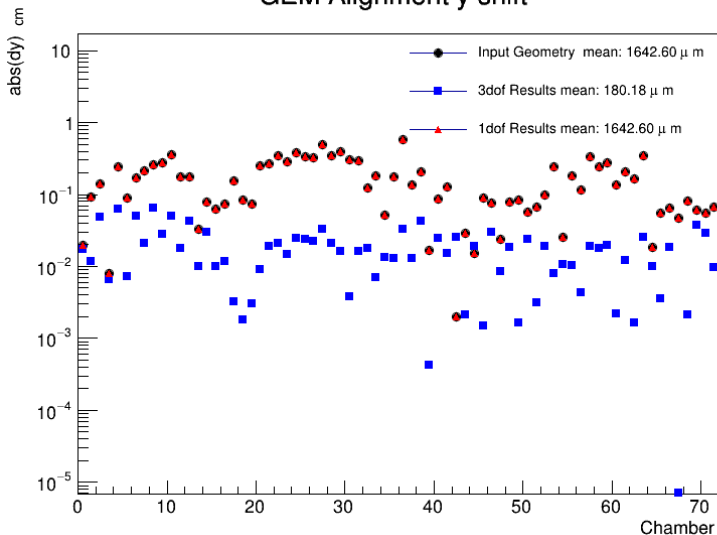


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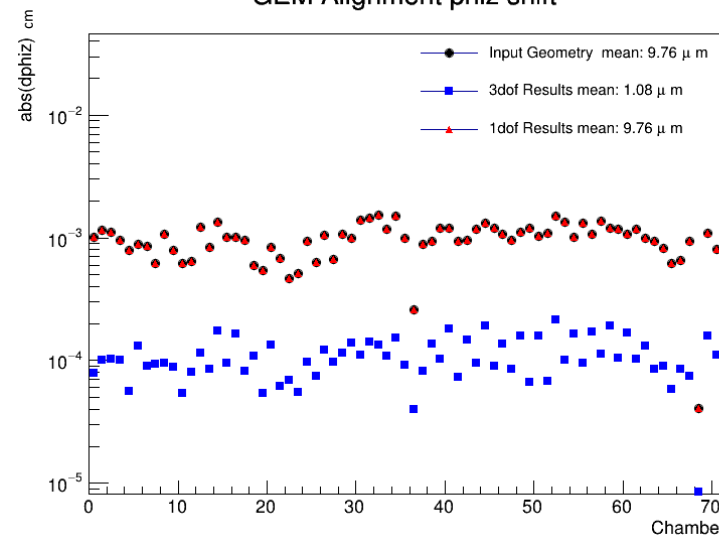
GEM Alignment x shift



GEM Alignment y shift



GEM Alignment phi_z shift



1. MC Simulation created at ideal geometry
 - True chamber position at (0,0,0)
 2. Calculate residuals with misaligned GEM geometry
 3. Minimize likelihood to find best alignment parameters
- Plots show performance for 72 chambers
 - Black: Initial misaligned GEM position
 - Blue: 3 degree of freedom (δx , δy , and $\delta \phi_z$)
 - Red: 1 degree of freedom (δx)
 - Red and black points are identical for δy and $\delta \phi_z$
 - Geometry corrected by the alignment algorithm is close to true positions of 0

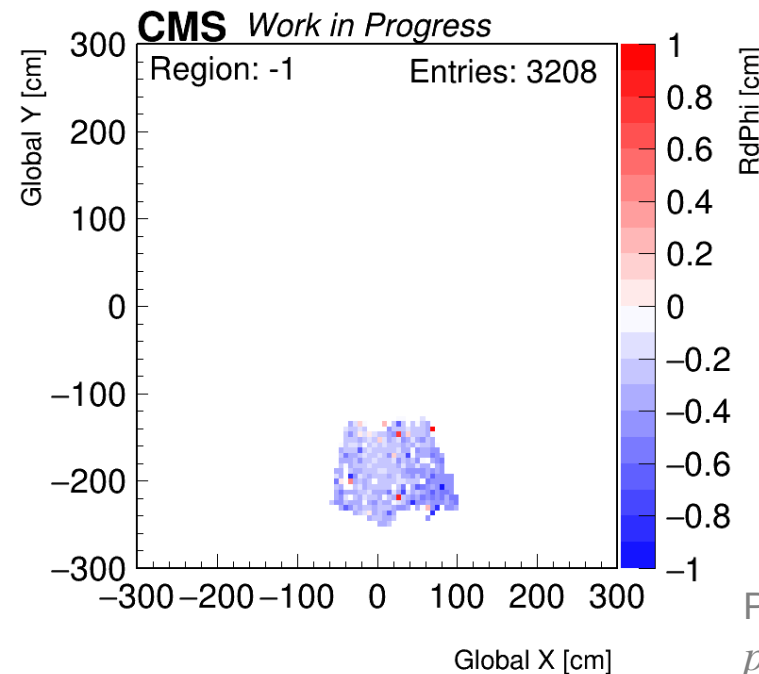
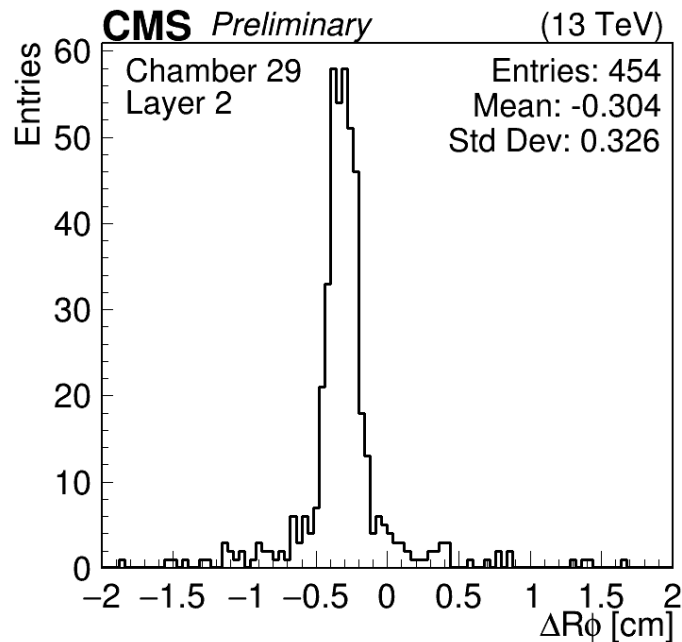
- With the alignment method shown to work for simulation, we have also tested the performance on real data
- Datasets include:
 - 2018 Slice Test — Collision data taken with 4 GE1/1 chambers installed during Run2 — 1dof alignment and 3dof alignment
 - 2021 CRUZET — Cosmic ray muons at 0T magnetic field — 3dof alignment and Global Shifts
 - 2022 CRAFT — Cosmic ray muons at 4T magnetic field — 3dof alignment, 4dof alignment ($+\delta z$), and 6dof alignment
- With these datasets our goal is to test GEM alignment performance and explore any other misalignment cases that can happen

2018 Slice Test — Initial



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- In 2018 four GEM Detectors were installed on the negative endcap during Run2
- First data taken with new GEMs
- Collision Data
- Can compare multiple degrees of freedom with real collision data



- Left Plot:

- Histogram of $\Delta R\phi$ (~local x) for chamber 29 Layer 2
- Only layer 2 due to problems with layer 1

- Right Plot:

- Color: Residual $\Delta R\phi$
- Position: Propagated hit position on the endcap — 4 installed chambers all at 6 o'clock position

Plot Cuts: $|\Delta R\phi| < 5\text{cm}$,
 $p_T > 30\text{GeV}$, nME1/1 segments = 1

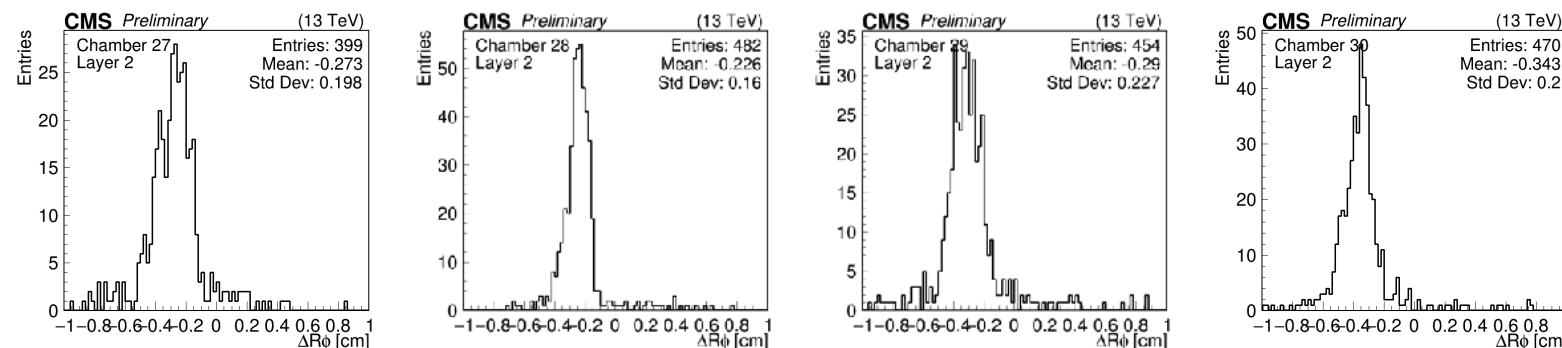
2018 Slice Test — Align Chambers



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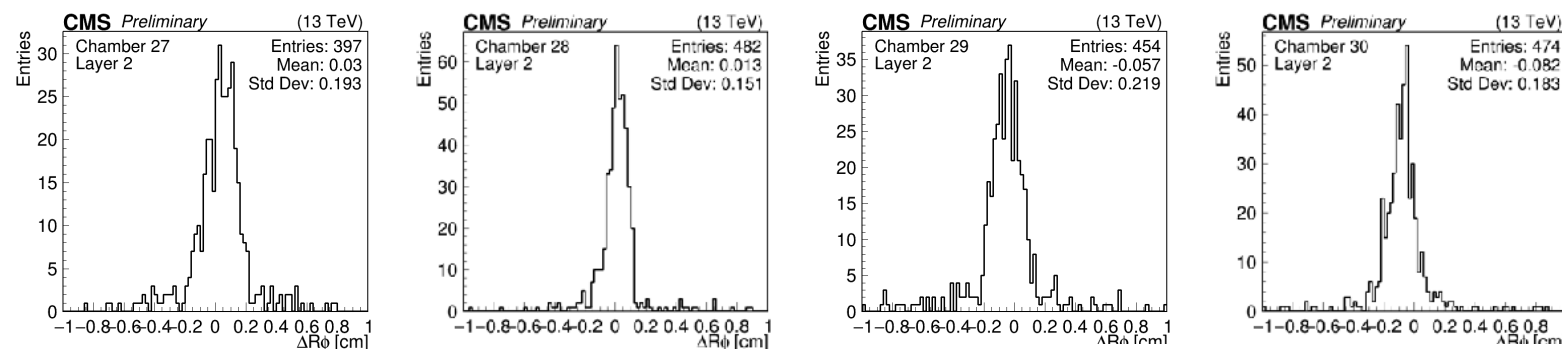
- With only 4 chambers, it is easy to view separate alignments

Before alignment



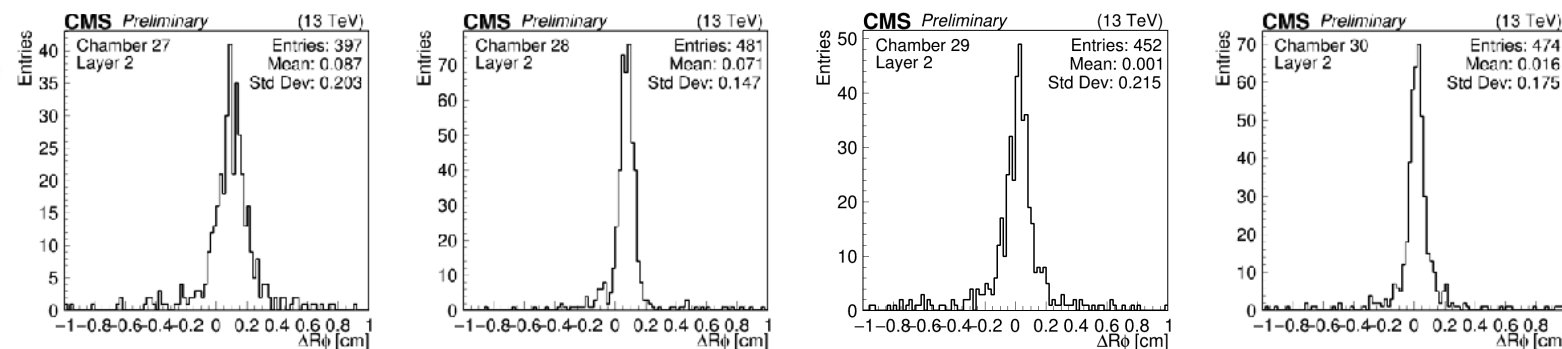
- δx can apply a linear shift, but cannot change residual shape

After 1dof alignment



- δy and $\delta\phi_z$ can change the residual distribution to be more narrow

After 3dof alignment

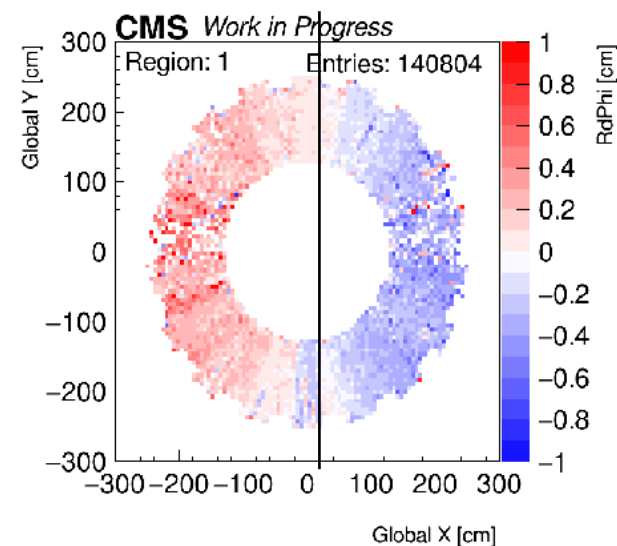
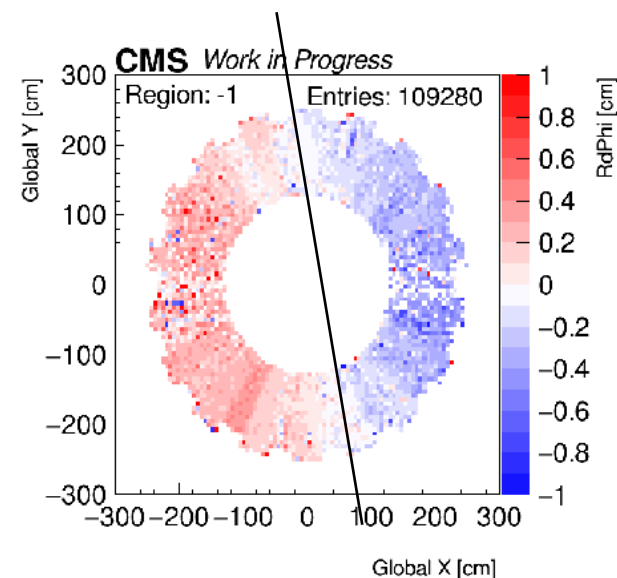


Plot Cuts: $|\Delta R\phi| < 5\text{cm}$,
 $30\text{GeV} < p_T < 200\text{GeV}$, nME1/1 segments = 1

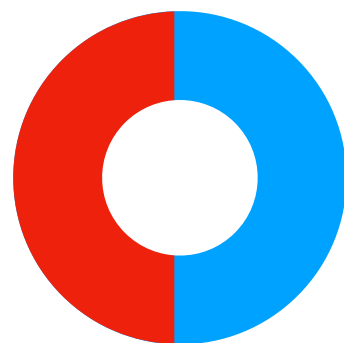
CRUZET — Initial



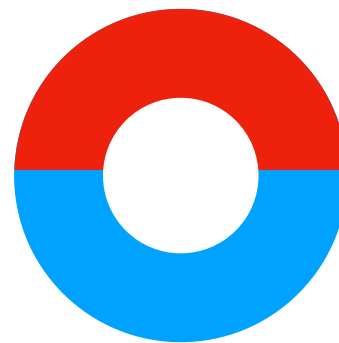
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Global Y Shift



Global X Shift



Typical Global Shifts

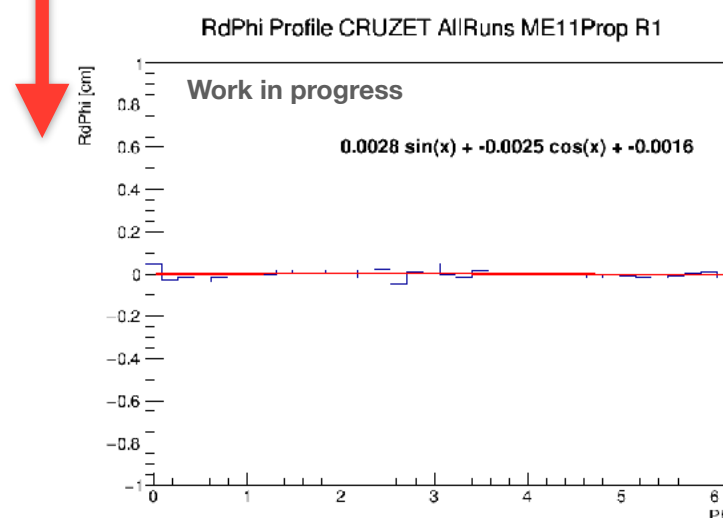
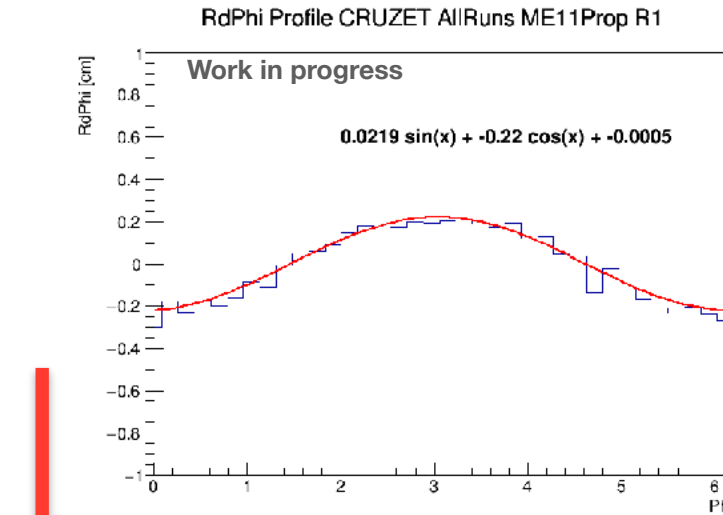
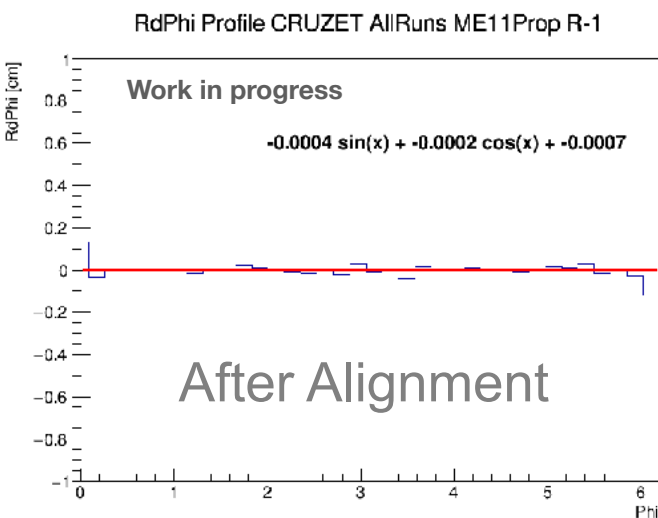
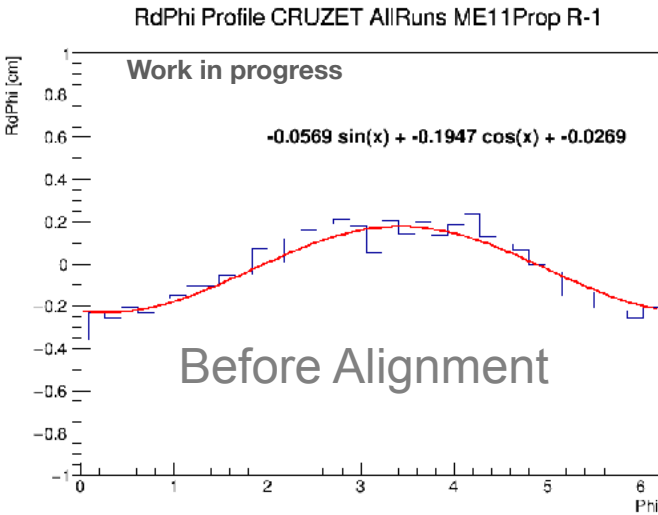
Plot Cuts: $|\Delta R\phi| < 5\text{cm}$,
nME1/1 segments = 1

- 2021 CRUZET (Cosmic Run Under Zero Tesla) data
 - All 72 GE1/1 Chambers installed
 - Top: Negative Endcap
 - Bottom: Positive Endcap
 - Plots show residual $\Delta R\phi$ vs global propagated position
 - ME1/1 Position used was Ideal
 - This allows us to test GE1/1 performance directly with little interference
- Color is 'local x'
 - Red: Clockwise Rotation
 - Blue: Counter Clockwise Rotation
- These show Global Shifts of the entire endcap along drawn axis

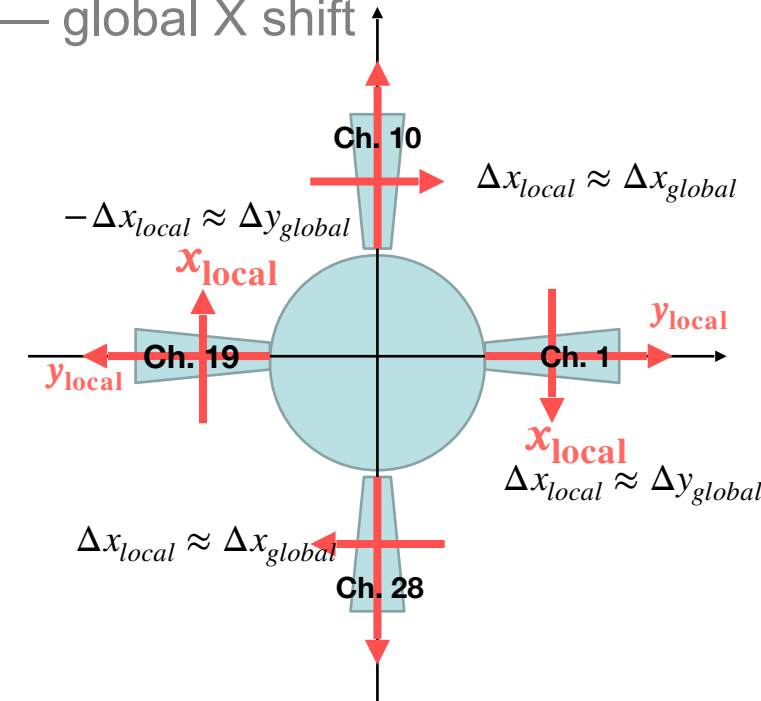
CRUZET — Global Shifts



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- Plots show residual $\Delta R\phi$ by global ϕ
 - Sin fit — global Y shift
 - Cos fit — global X shift



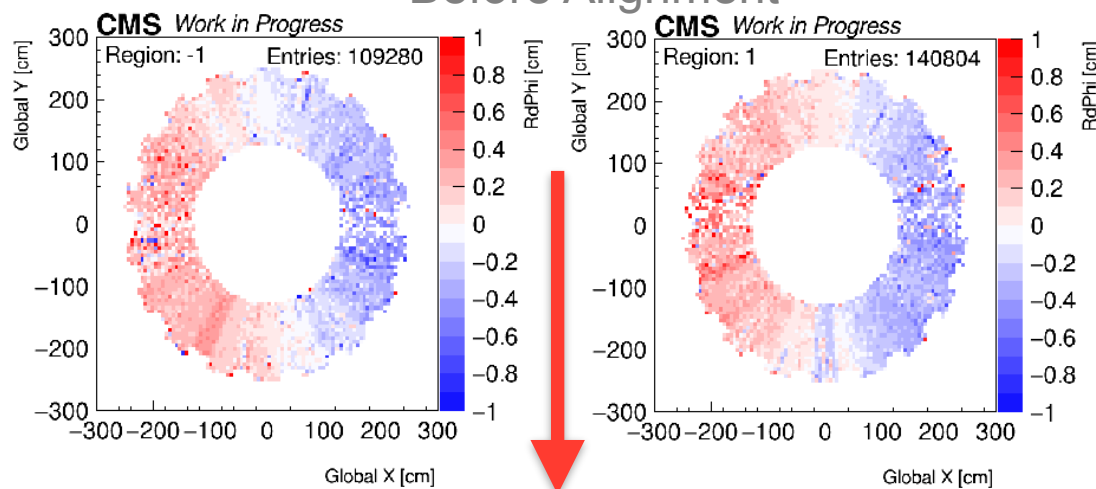
- From these shifts we see:
R-1: X 0.0569cm Y -0.1947cm
R+1: X -0.0219cm Y - 0.22cm

CRUZET — Results

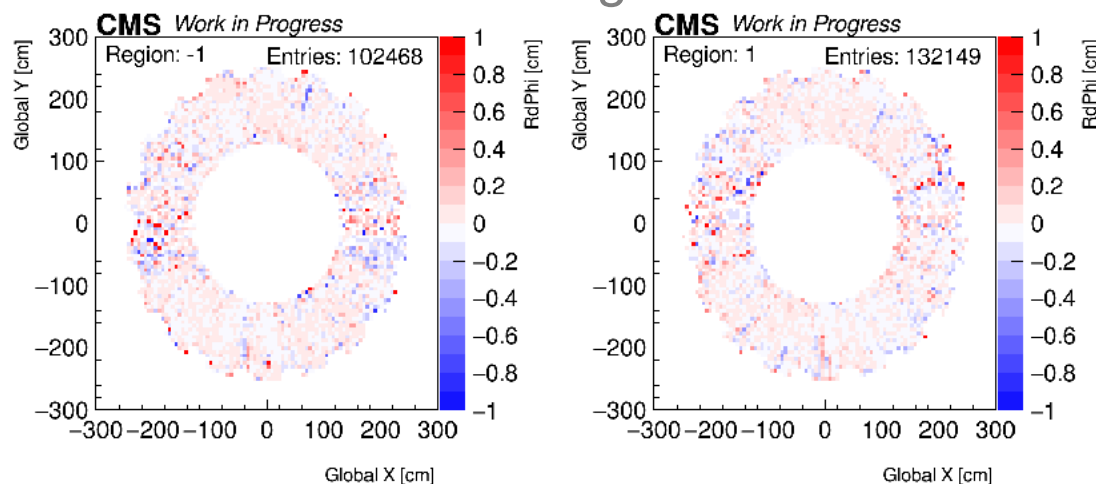


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Before Alignment



After Alignment



- CRUZET runs provided enough statistics to test GEM alignment
- Alignments showed GEM ring level shifts

	Global X Shift [cm]	Global Y Shift [cm]
GE-1/1	-0.0613	0.2562
GE+1/1	0.0304	0.288

- GEM Chambers are roughly
 - 30 cm wide
 - 100 cm tall
- Total alignments done:
 - 2 Ring Level
 - 1 Chamber Level

CRAFT Alignment

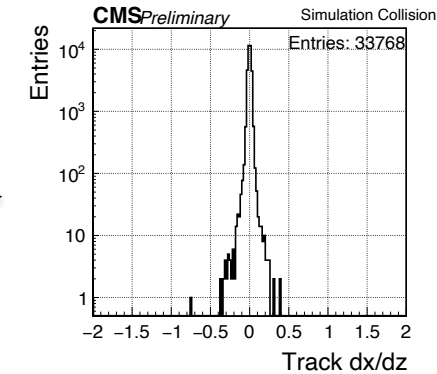


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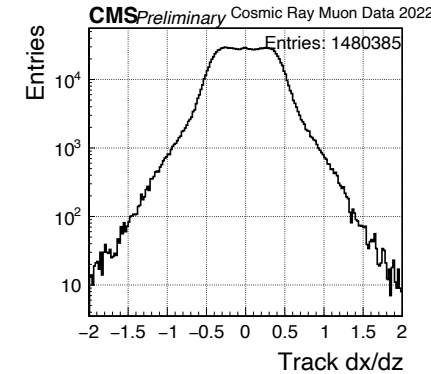
- 2022 CRAFT (Cosmic Run at Four Tesla)
- **Cosmic data with magnetic field** — introduces new challenges:
 - Cosmic ray muons have low poorly measured p_T
 - Cosmic ray muons have extreme directions not normally seen in collision $\left(\frac{dx}{dz}\right)$
 - Magnetic field causes charged particles (muons) used for alignment to bend
 - **To create a real geometry for Run3, we need to handle alignment very carefully**



Collision Simulation

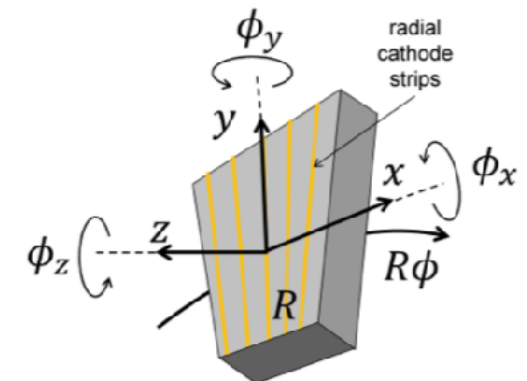


CRAFT Data



- GEM alignment currently has only measured δx , δy , and $\delta \phi_z$
 - δz , $\delta \phi_x$, and $\delta \phi_y$ have next order impact on residuals
 - Due to cosmic ray muons having extreme $\frac{dx}{dz}$ angles, δz has a large impact on residuals
 - ME1/1 has a δz value $\sim 2\text{cm}$
 - GEM-CSC distance is $\sim 21.94(49.32)\text{cm}$ for close(far) chambers
 - CRAFT shows $\frac{dx}{dz}$ values of ± 0.5
 - $\Delta R\phi = -\frac{dx}{dz}\delta z$
 - **Potential to have $\Delta R\phi \approx 1\text{cm}$**
 - **Solution is to use this large cosmic angle to do 6dof alignment**

$$\Delta(R\phi) = \delta x + \left[-\frac{x}{R} + 3 \left(\frac{x}{R} \right)^3 \right] \delta y - \frac{dx}{dz} \delta z - y \frac{dx}{dz} \delta \phi_x + x \frac{dx}{dz} \delta \phi_y - y \delta \phi_z$$

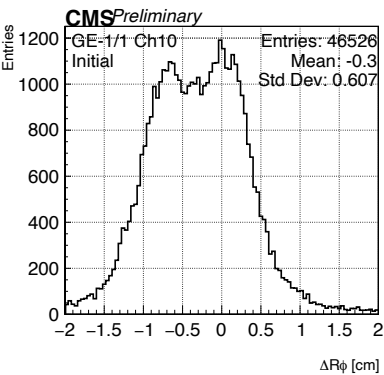


CRAFT — Single Chamber Results

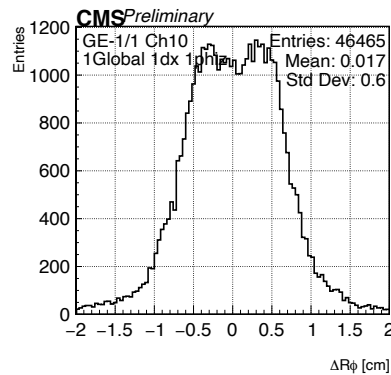


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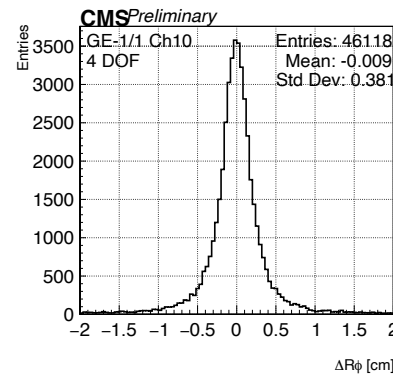
Initial



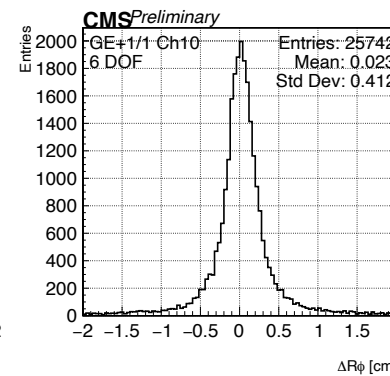
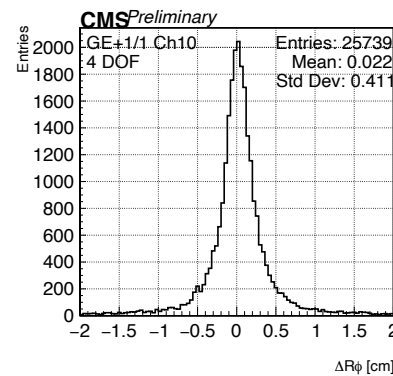
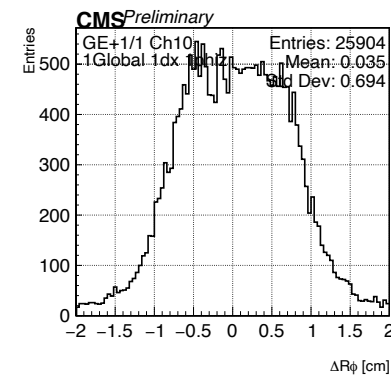
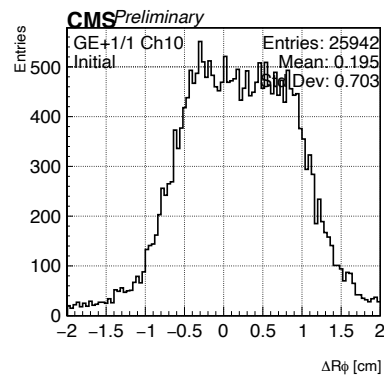
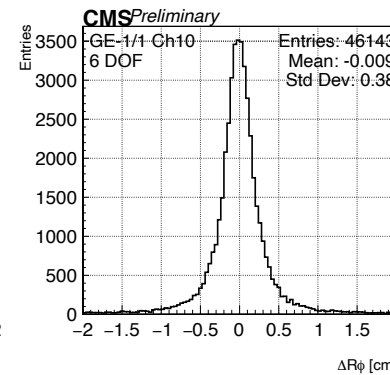
3dof



4dof
+ δz



6dof



Plot Cuts: $|\Delta R\phi| < 5\text{cm}$, $p_T > 10\text{GeV}$, nME1/1 segments = 1

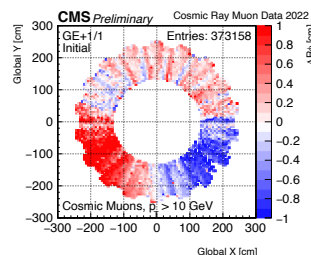
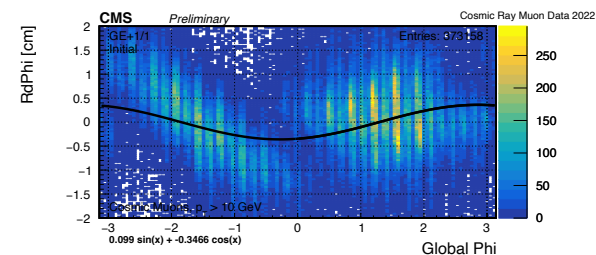
- Cosmic muons have large $\frac{dx}{dz}$ values
 - This does not happen for collision muons
 - δz is a dominant alignment factor for high angled cosmic muons
- δz shows large improvement in alignment
 - $\delta\phi_x$ and $\delta\phi_y$ show less overall improvement
- Chamber 10 Layer 1
 - Bold numbers are the new alienable
 - Red numbers are not used

Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
3DOF	0.145 +/- 0.0045	1.647 +/- 0.112	0.0	0.0	0.0	-1.231E-03 +/- 1.575E-04
4DOF	0.181 +/- 0.0045	1.927 +/- 0.112	1.544 +/- 0.0142	0.0	0.0	-8.767E-04 +/- 1.578E-4
5DOF	0.181 +/- 0.0045	1.936 +/- 0.113	1.543 +/- 0.0142	0.0	2.770E-03 +/- 1.883E-03	-8.6294E-04 +/- 1.578E-4
6DOF	0.179 +/- 0.0045	1.835 +/- 0.113	1.506 +/- 0.0144	8.275E-03 +/- 5.061E-04	3.606E-03 +/- 1.883E-03	-1.017E-03 +/- 1.581E-04

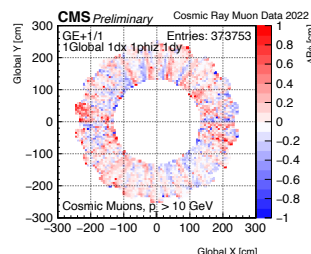
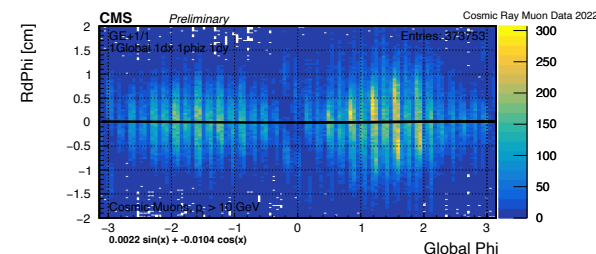
CRAFT — All Chambers Results



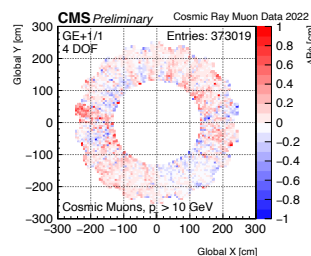
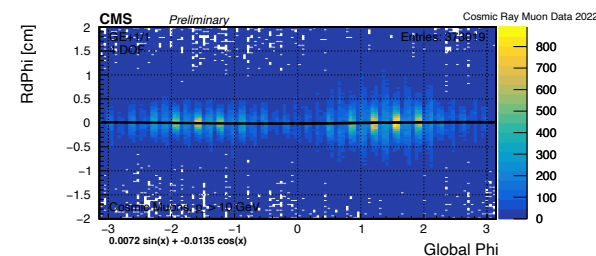
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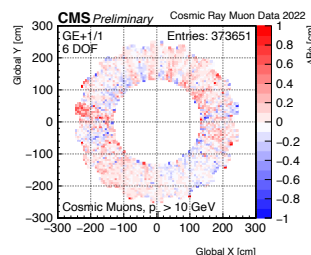
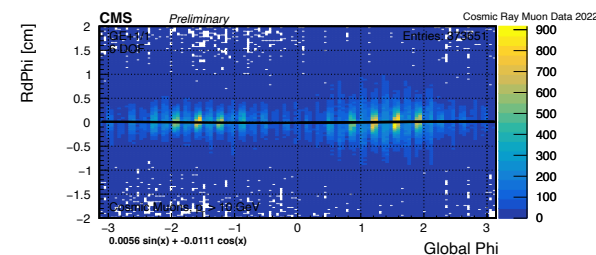
Initial



3dof



4dof
(+ δz)

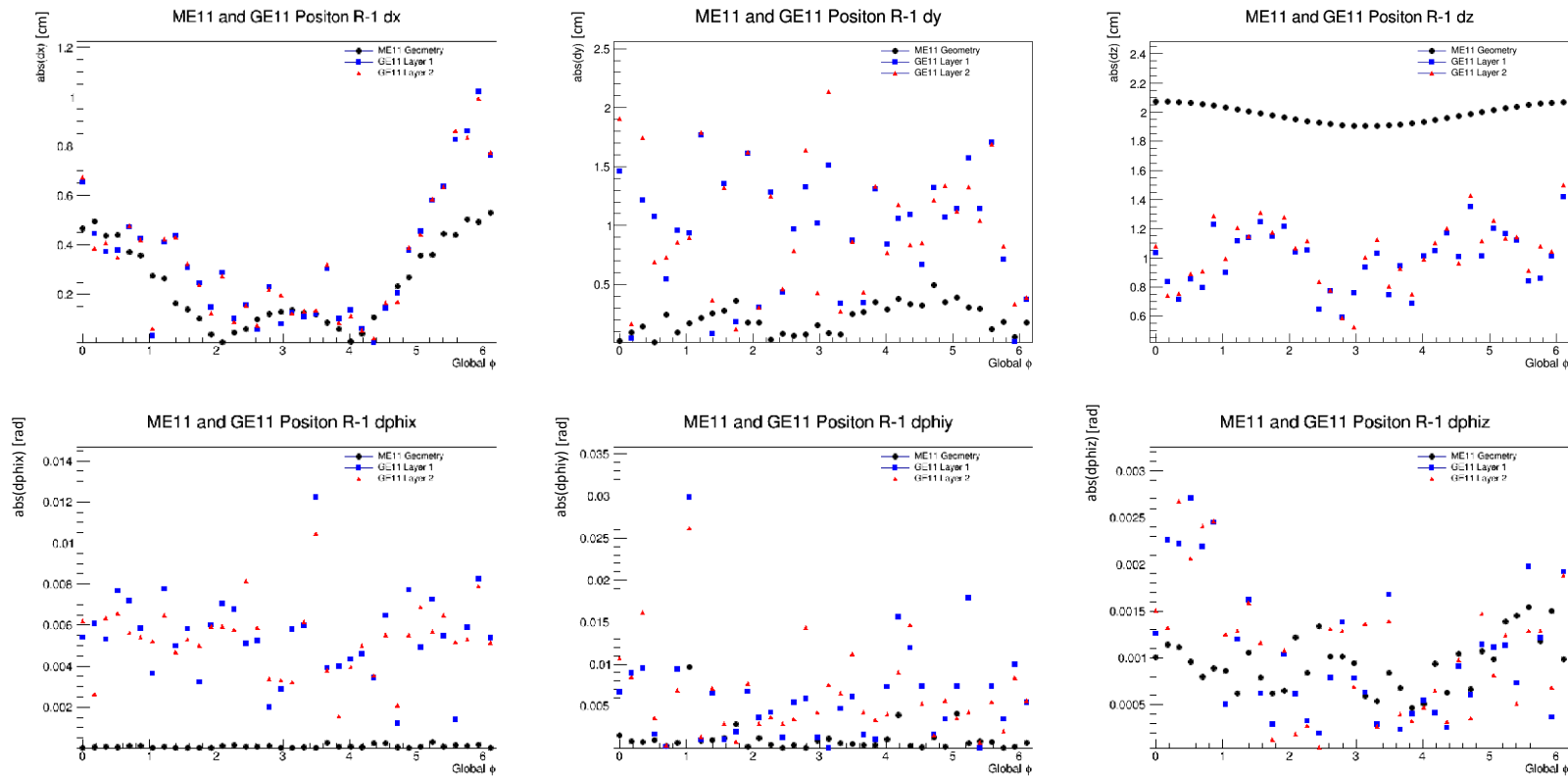


6dof

- Over all chambers, performance of 4DOF is easy to see
 - 3DOF could not fix the smearing due to vulnerability to δz
 - 6DOF shows some further improvement
- To compare if these values of alignment are reasonable, we can check with existing ME1/1 alignment values

Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
3DOF	0.145 +/- 0.0045	1.647 +/- 0.112	0.0	0.0	0.0	-1.231E-03 +/- 1.575E-04
4DOF	0.181 +/- 0.0045	1.927 +/- 0.112	1.544 +/- 0.0142	0	0	-8.767E-04 +/- 1.578E-4
5DOF	0.181 +/- 0.0045	1.936 +/- 0.113	1.543 +/- 0.0142	0	2.770E-03 +/- 1.883E-03	-8.6294E-04 +/- 1.578E-4
6DOF	0.179 +/- 0.0045	1.835 +/- 0.113	1.506 +/- 0.0144	8.275E-03 +/- 5.061E-04	3.606E-03 +/- 1.883E-03	-1.017E-03 +/- 1.581E-04

CRAFT — Alignments compared to existing ME1/1



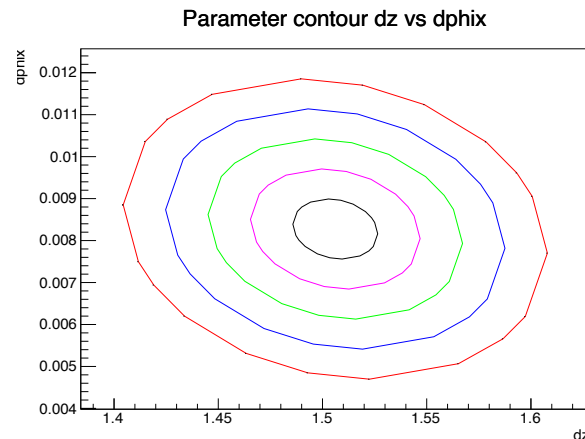
- Plots show absolute value of alignable for
 - ME11 — Black
 - GE11 Layer 1 — Blue
 - GE11 Layer 2 — Red
 - Layers 1 and 2 should be close to each other
- Comparing GEM alignment to existing ME1/1
 - $\delta x, \delta \phi_z$ are similar
 - GE1/1 $\delta y \approx 4 \cdot \text{ME1/1}$
 - GE1/1 $\delta z \approx 0.5 \cdot \text{ME1/1}$
 - GE1/1 $\delta \phi_x \approx 100 \cdot \text{ME1/1}$
 - GE1/1 $\delta \phi_y \approx 10 \cdot \text{ME1/1}$

CRAFT — Misalignment impact on $\Delta R\phi$



Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
6DOF	0.179 +/- 0.0045	1.835 +/- 0.113	1.506 +/- 0.0144	8.275E-03 +/- 5.061E-04	3.606E-03 +/- 1.883E-03	-1.017E-03 +/- 1.581E-04
Maximum Residual (Hold other variables at ideal)	0.179	-0.135	-0.753	-0.207	0.027	0.051

$$\Delta(R\phi) = \delta x + \left[-\frac{x}{R} + 3 \left(\frac{x}{R} \right)^3 \right] \delta y - \frac{dx}{dz} \delta z - y \frac{dx}{dz} \delta \phi_x + x \frac{dx}{dz} \delta \phi_y - y \delta \phi_z$$



- Estimate residual size a misalignment could cause at maximum conditions

- Set other misalignments to 0

- Examples:

$$\Delta R\phi_{\delta z} = -\frac{dx}{dz_{max}} \delta z = -0.5 \cdot 1.506 \text{ cm} = -0.753 \text{ cm}$$

$$\Delta R\phi_{\delta \phi_x} = -y_{max} \frac{dx}{dz_{max}} \delta \phi_x = -50 \text{ cm} \cdot 0.5 \cdot 0.008275 = -0.207 \text{ cm}$$

- Use maximum positions on chamber for estimation

- $x_{max} = 15 \text{ cm}$

- $y_{max} = 50 \text{ cm}$

- $\frac{dx}{dz_{max}} = 0.5$

- Estimations show δz and $\delta \phi_x$ potentially have a large impact on residuals

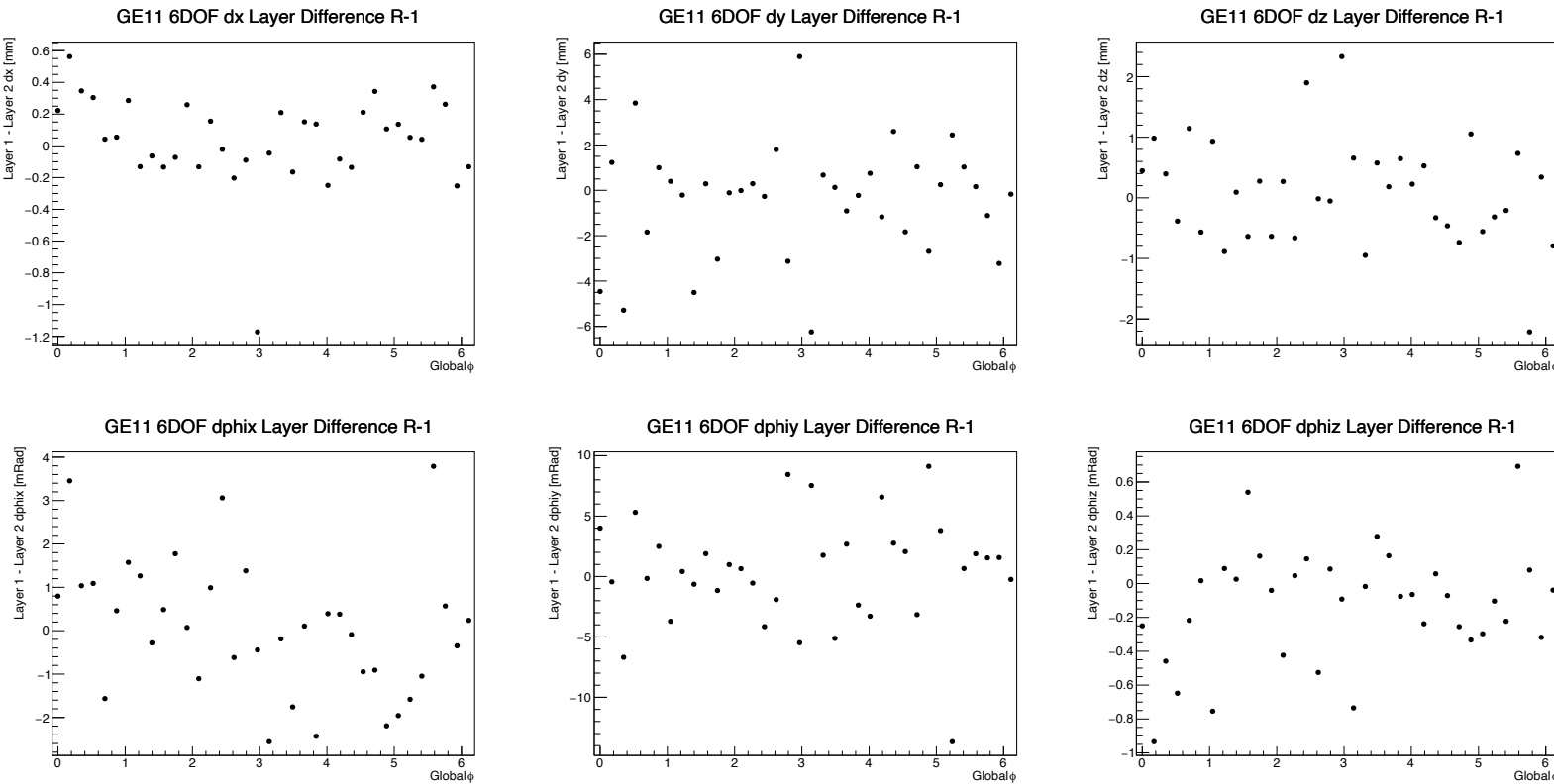
- We also see these 2 alignment parameters are correlated

- Estimations are not constant for every muon, but show the potential residual that could be caused

CRAFT — Layer 1 vs Layer 2 Alignments



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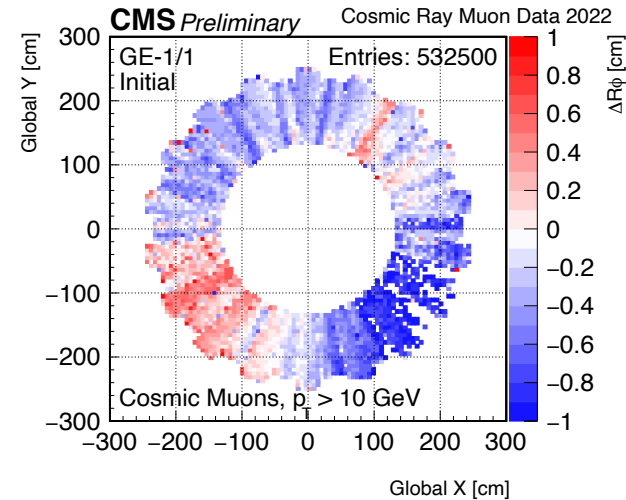
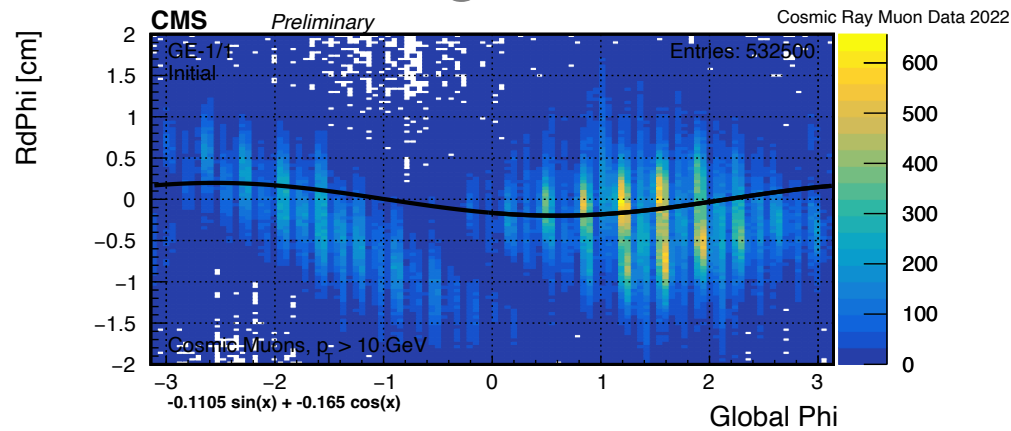
- GEM Chambers include 2 layers mounted to each other
 - Layer pairs should have similar alignments
- Maximum Differences
 - $\Delta(\delta x) \approx 0.5\text{mm}$ (Ignoring -1.2 outlier due to low statistics)
 - $\Delta(\delta y) \approx 6\text{mm}$
 - $\Delta(\delta z) \approx 2\text{mm}$
 - $\Delta(\delta\phi_x) \approx 4\text{mRad} \approx 0.23^\circ$
 - $2\text{mm } \delta z$ at edges
 - $\Delta(\delta\phi_y) \approx 10\text{mRad} \approx 0.57^\circ$
 - $1.5\text{mm } \delta z$ at edges
 - $\Delta(\delta\phi_z) \approx 1\text{mRad} \approx 0.06^\circ$
 - $0.5\text{mm } \delta x$ at edges
 - $0.15\text{mm } \delta y$ at edges
- Since we do not constrain alignments to be similar for the 2 layers, this performance validates that these alignments are reasonable.

CRAFT — Negative Endcap



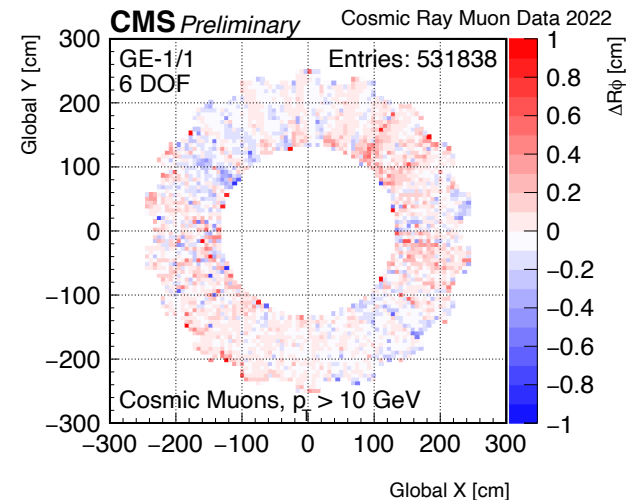
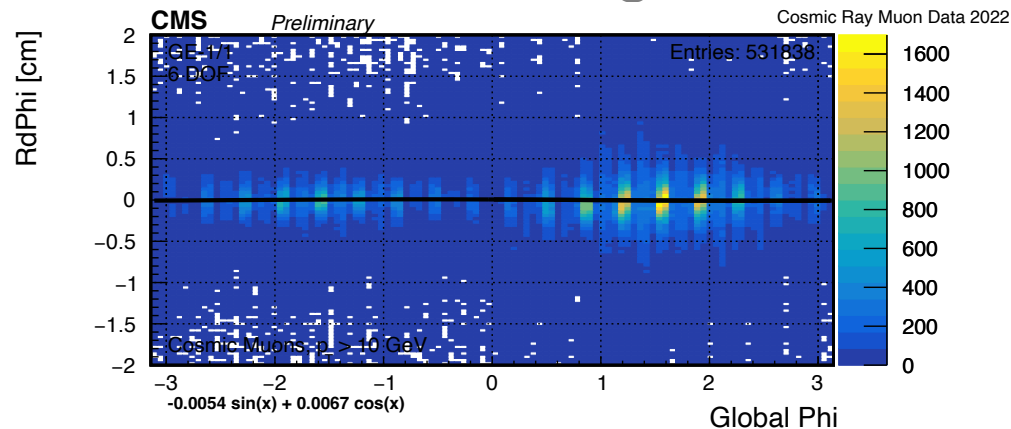
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Before Alignment



- 6DOF Alignment shows large improvement
- CRAFT's extreme $\frac{dx}{dz}$ values are very useful in aligning a previously weak alignment parameter

After 6 DOF Alignment



- With this alignment we have created the first official GE1/1 geometry, and will be used at the start of Run3

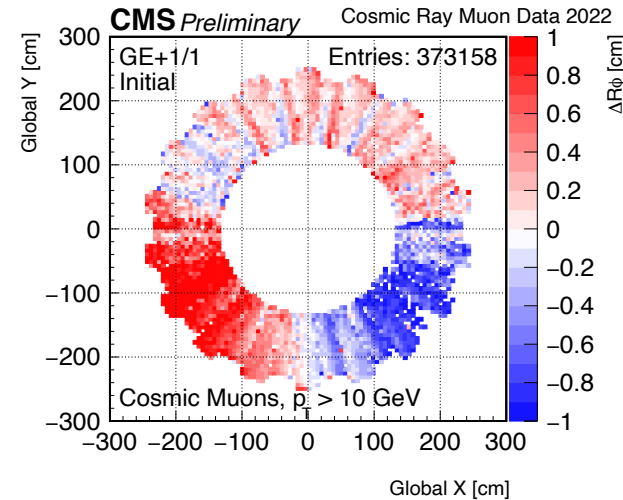
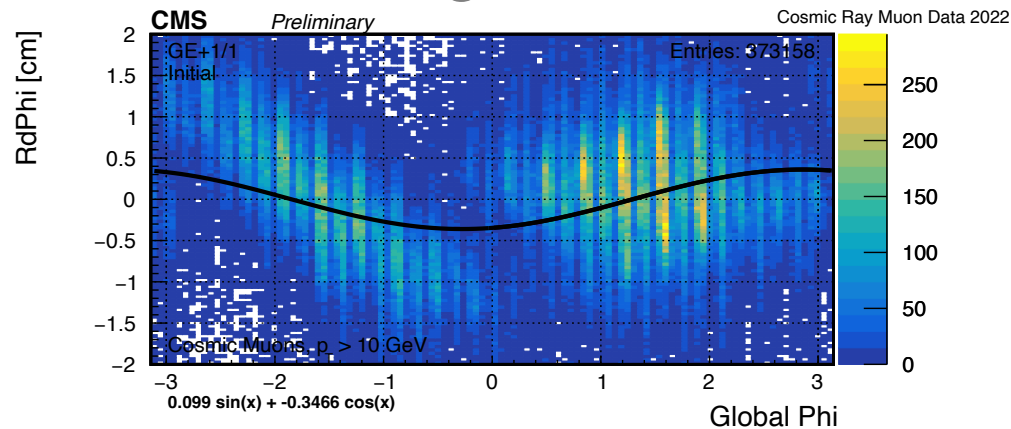
Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
6DOF	0.179 +/- 0.0045	1.835 +/- 0.113	1.506 +/- 0.0144	8.275E-03 +/- 5.061E-04	3.606E-03 +/- 1.883E-03	-1.017E-03 +/- 1.581E-04

CRAFT — Positive Endcap



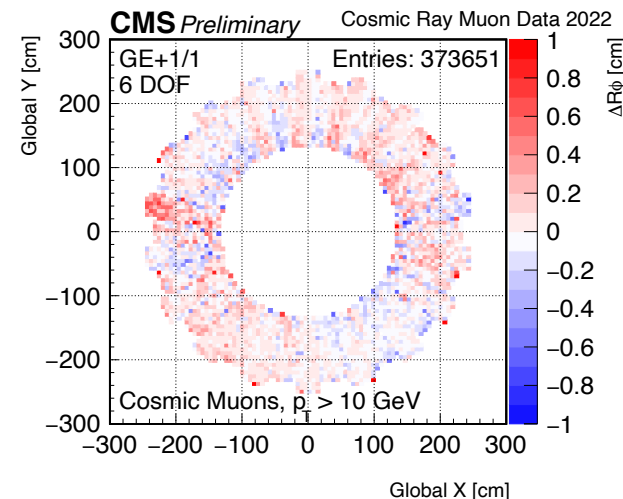
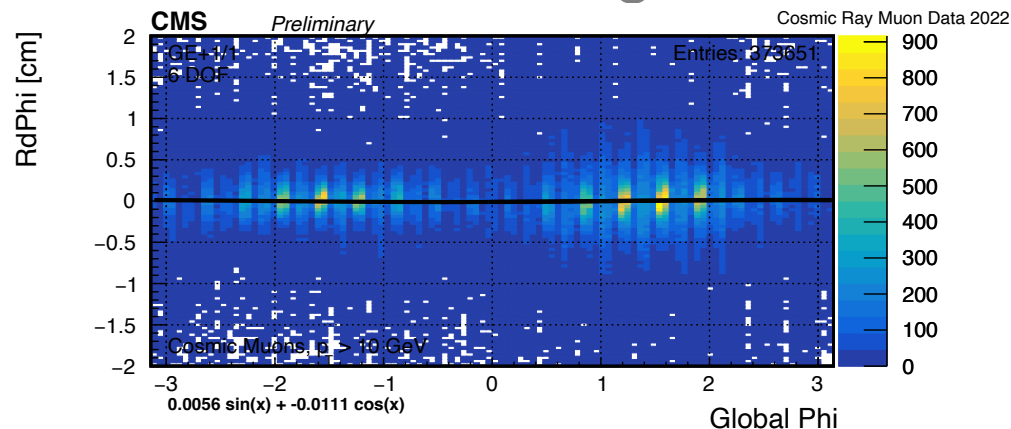
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Conclusion



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- We have created a method and recipe for aligning new GEM Detectors
- GEM Alignment features a new method of propagation and residual calculation
- Our procedure has been tested with 2018 collision data, 2022 cosmic data, and simulation
- With 2022 cosmic data we used the extreme angles of cosmic muons to align previously unaligned parameters
- This alignment will directly affect CMS object reconstruction and will benefit every analysis
- We are ready for Run 3 to start and perform GEM Alignment with the rest of the Muon System
- Future GEM Plans
 - We will also be installing GE2/1 and ME0 in the future — this will require new algorithms to be developed

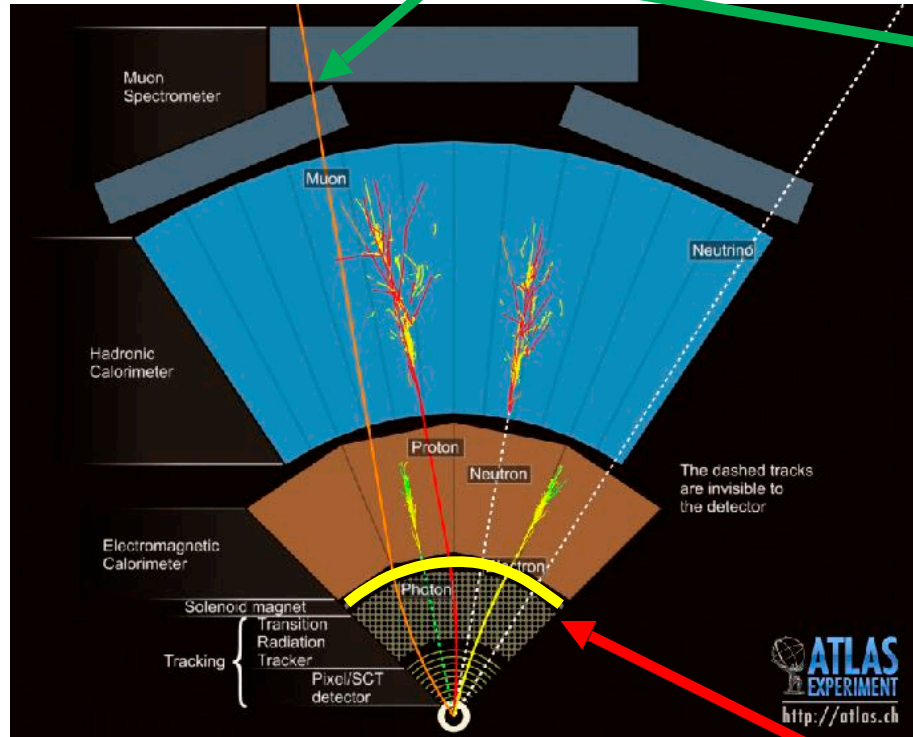


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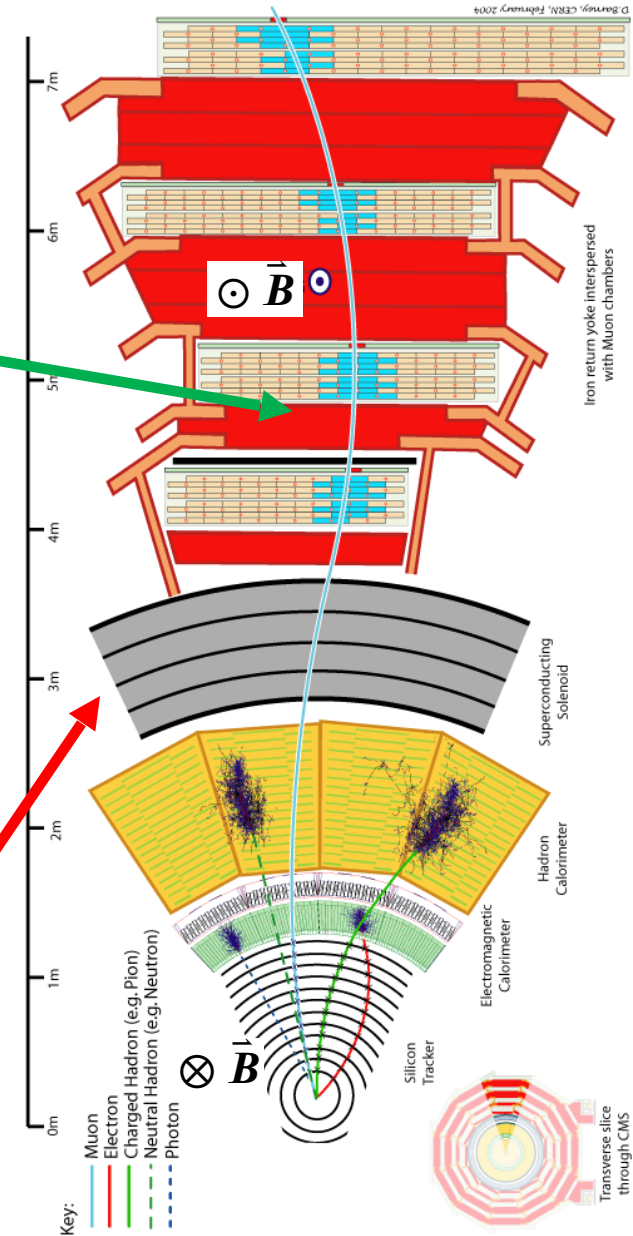
Backup

ATLAS and CMS

Similar, but clearly two main differences in (a) its detector configurations and (b) its curvature of muons (See Appendix).



magnet

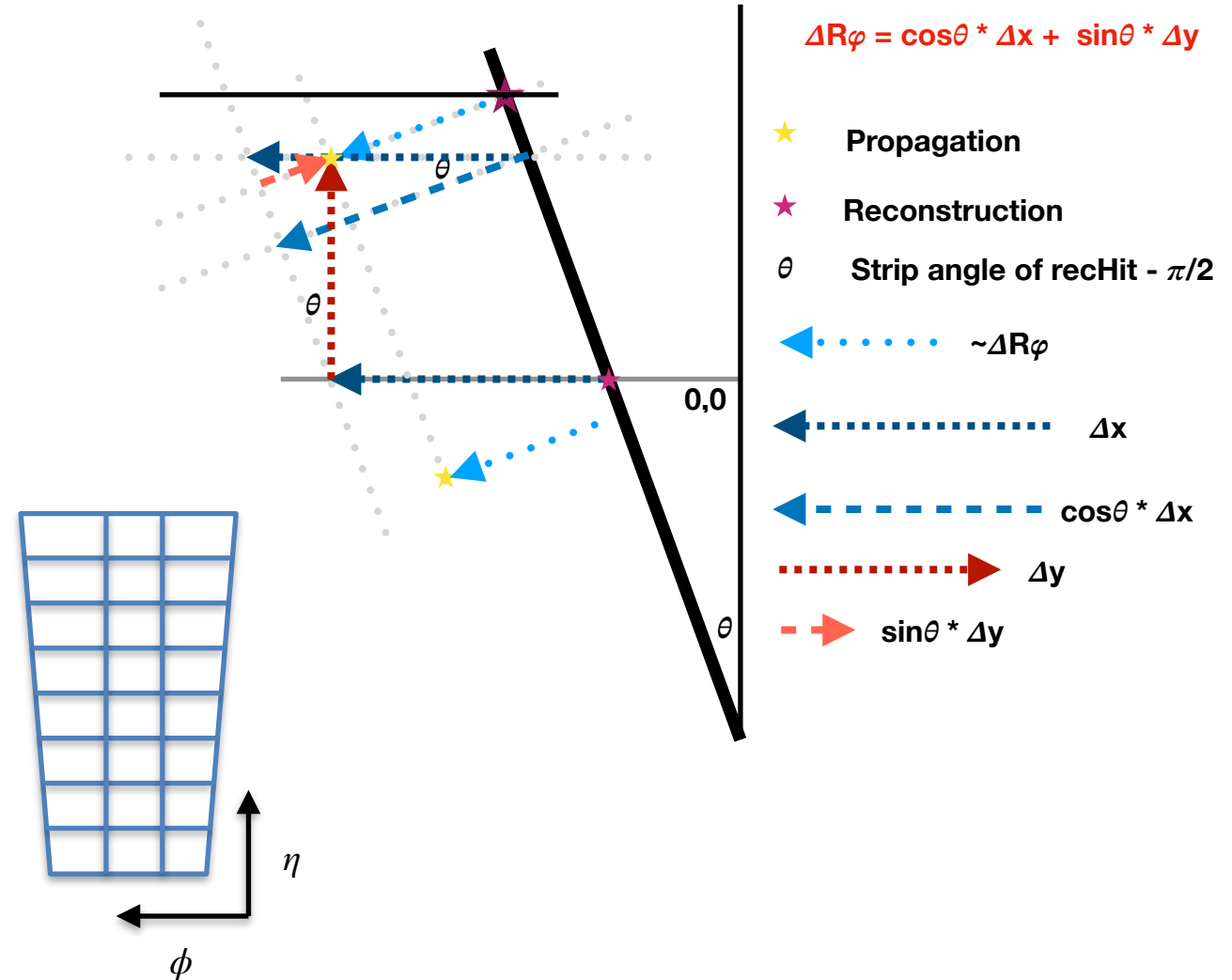


GEM Residual Measurement

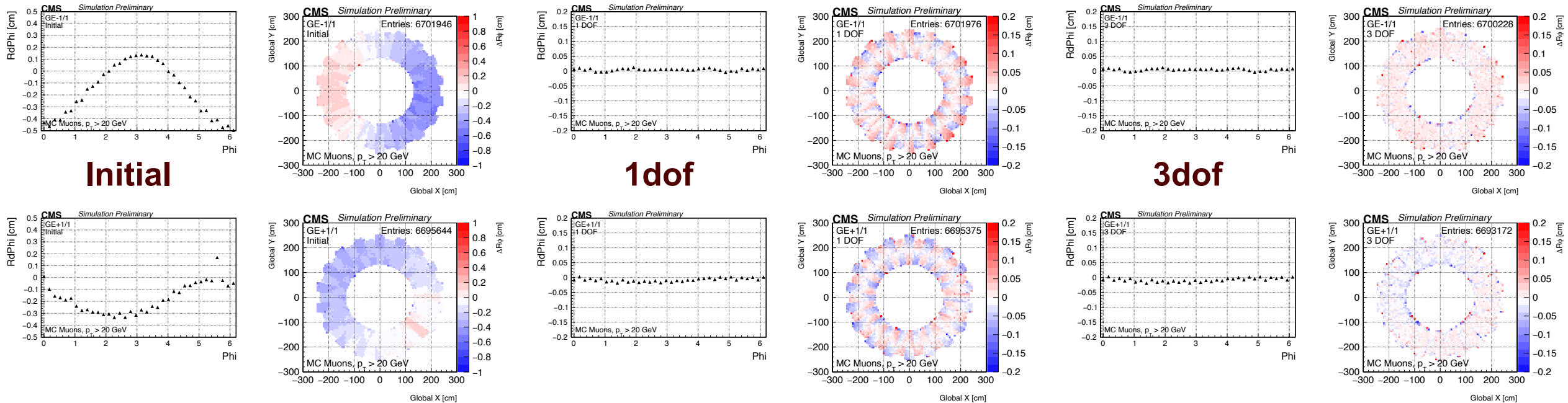


- GEM Detectors

- 8 η partitions
- 3 VFATs
- 128 Radial Strips — Strips are angled
- Propagated position can be anywhere
- Hit position is always at center of strip
- Cannot use Δx alone
- $\Delta R\phi$ calculates distance to the strip (light-blue arrow) from propagated position
- GEM local y is always in the center of the η partition — poor y resolution (η partitions ~ 20 cm tall)



Simulation — Results



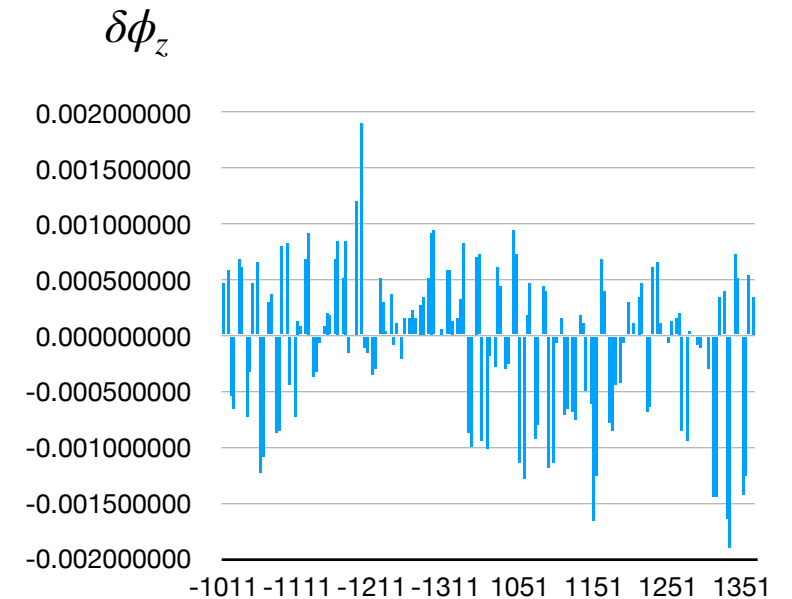
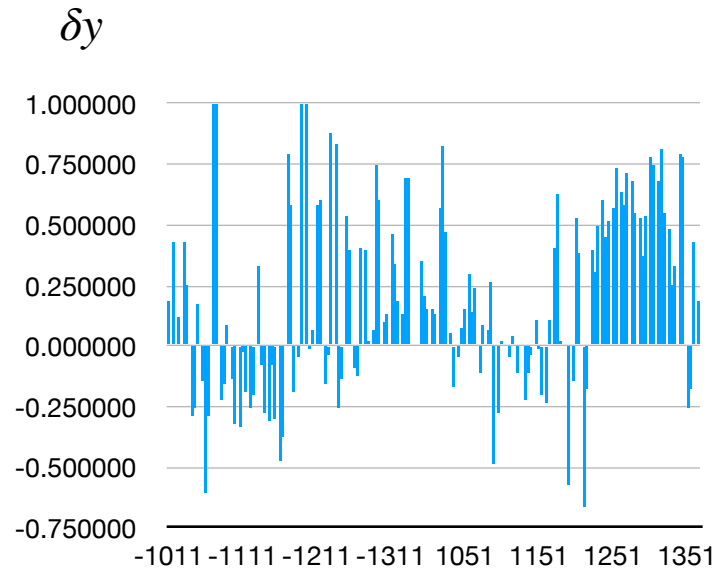
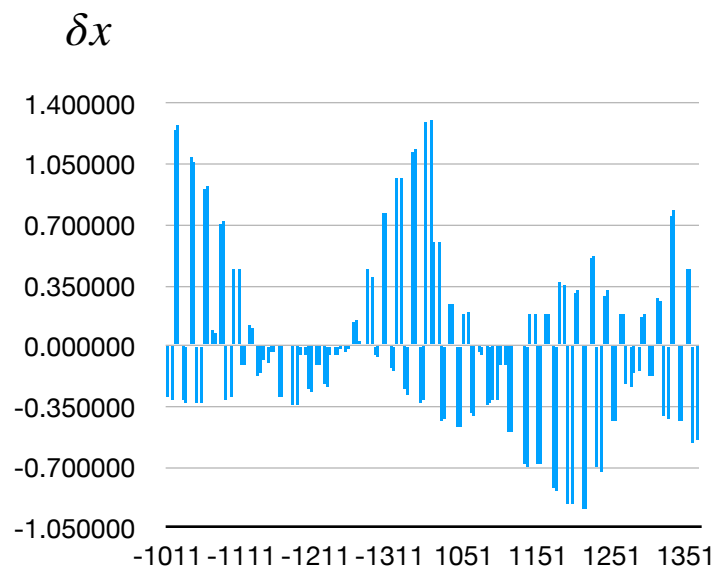
- Top: Negative Endcap Bottom: Positive Endcap
- 1 dof (δx) performance shows alignment, but **causes top/bottom effect on each chamber**
- 3 dof ($\delta x, \delta y, \delta \phi_z$) performance shows a **removal of this top/bottom effect**
- This shows the performance of the $\delta \phi_z$ alignment. This is our second order misalignment value, so it requires more statistics than δx . **This MC test shows our 3dof fit is working**

CRAFT Alignment



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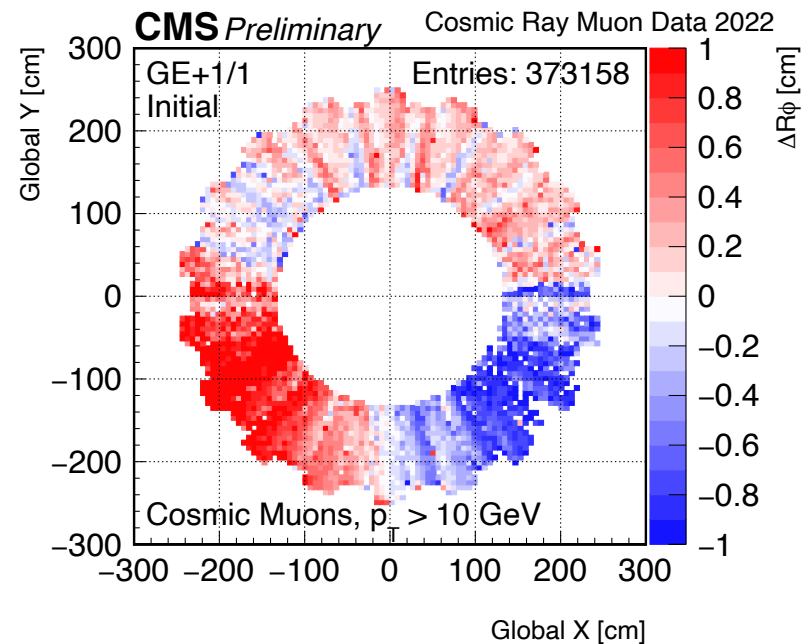
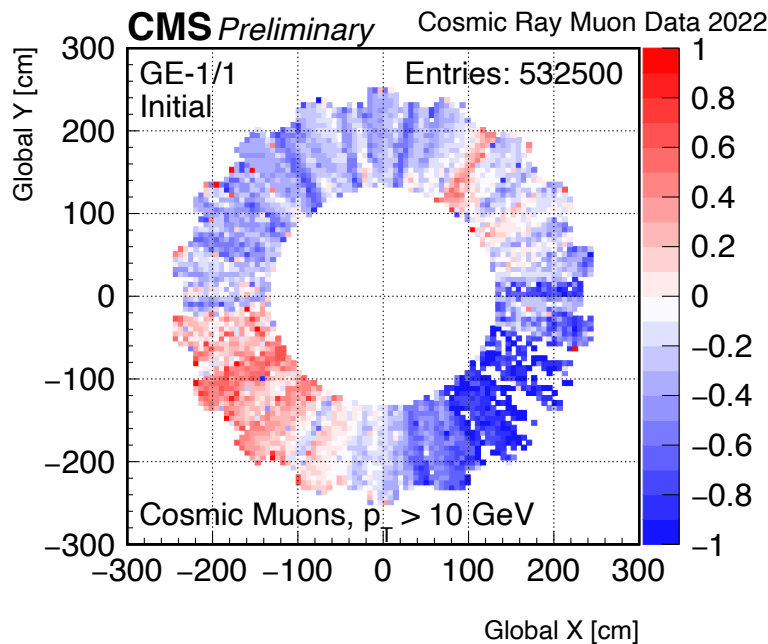
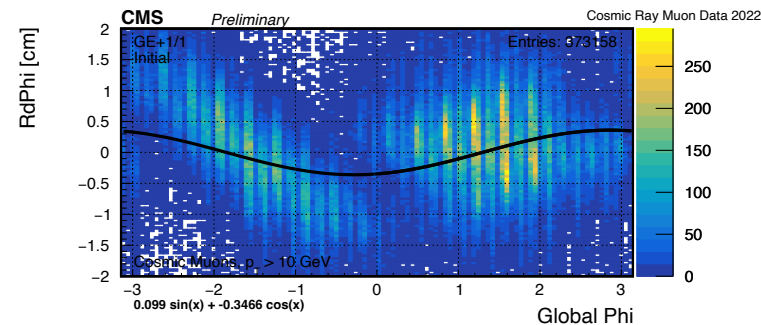
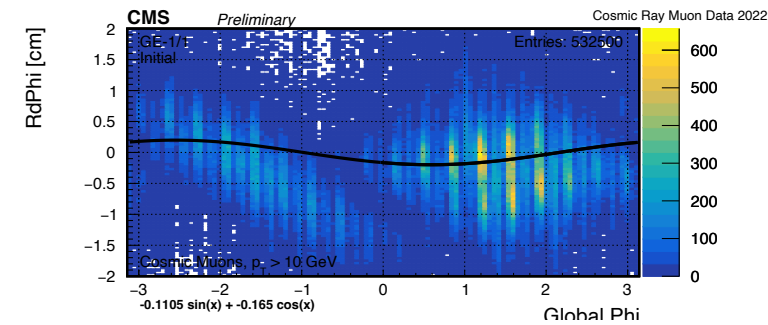
- Alignment values from 3DOF fit of CRAFT data using ME1/1 GT alignment -> GE1/1 initial position



CRAFT Alignment GEM from GT



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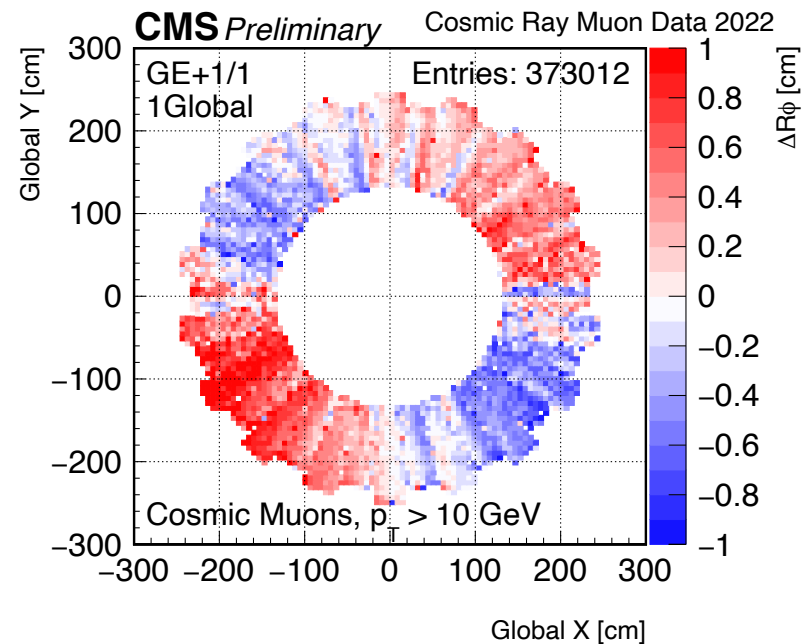
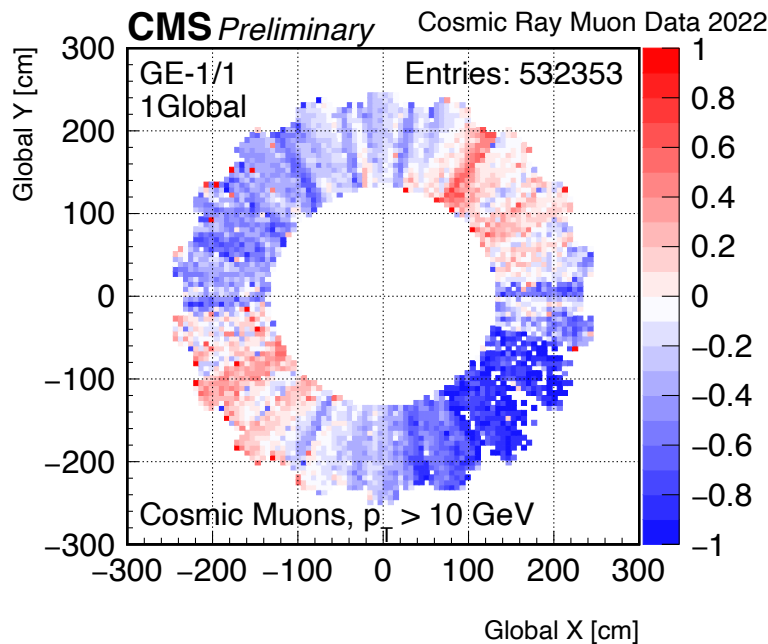
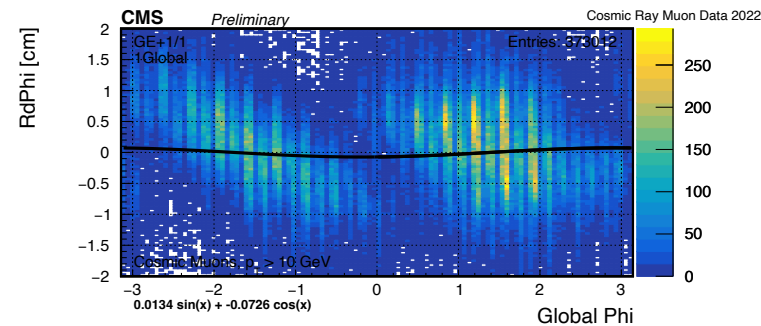
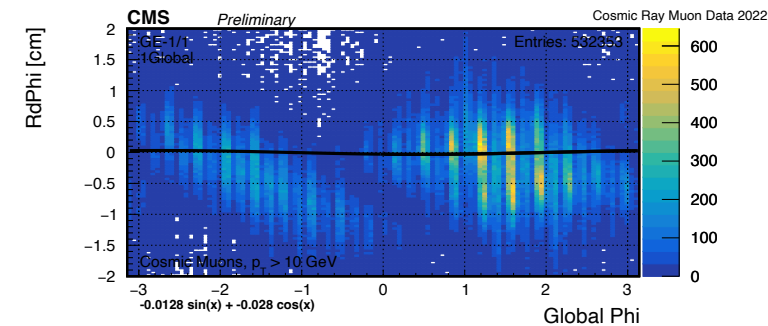


- CRAFT
- GE1/1 using the global tag
run3_data_prompt
(GE1/1 ideal)

CRAFT Alignment GEM from GT



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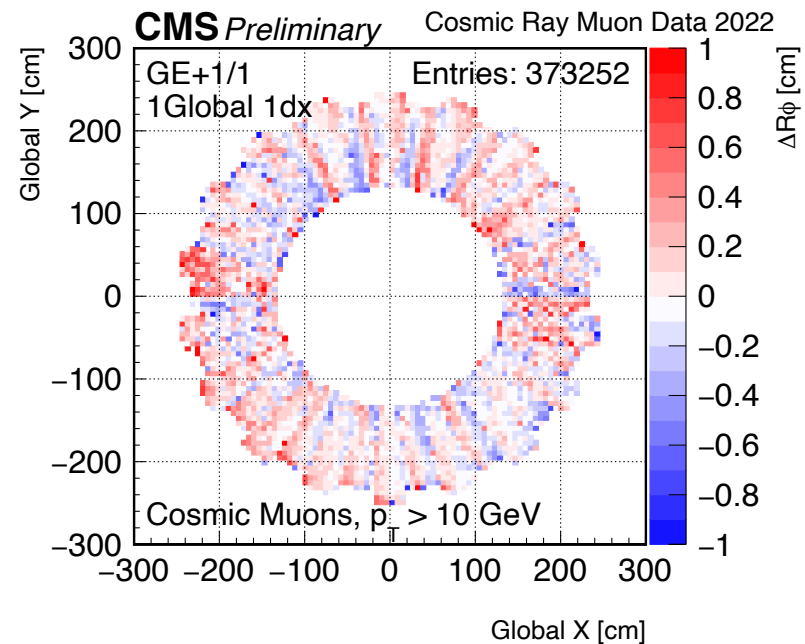
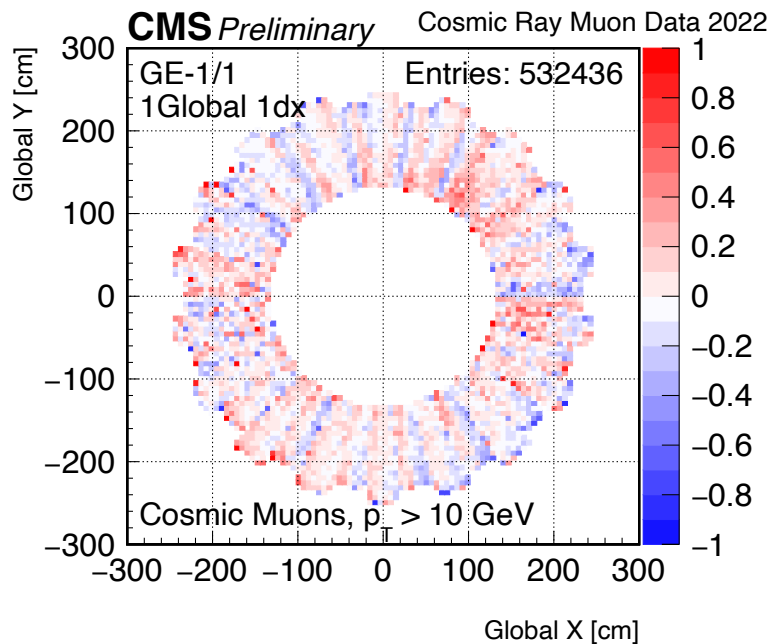
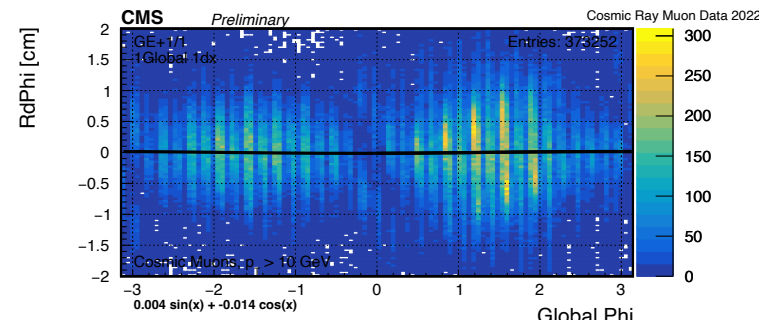
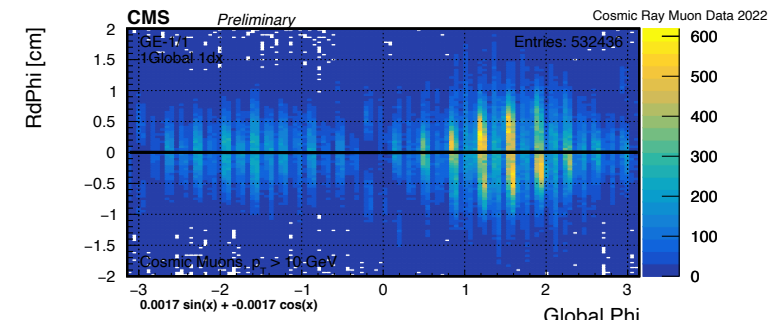


- CRAFT
- GE1/1 using the global tag
run3_data_prompt
(GE1/1 ideal)
- 1 Global shift
alignment

CRAFT Alignment GEM from GT



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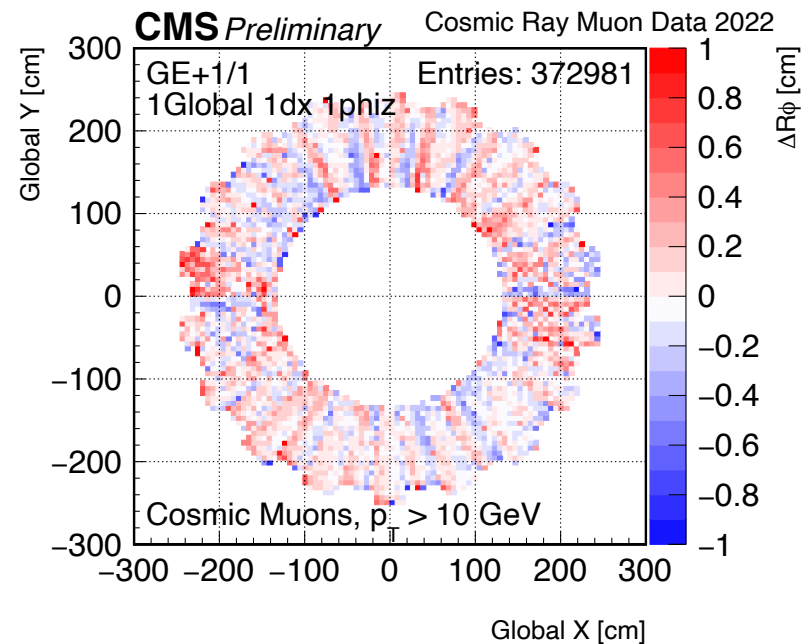
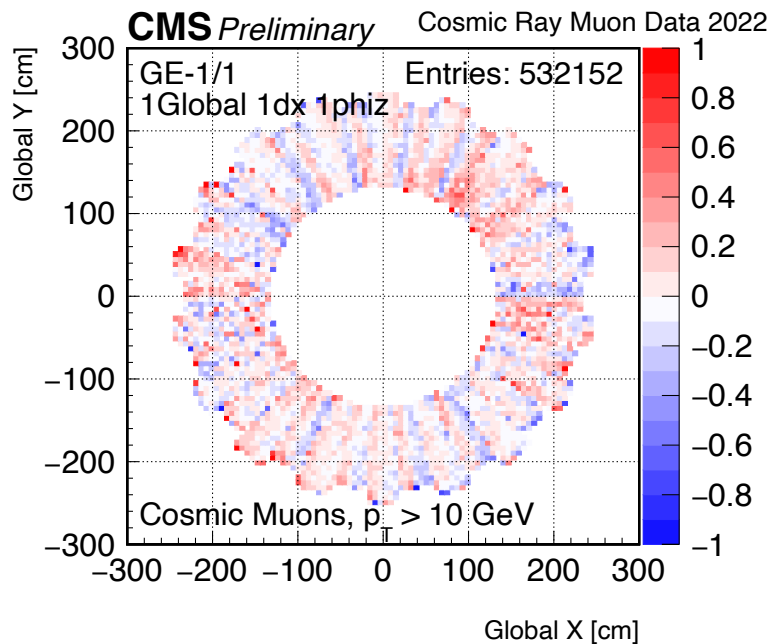
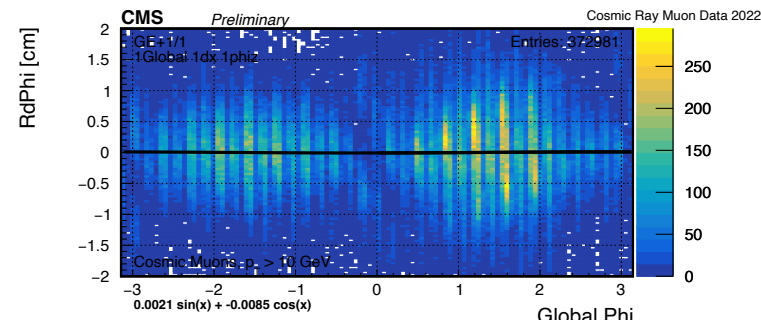
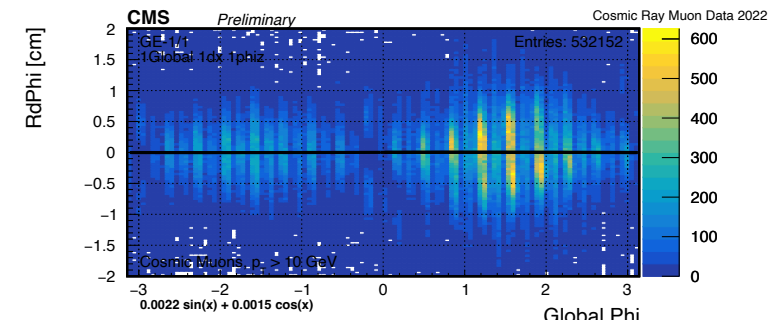


- CRAFT
- GE1/1 using the global tag
run3_data_prompt
(GE1/1 ideal)
- 1 Global shift
alignment + 1 dx

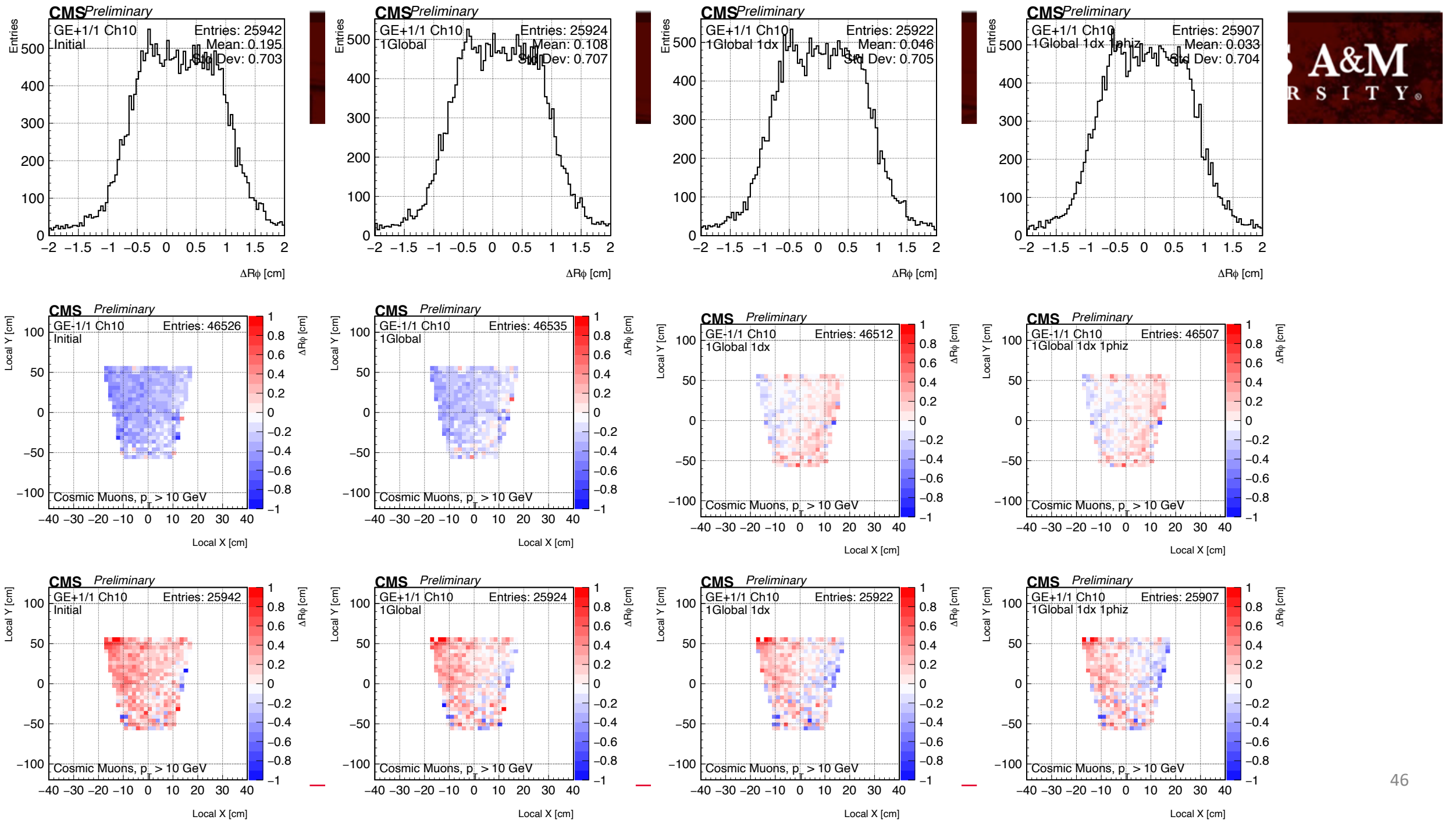
CRAFT Alignment GEM from GT



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- CRAFT
- GE1/1 using the global tag run3_data_prompt (GE1/1 ideal)
- 1 Global shift alignment + 1 dx + 1 phiz

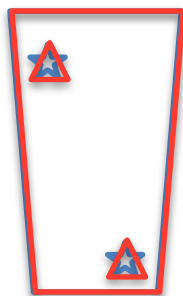


GEM Residual Measurement

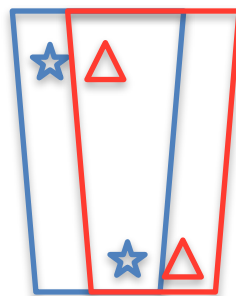


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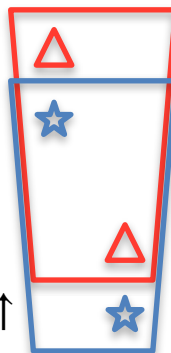
Ideal



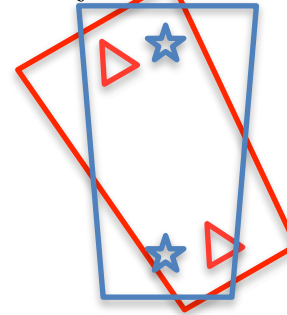
1cm $\delta x \rightarrow$



5cm $\delta y \uparrow$



0.02rad $\delta\phi_z \curvearrowright$



- △ Measured Hit Location
- ★ Expected Hit Location

GEM Detectors

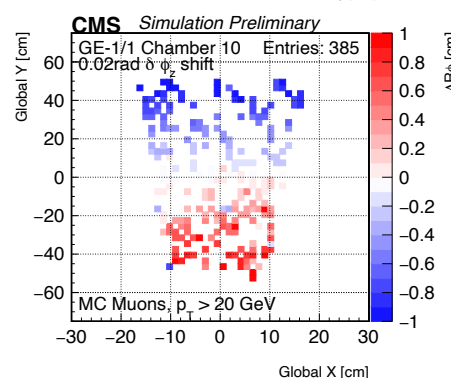
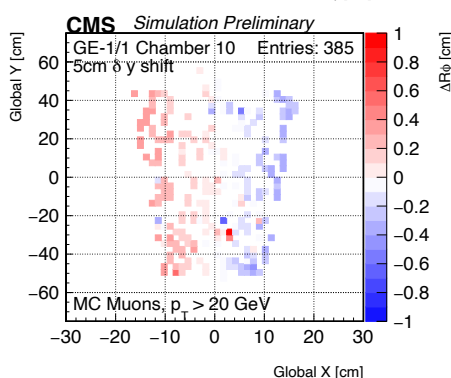
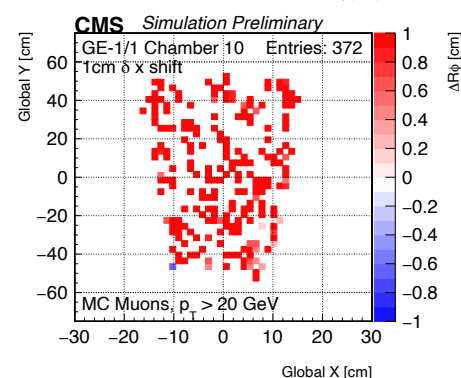
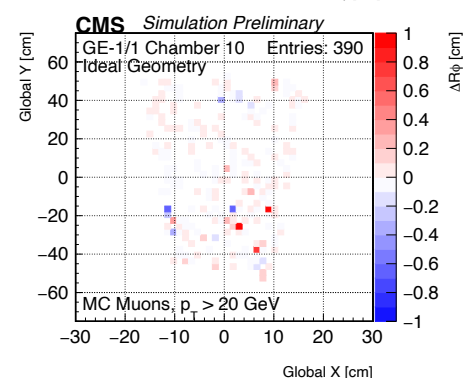
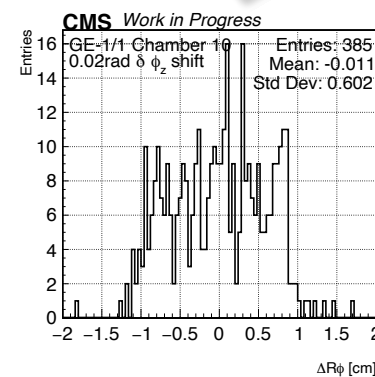
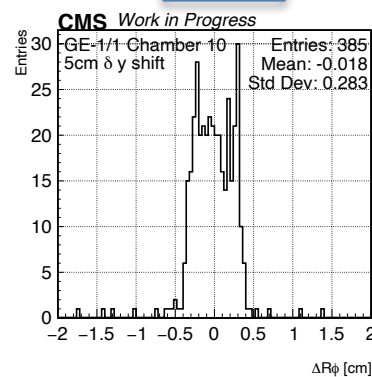
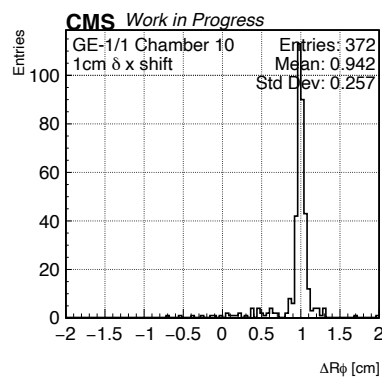
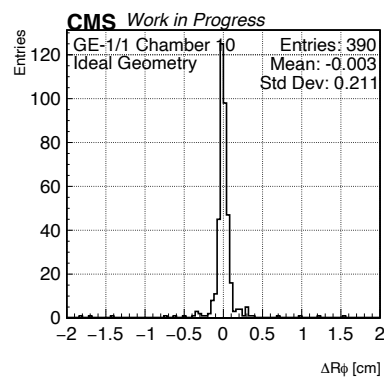
- 8 η partitions / 3 VFATs
- 128 Radial Strips
- Cannot use Δx alone due to strip angles

• $\Delta R\phi = \cos(\theta)\Delta x + \sin(\theta)\Delta y$
Where θ is the strip angle

- GEM local y is reported at the center of the η partition — poor y resolution (η partitions ~ 20 cm tall)

Some basic misalignment patterns and their $\Delta R\phi$ shapes are shown

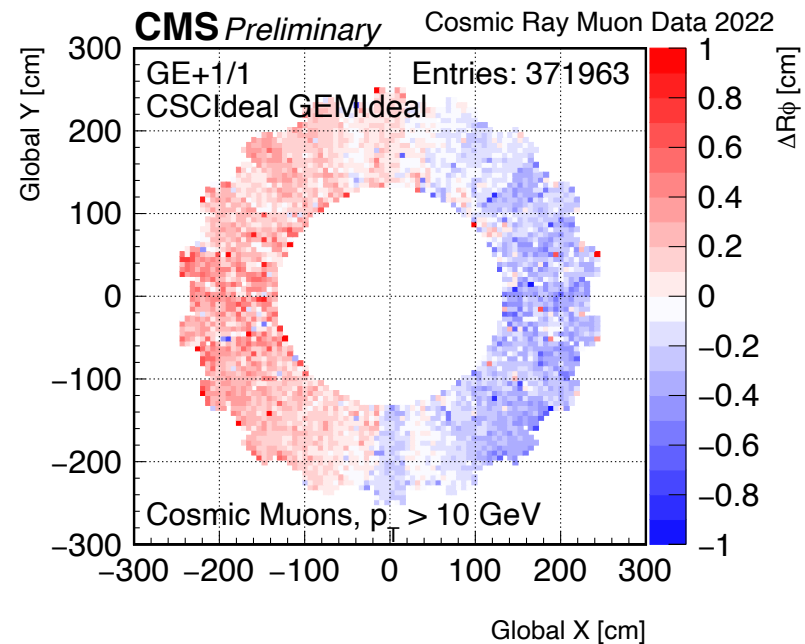
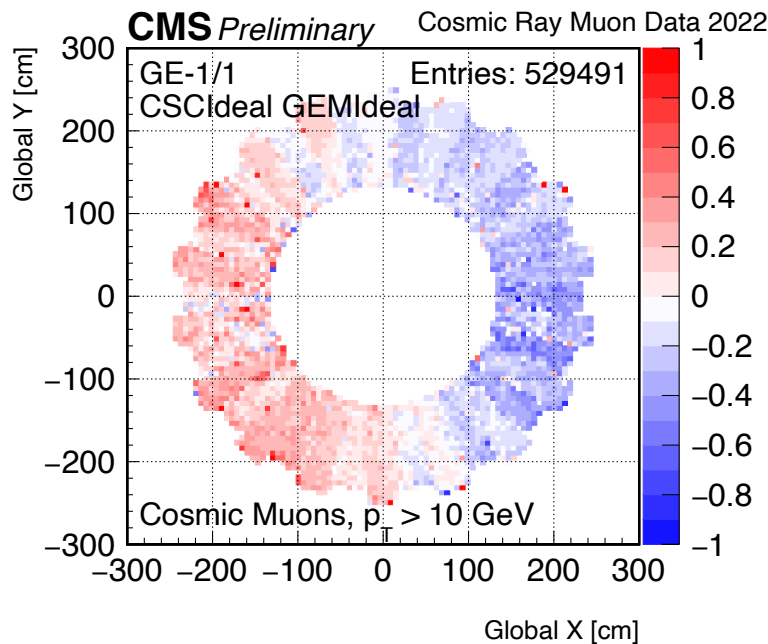
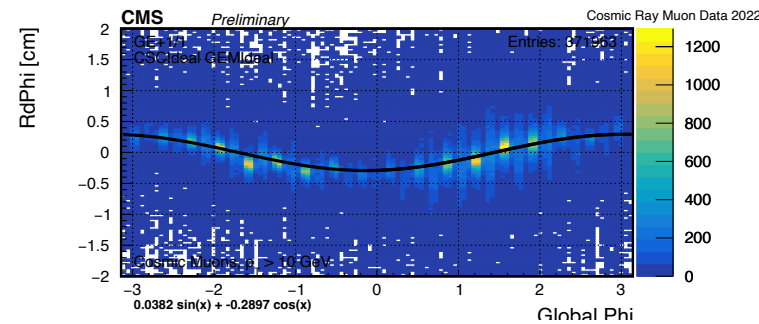
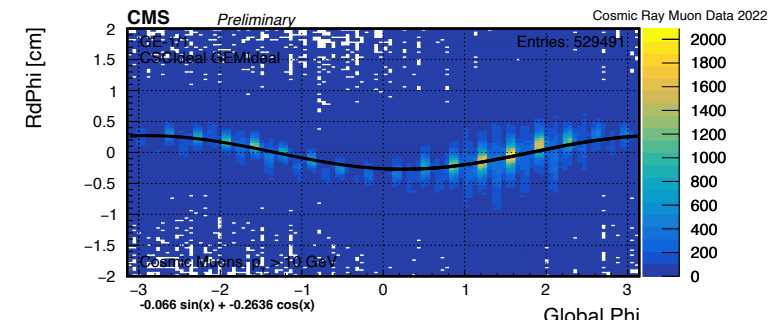
- Plots show simulation with ideal, 1cm δx , 5cm δy , and 0.02rad $\delta\phi_z$ misalignments
- Middle plots show histogram of residuals $\Delta R\phi$
- Bottom plots show residuals $\Delta R\phi$ and where they are on the detector



CRAFT Alignment Both Ideal



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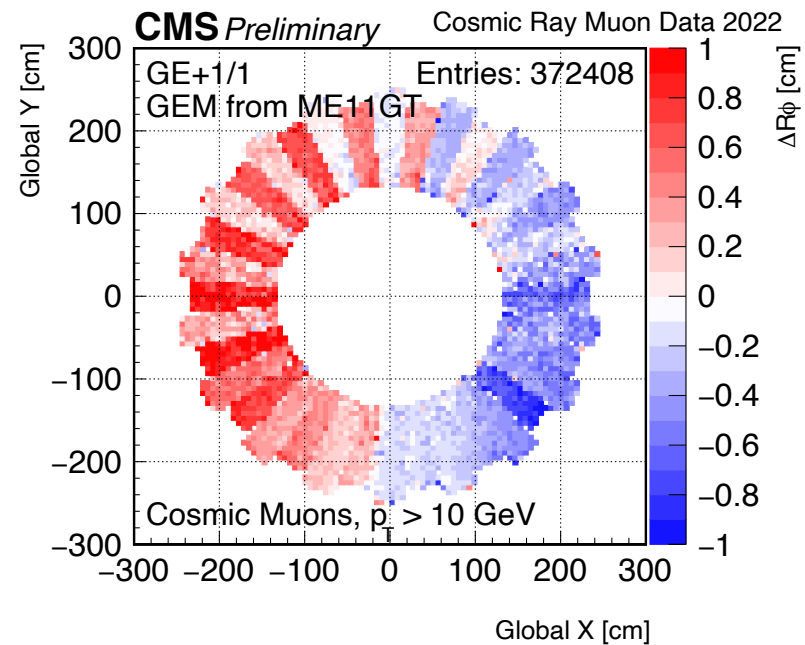
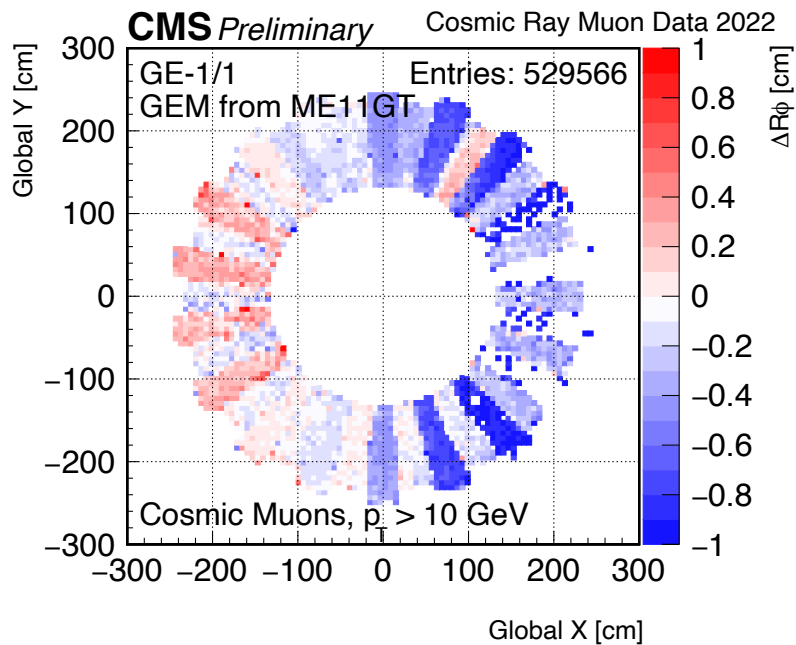
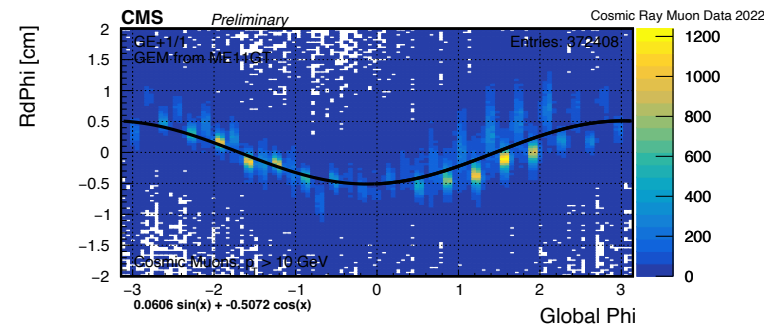
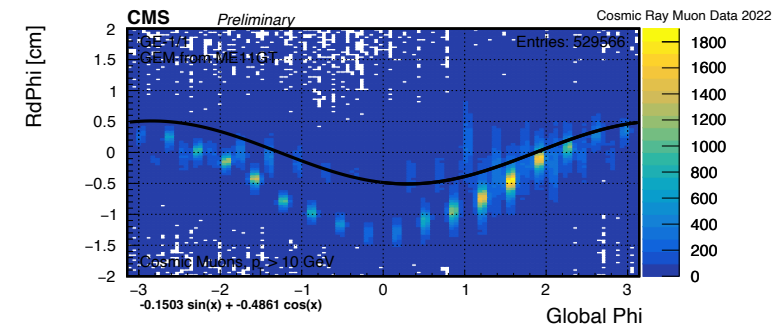


- CRAFT
- GE1/1 and ME1/1 at ideal geometry (much better distribution)
- This was smoking gun
- Maybe GEM should be compared to CSC Geometry

CRAFT Alignment GEM ME1/1 GT



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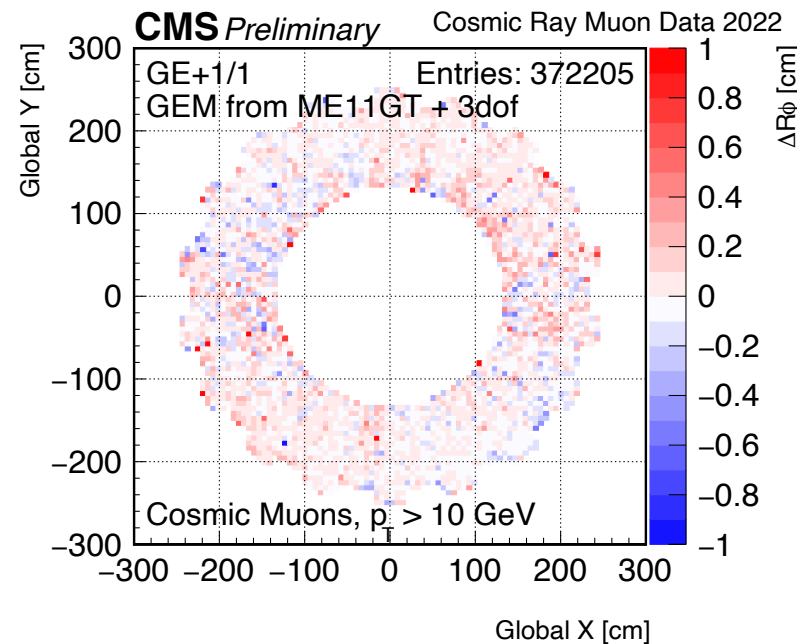
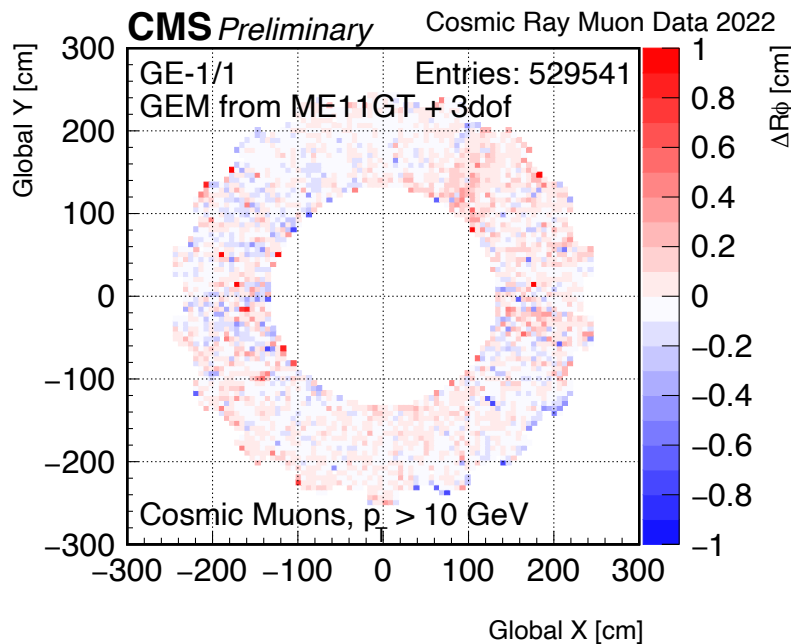
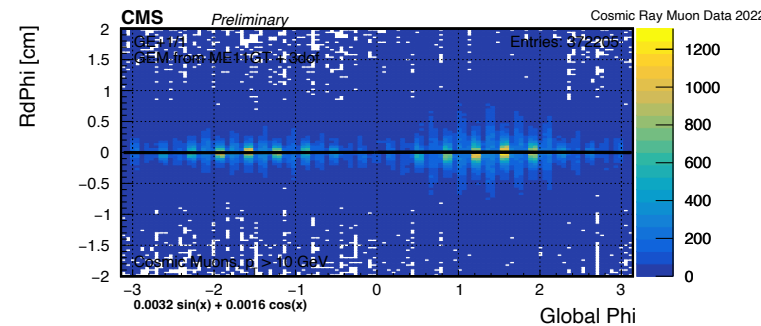
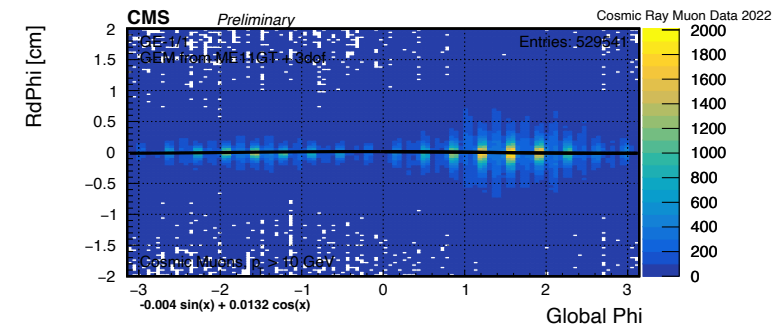


- CRAFT
- GE1/1 using the alignments from ME1/1 in Global Tag 'run3_data_prompt'

CRAFT Alignment GEM ME1/1 GT



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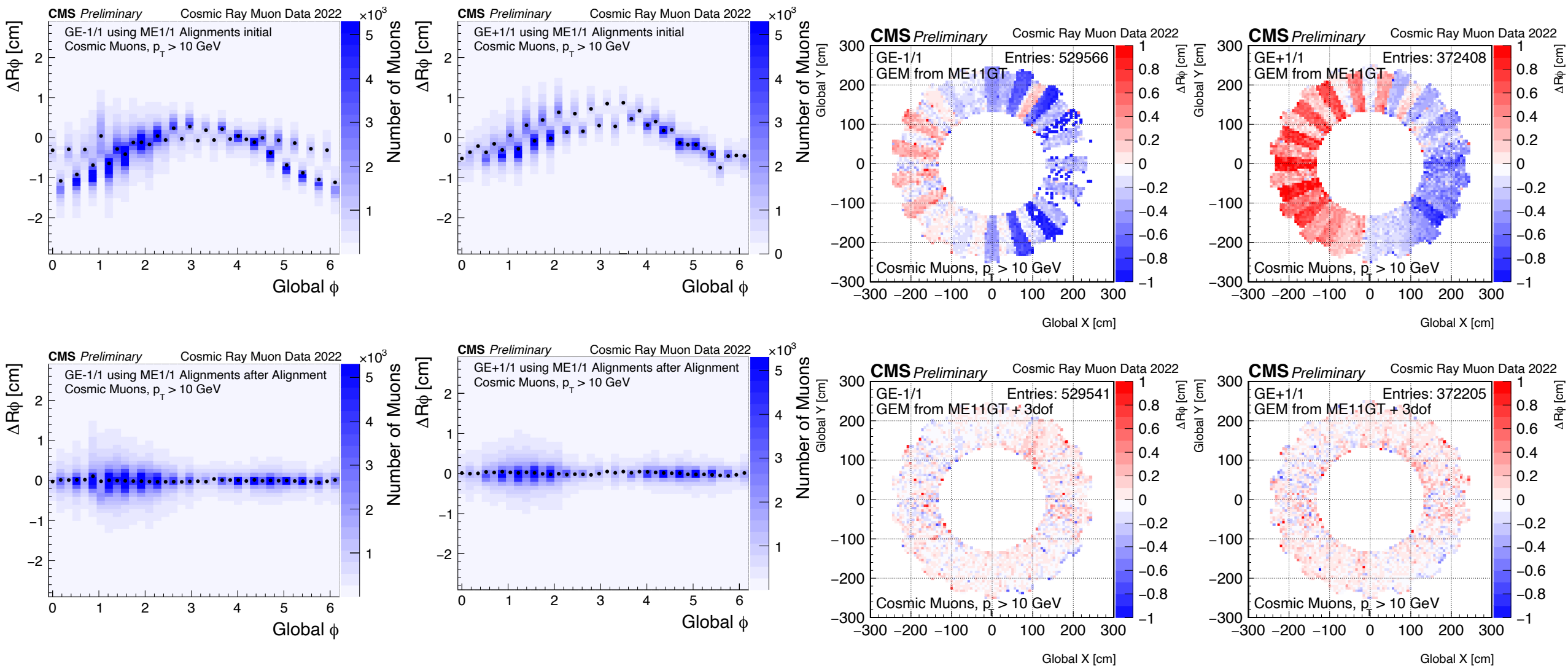


- CRAFT
- GE1/1 using the alignments from ME1/1 in Global Tag 'run3_data_prompt'
- 3DOF alignment, shows better results than when starting from GE1/1 at ideal

Plots for Teruki



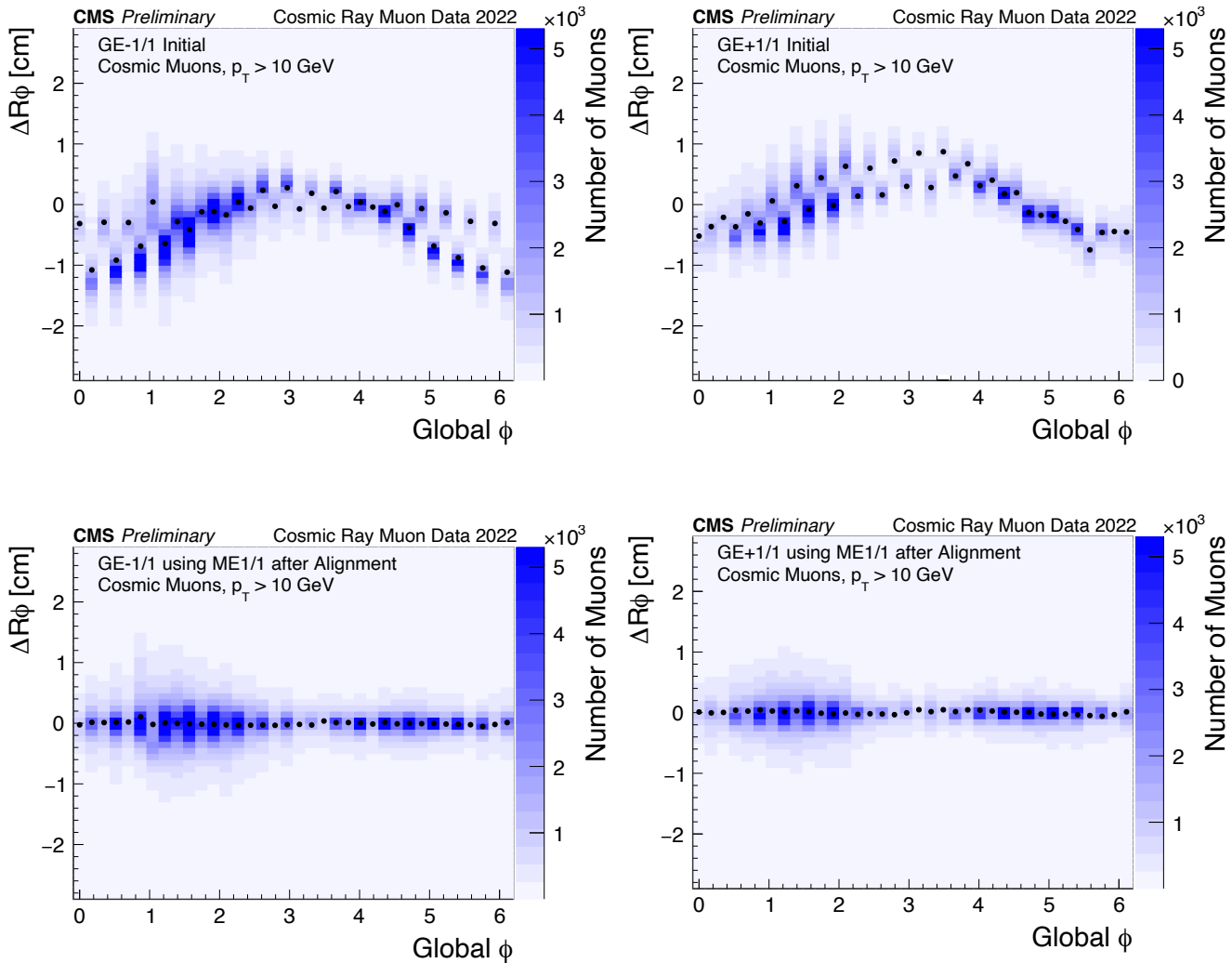
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Plots for Teruki



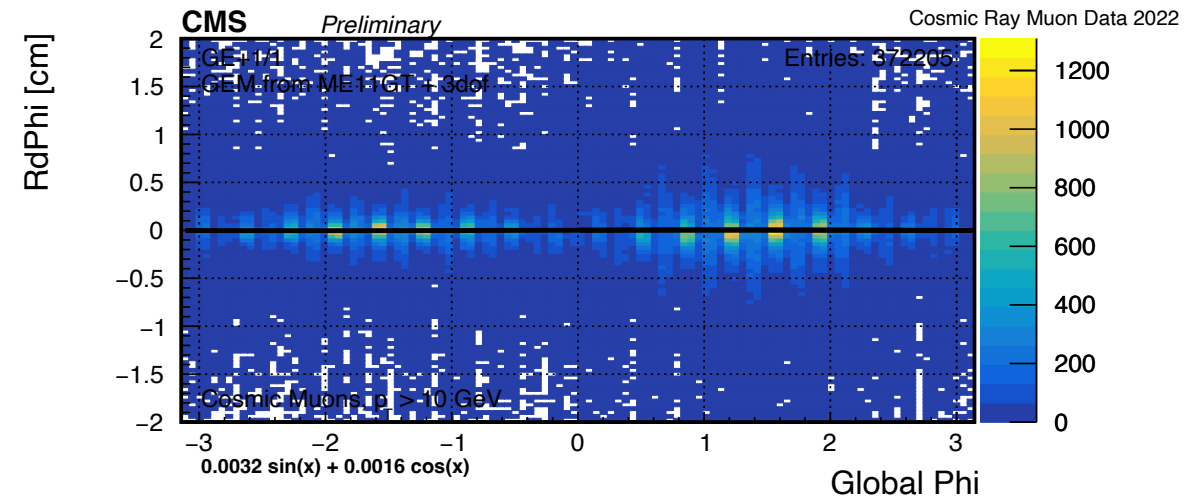
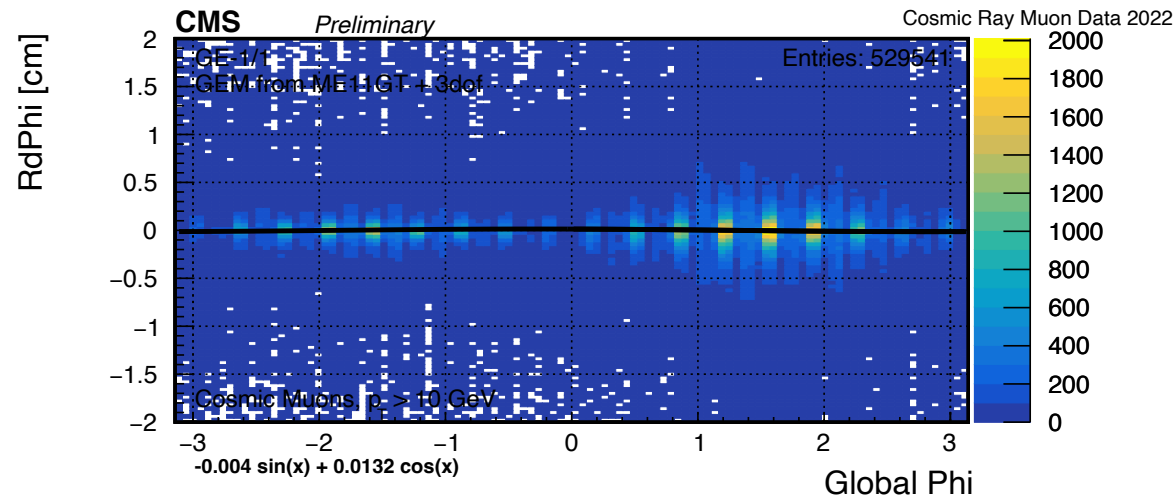
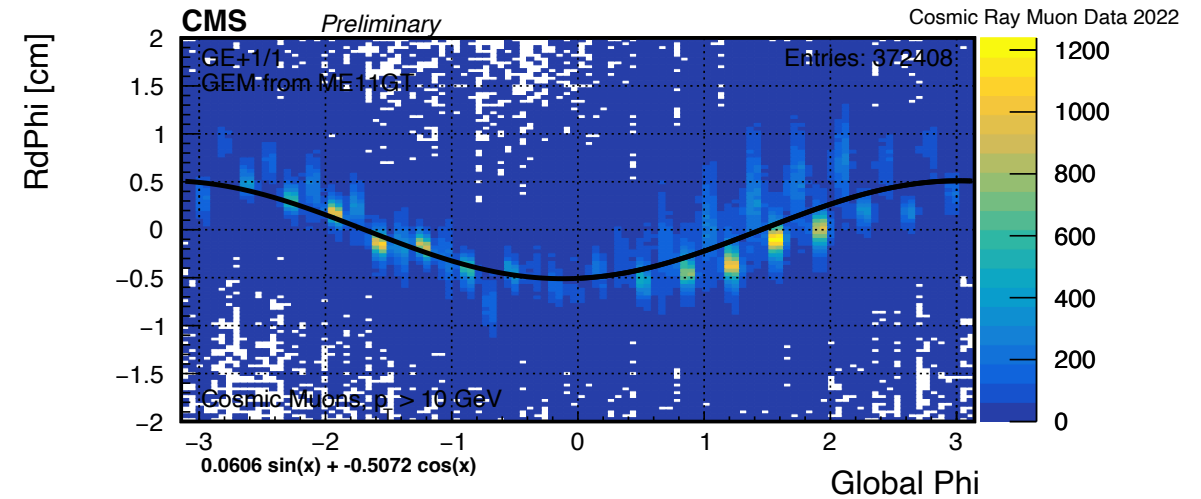
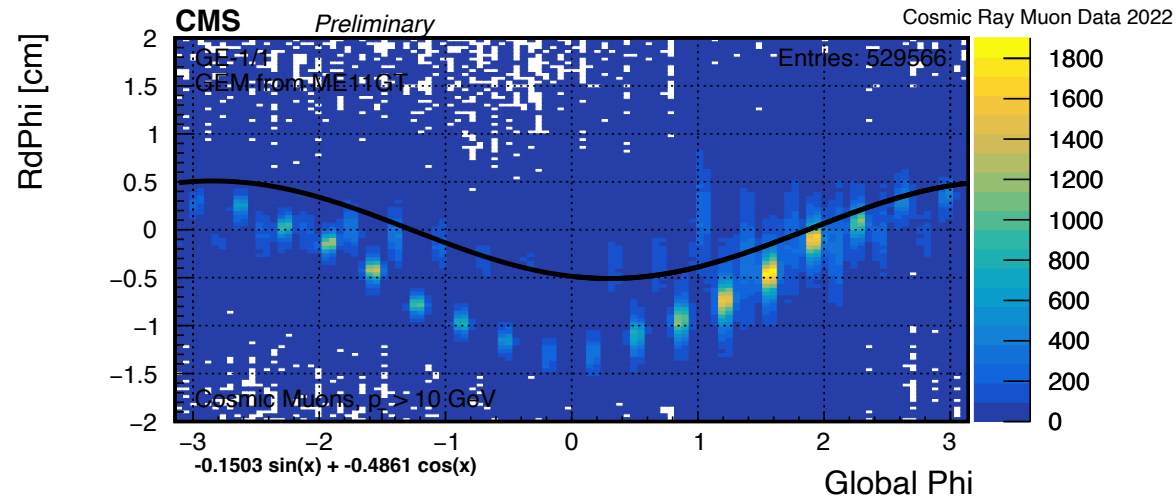
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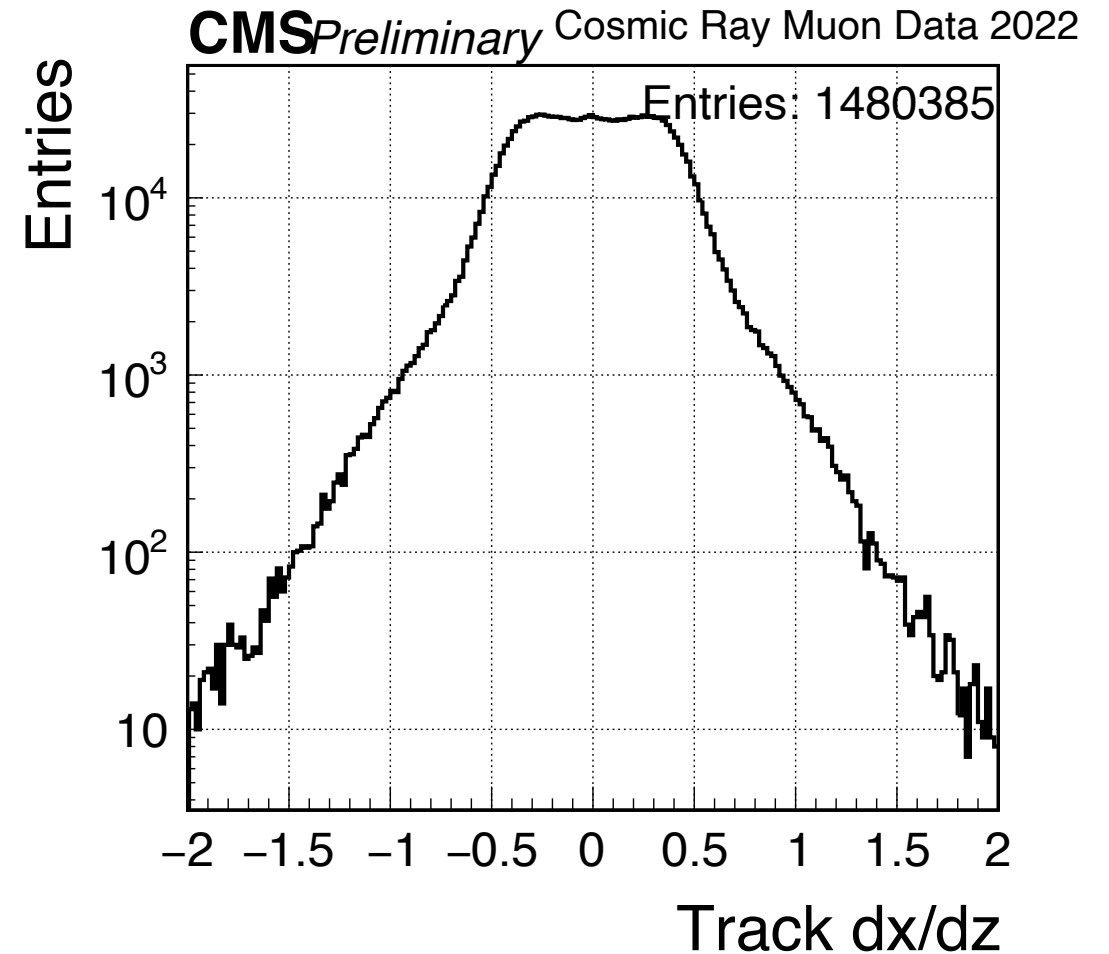
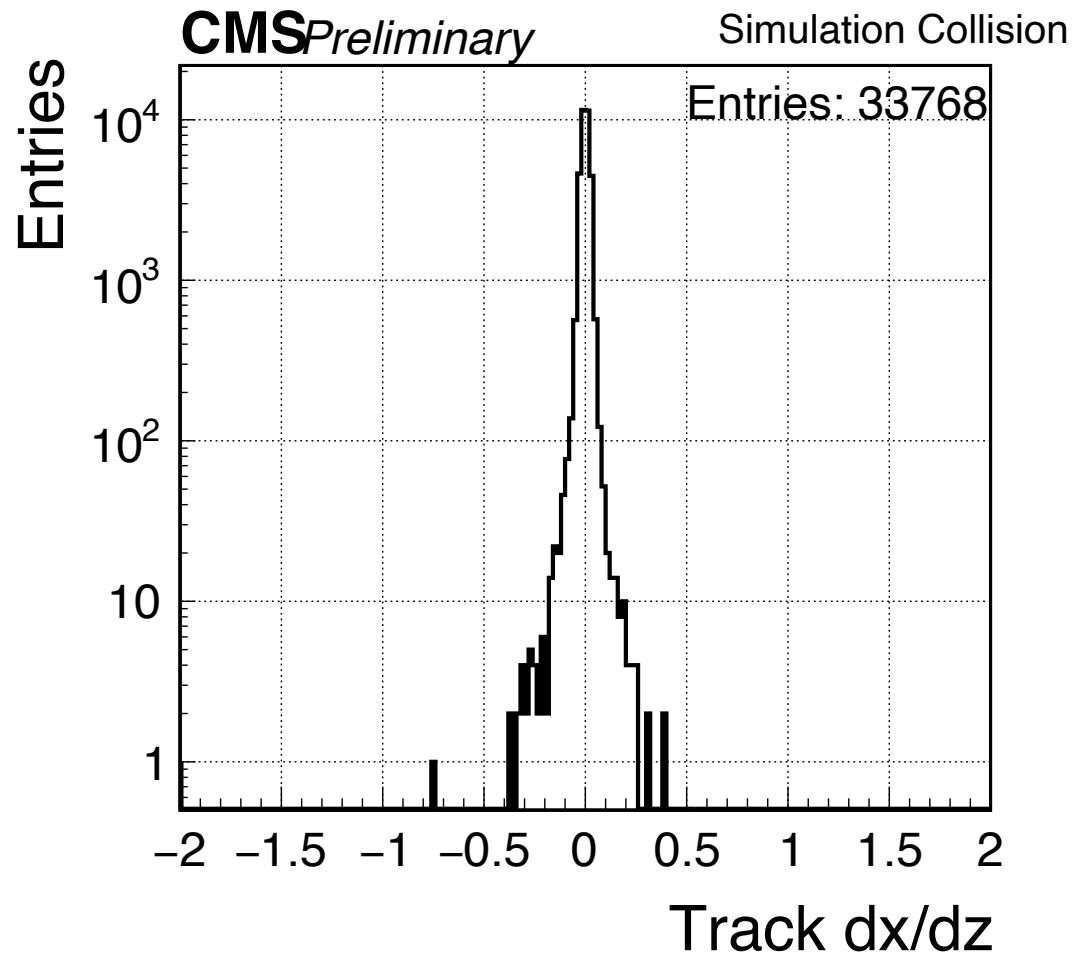


Plots for Teruki

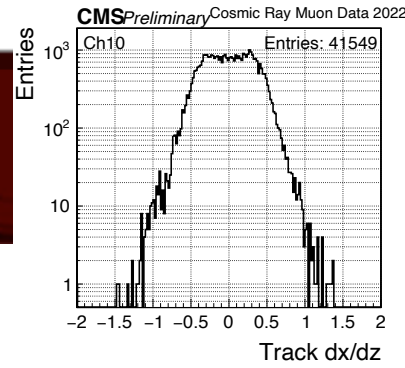
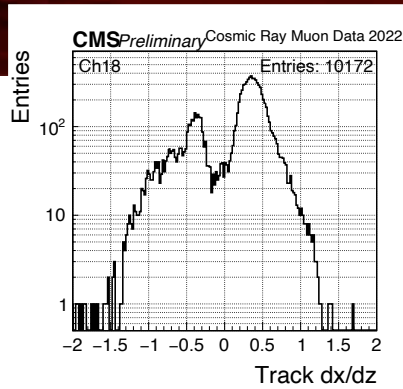


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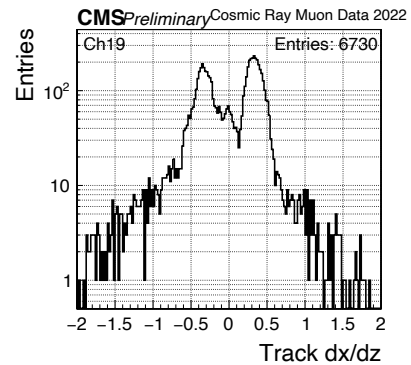
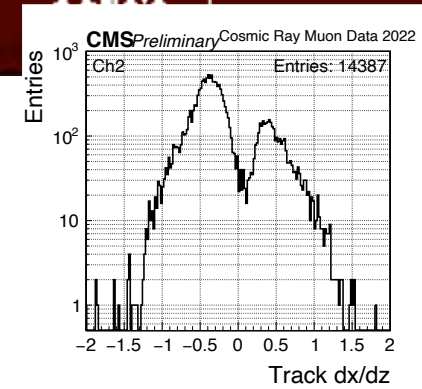




DXDZ

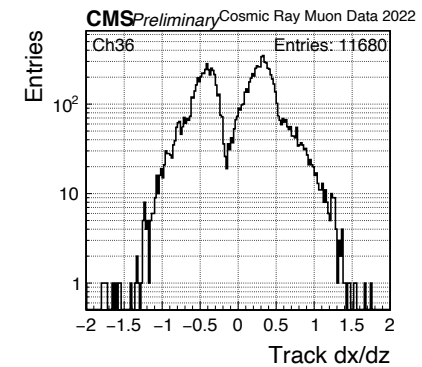
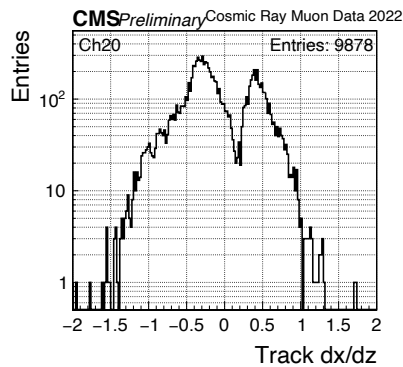
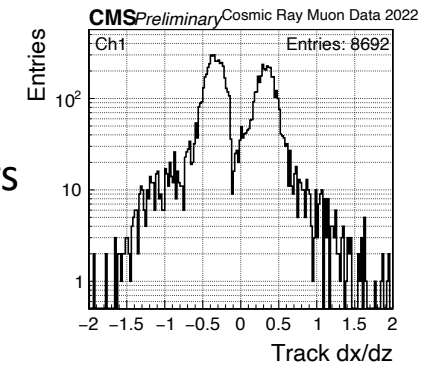


12 o'clock chamber



9 o'clock chambers

3 o'clock chambers



CRAFT — Alignment with/without DY maximum



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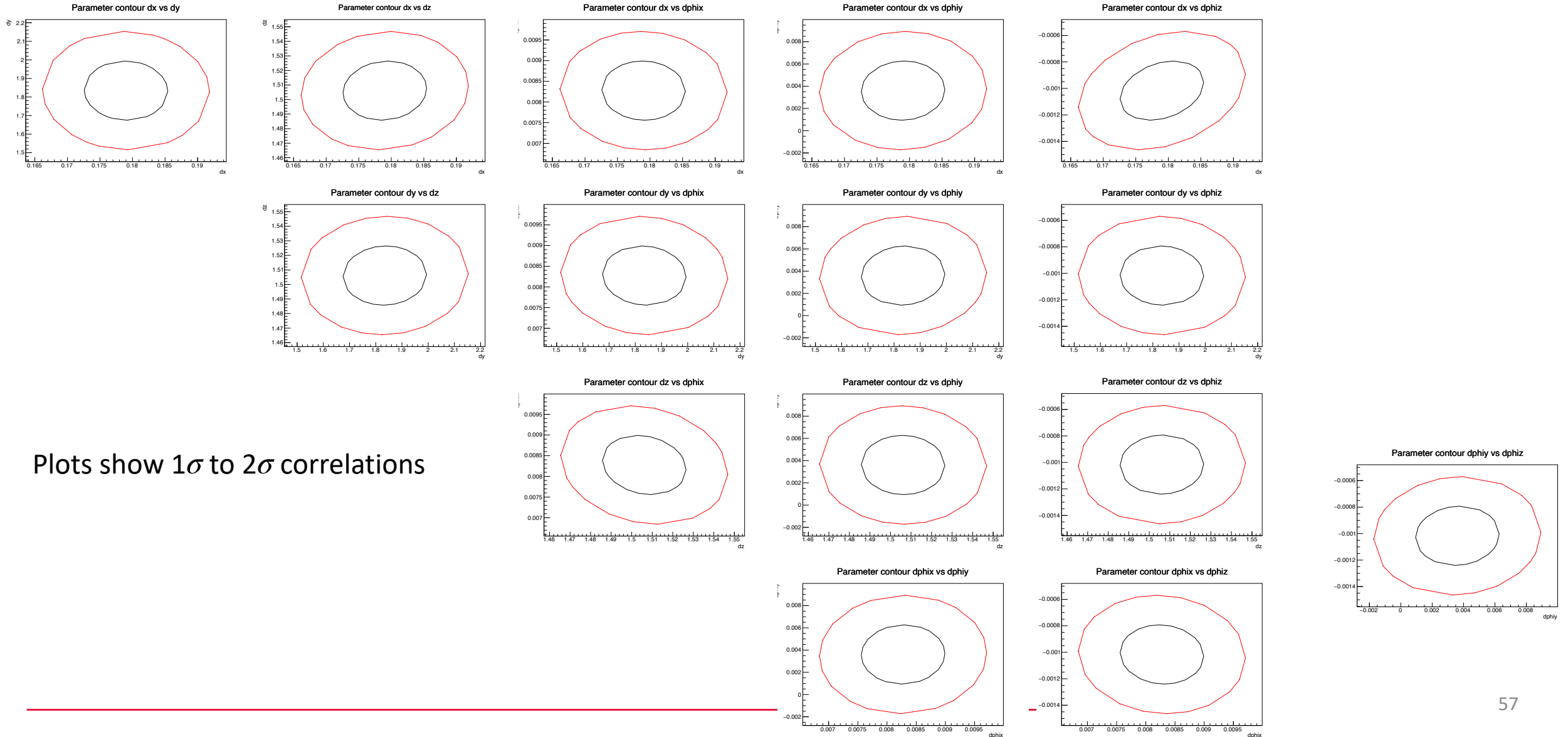
Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
3DOF	0.146 +/- 0.0045	1.0 +/- 0.0097	0.0	0.0	0.0	-1.197E-03 +/- 1.574E-4
4DOF	0.182 +/- 0.0045	1.0 +/- 0.0068	1.542 +/- 0.0142	0	0	-8.291E-04 +/- 1.578E-4
5DOF	0.182 +/- 0.0045	1.0 +/- 0.0068	1.541 +/- 0.0142	0	1.877E-03 +/- 1.880E-03	-8.194E-04 +/- 1.578E-4
6DOF	0.180 +/- 0.0045	1.0 +/- 0.0076	1.503 +/- 0.0144	8.482E-03 +/- 5.053E-04	2.834E-03 +/- 1.881E-03	-9.817E-04 +/- 1.580E-4
3DOF From ME1/1 Geometry	0.194	0.105	2.022	-9.687E-05	-5.739E-04	-1.252E-03
ME1/1 Starting Point	0.304	0.089	2.022	-9.687E-05	-5.739E-04	-1.191E-03

Chamber 10 Layer 1	dx [cm]	dy [cm]	dz [cm]	dphix [rad]	dphiy [rad]	dphiz [rad]
3DOF	0.145 +/- 0.0045	1.647 +/- 0.112	0.0	0.0	0.0	-1.231E-03 +/- 1.575E-04
4DOF	0.181 +/- 0.0045	1.927 +/- 0.112	1.544 +/- 0.0142	0	0	-8.767E-04 +/- 1.578E-4
5DOF	0.181 +/- 0.0045	1.936 +/- 0.113	1.543 +/- 0.0142	0	2.770E-03 +/- 1.883E-03	-8.6294E-04 +/- 1.578E-4
6DOF	0.179 +/- 0.0045	1.835 +/- 0.113	1.506 +/- 0.0144	8.275E-03 +/- 5.061E-04	3.606E-03 +/- 1.883E-03	-1.017E-03 +/- 1.581E-04
6DOF Iter2	0.182 +/- 0.0047	1.034 +/- 0.119	1.824 +/- 0.0156	5.936E-03 +/- 5.690E-04	6.415E-03 +/- 2.016E-03	-1.3992E-03 +/- 1.716E-04
3DOF From ME1/1 Geometry	0.194	0.105	2.022	-9.687E-05	-5.739E-04	-1.252E-03
ME1/1 Starting Point	0.304	0.089	2.022	-9.687E-05	-5.739E-04	-1.191E-03

CRAFT — Alignment Correlations



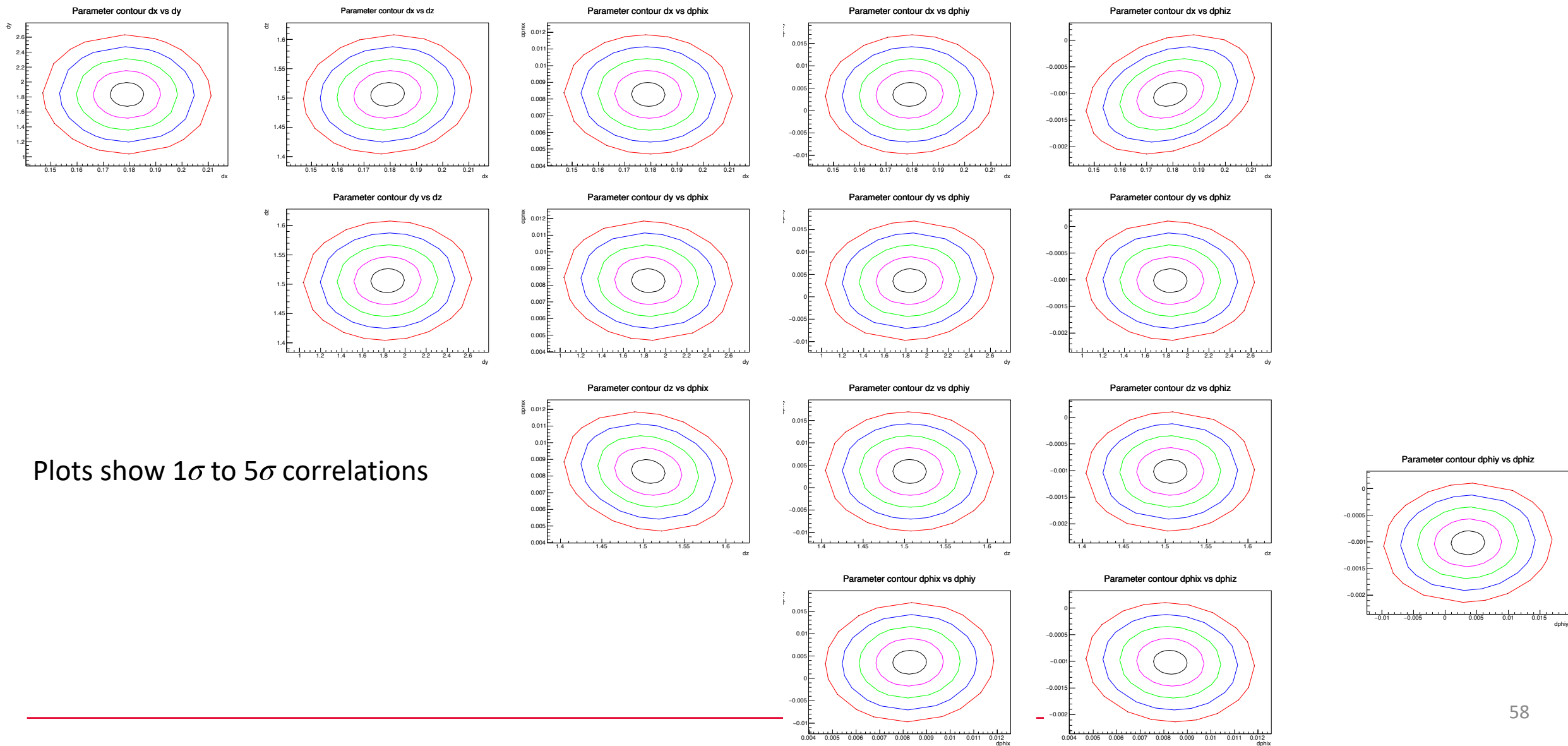
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CRAFT — Alignment Correlations Chamber 10 Layer 1



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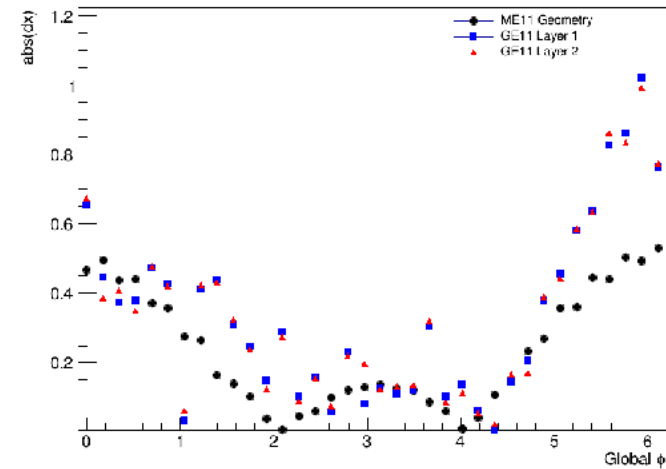


CRAFT — Alignment Relative to ME1/1 Negative Endcap

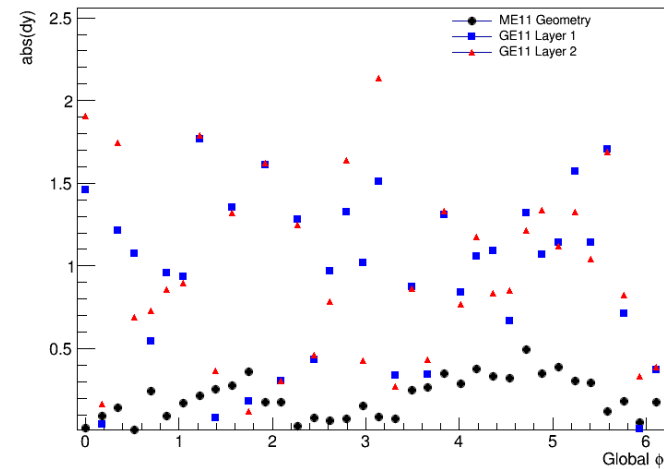


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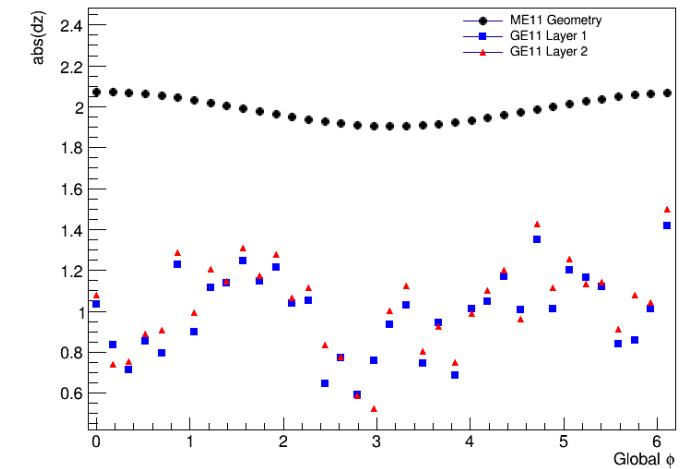
ME11 and GE11 Positon R-1 dx



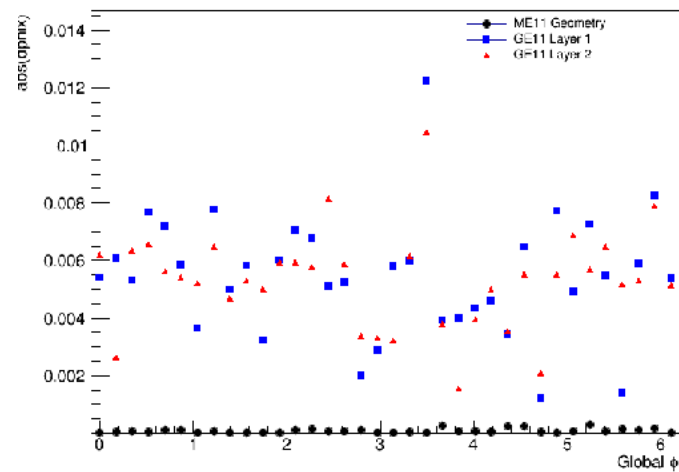
ME11 and GE11 Positon R-1 dy



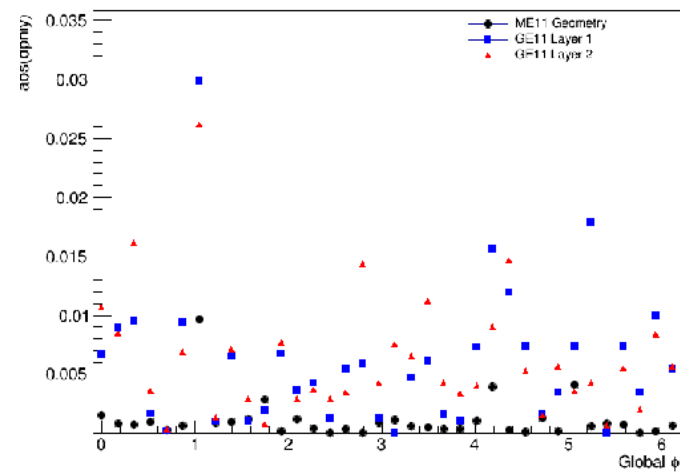
ME11 and GE11 Positon R-1 dz



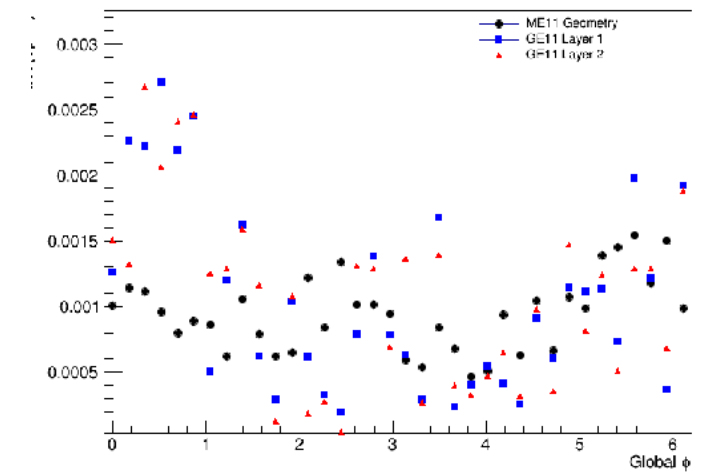
ME11 and GE11 Positon R-1 dphiX



ME11 and GE11 Positon R-1 dphiY



ME11 and GE11 Positon R-1 dphiZ

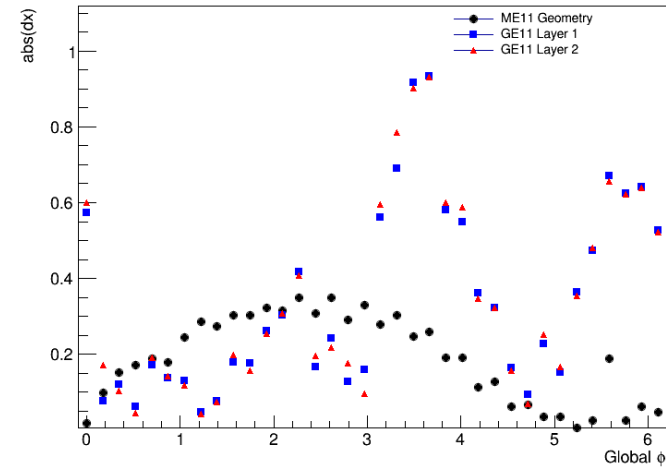


CRAFT — Alignment Relative to ME1/1 Positive Endcap

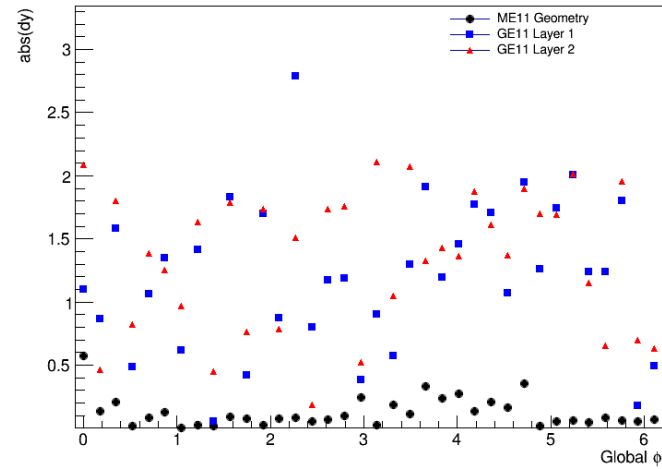


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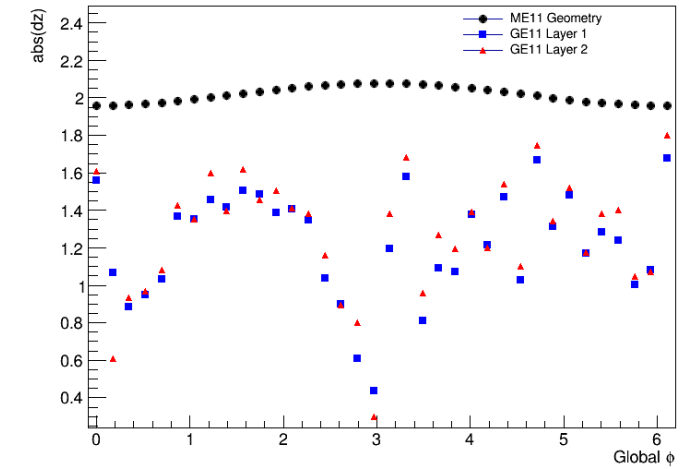
ME11 and GE11 Positon R1 dx



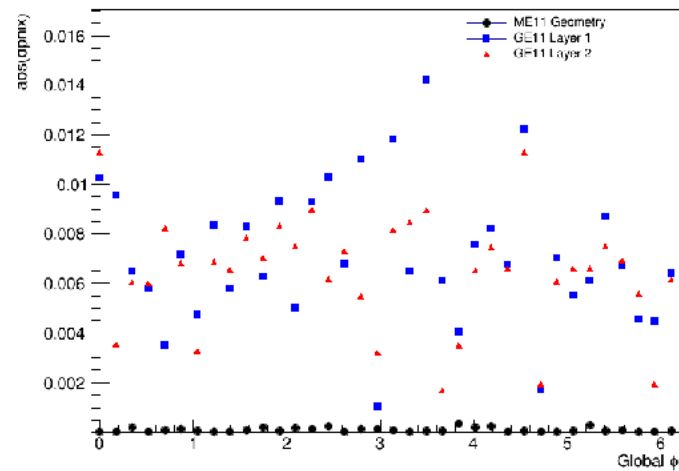
ME11 and GE11 Positon R1 dy



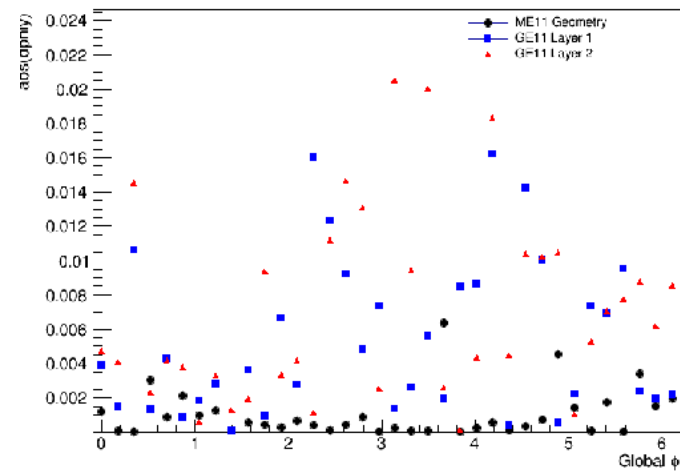
ME11 and GE11 Positon R1 dz



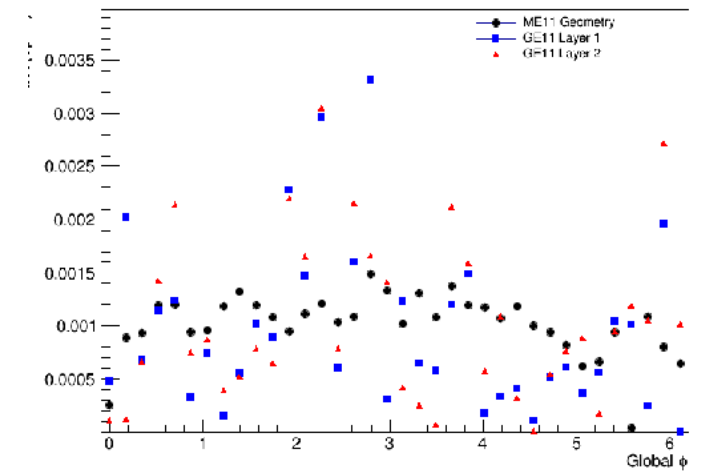
ME11 and GE11 Positon R1 dphix



ME11 and GE11 Positon R1 dphiy



ME11 and GE11 Positon R1 dphiz

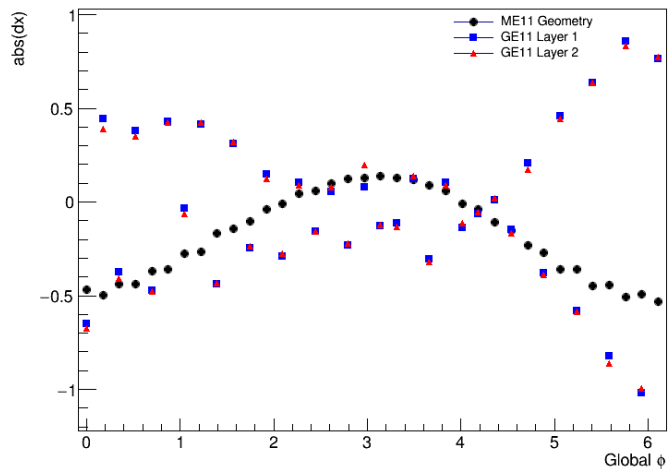


CRAFT — Alignment Relative to ME1/1 Negative Endcap

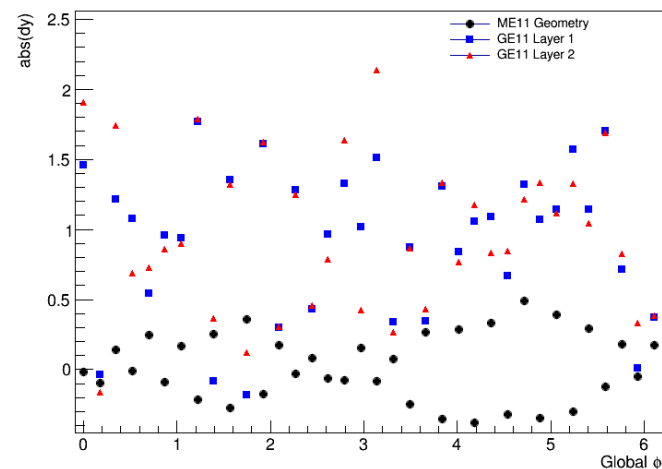


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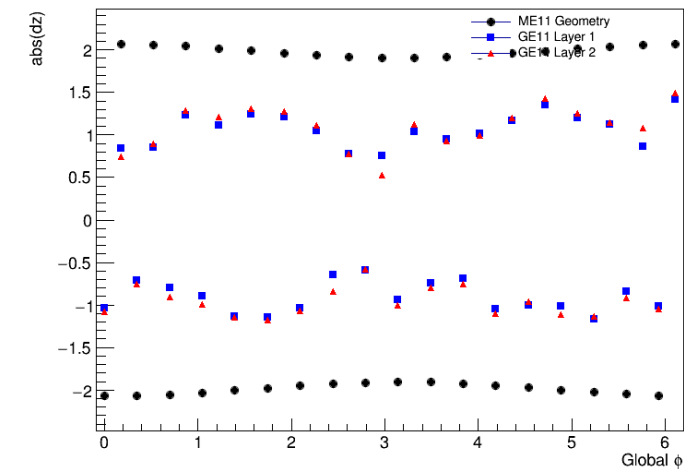
ME11 and GE11 Positon R-1 dx



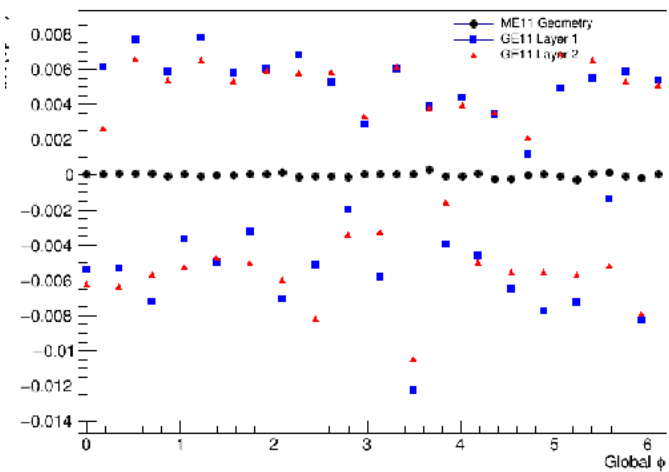
ME11 and GE11 Positon R-1 dy



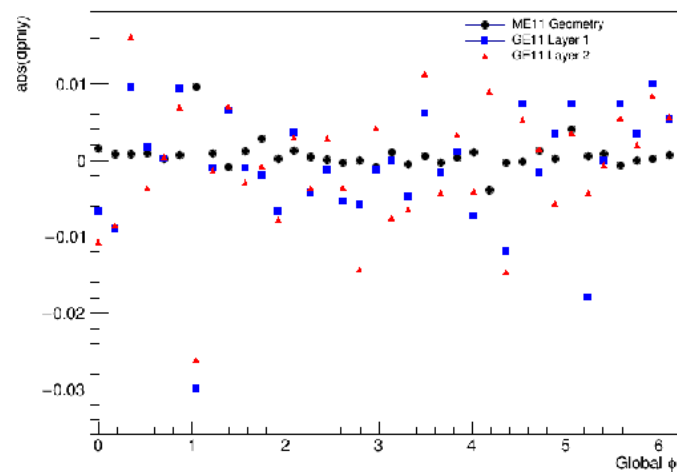
ME11 and GE11 Positon R-1 dz



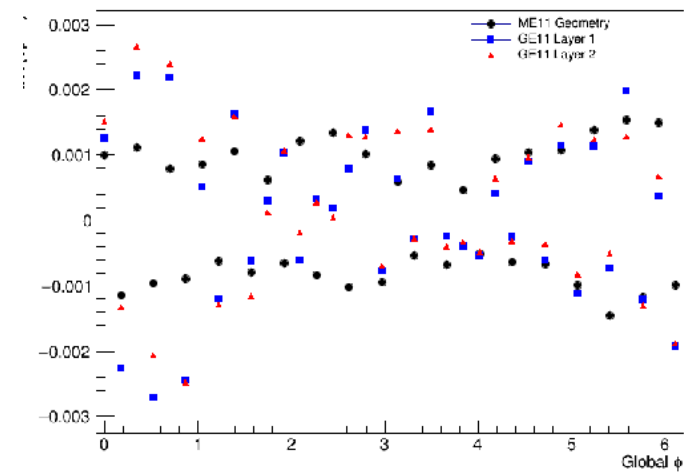
ME11 and GE11 Positon R-1 dphi



ME11 and GE11 Positon R-1 dphiy



ME11 and GE11 Positon R-1 dphiz

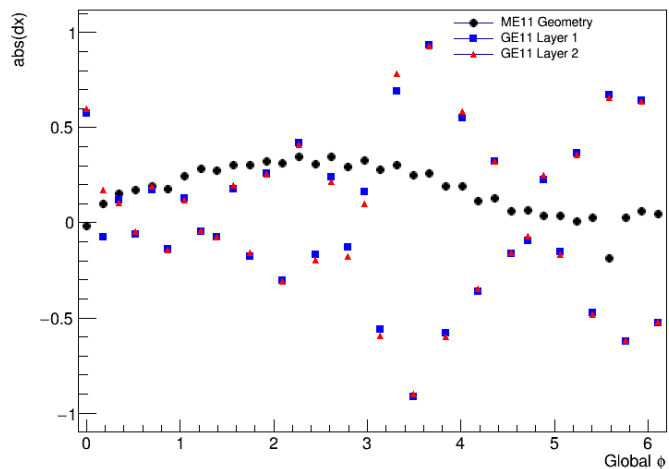


CRAFT — Alignment Relative to ME1/1 Positive Endcap

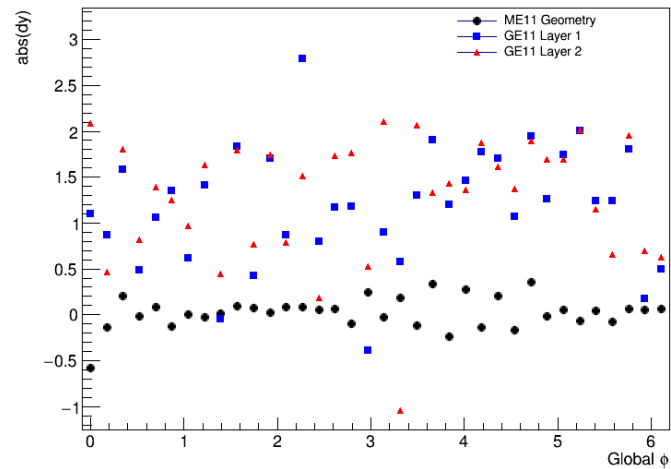


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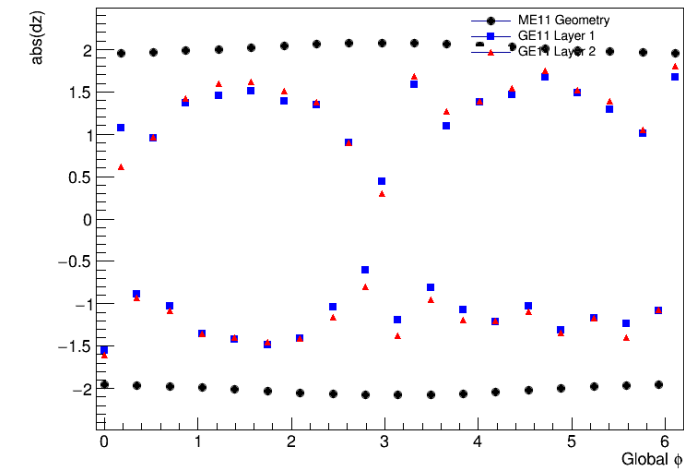
ME11 and GE11 Positon R1 dx



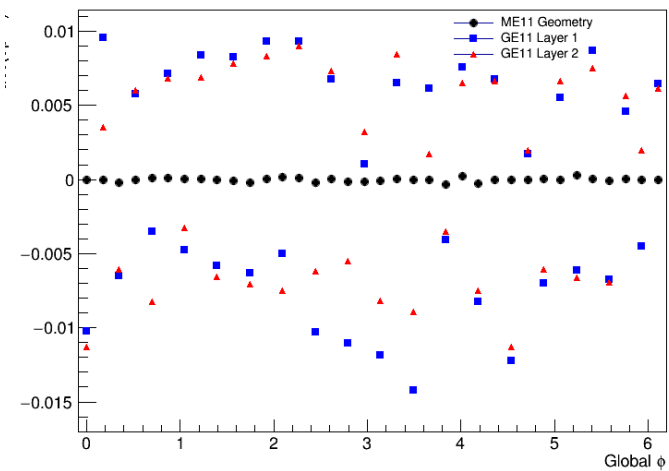
ME11 and GE11 Positon R1 dy



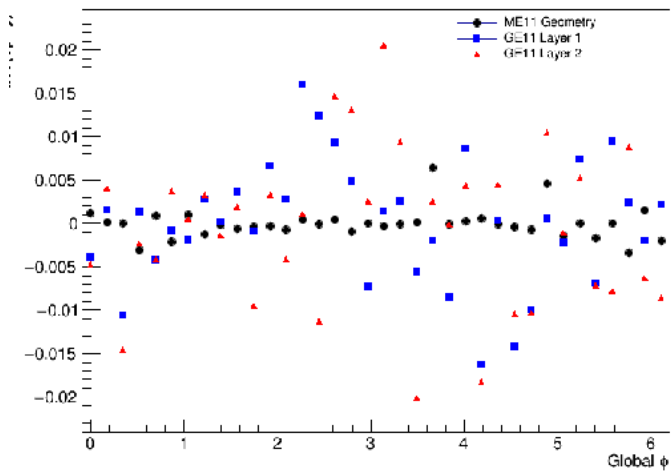
ME11 and GE11 Positon R1 dz



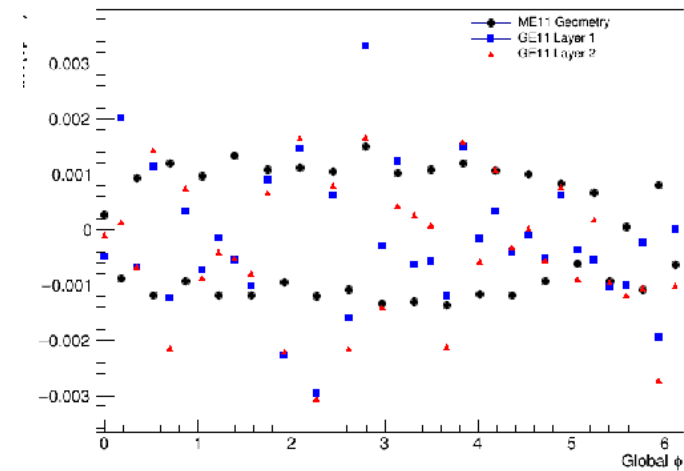
ME11 and GE11 Positon R1 dphix



ME11 and GE11 Positon R1 dphiy



ME11 and GE11 Positon R1 dphiz



GEM Specs



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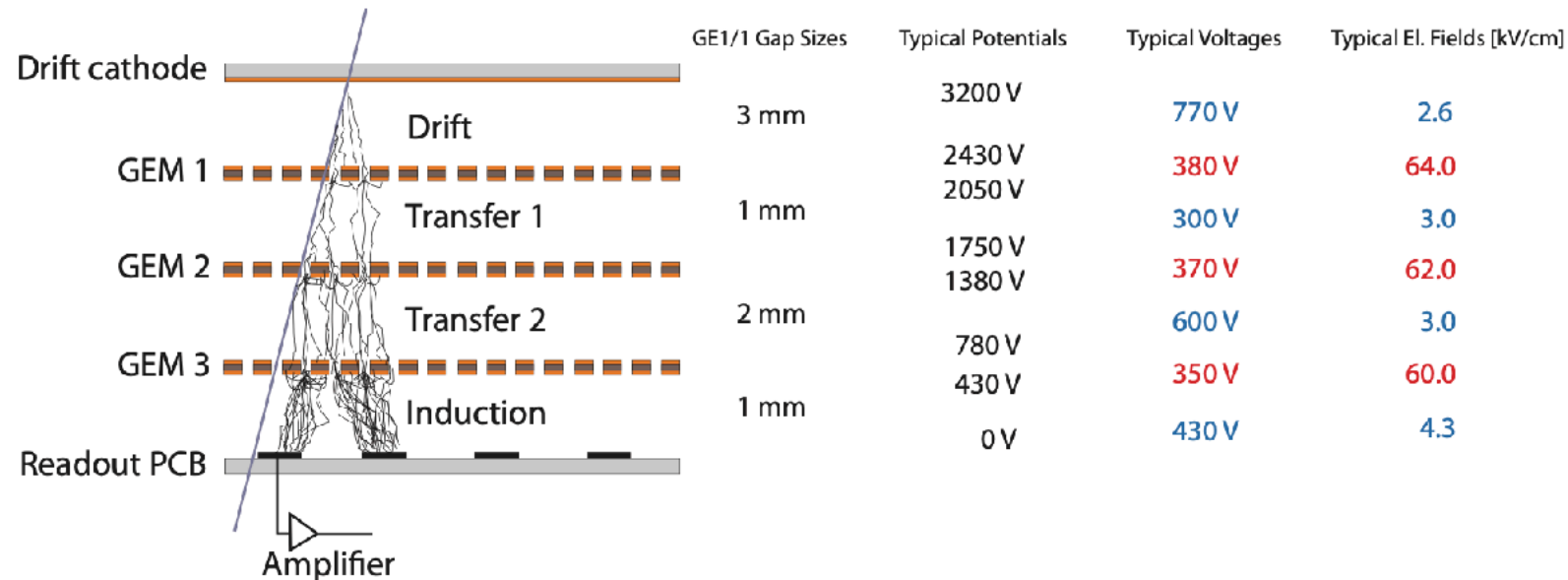


Figure 2.2: Principle of operation of a generic triple-GEM chamber and definition of drift, transfer, and signal induction gap regions within the detector [10]. The columns on the right give the actual gap sizes in the GE1/1. They also list typical values for electric potentials on the seven electrodes and typical values for voltages and electric fields across the four gaps (blue) and the three foils (red) if the nominal potential of 3200 V for operation in Ar/CO₂ 70:30 is applied to the drift cathode.

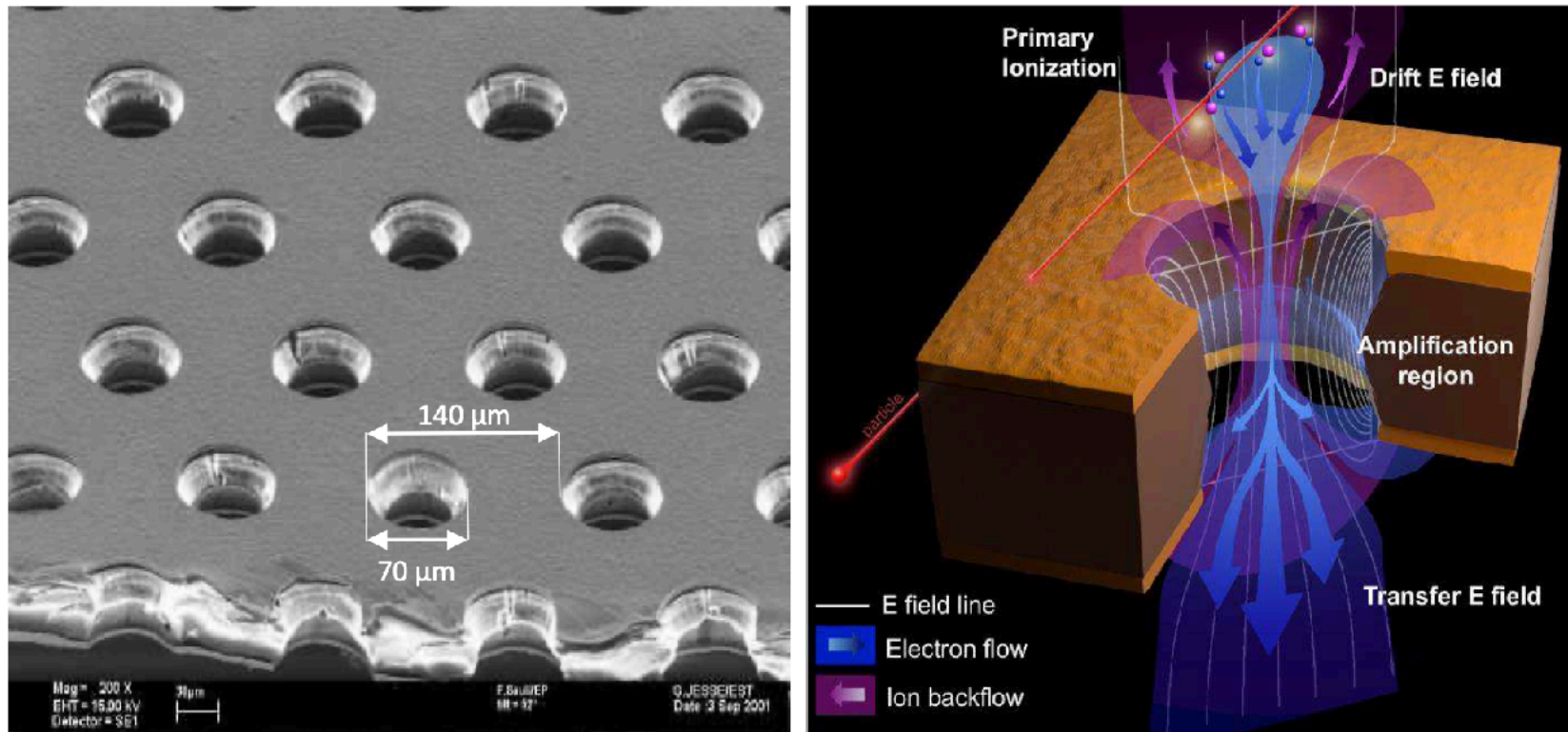


Figure 2.1: Scanning Electron Microscope (SEM) picture of a GEM foil (left) [10] and schematic view of the electric field lines (white), electron flow (blue), and ion flow (purple) through a bi-conical GEM hole (right). The outer diameters of the hole are 70 μm and the inner diameter is 50 μm ; the hole pitch is 140 μm .

Table 7.4: General specifications of the GE1/1 gas system.

Detector gas volume	Volume [cm ³]	Gas flow [Volume/h]	Gas mixture
Single GE1/1 detector - Long	3120	1	<i>Ar/CO₂/CF₄</i> 45:15:40
GE1/1 superchamber - Long	6240		
Endcap disk	224640		
Full installation	449280		

Long/Short



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